KINETIC ACCELERATION OF NON-EQUILIBRIUM STEADY STATES

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Abstract. We ask whether kinetic acceleration (lifting) survives away from equilibrium, and how the convergence rate deteriorates with the strength η of non-equilibrium driving. We study strongly continuous semigroup generators of the form $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ on a Hilbert space, under structural assumptions of microscopic dissipation, centered slow–fast coupling, flux regularity, and perturbation control; here $\gamma > 0$ is friction, $\mathcal{R}$ is self-adjoint and non-positive, $\mathcal{V}_{\text{skew}}$ is skew-adjoint, and $\eta \geq 0$ measures departure from detailed balance.

Within the Dolbeault–Mouhot–Schmeiser hypocoercivity framework [13, 14], we construct a gap-adapted Lyapunov functional and obtain an explicit rate lower bound with a computable perturbation threshold $\eta^*(\gamma)$ valid under these structural hypotheses. For model families whose structural constants remain uniformly controlled as the spectral gap $\lambda(Q_O)$ of the reversible backbone tends to zero, the gap-adapted choice of the DMS corrector recovers the kinetic scale $\mathcal{O}(\sqrt{\lambda(Q_O)})$ at optimal friction, improving on the diffusive $\mathcal{O}(\lambda(Q_O))$ scale. A complementary conditional overdamped ceiling shows that this acceleration is genuinely a finite-friction effect and that the $\sqrt{\lambda(Q_O)}$ exponent is optimal whenever the ceiling assumptions hold. We also give a constructive reverse-adiabatic-elimination recipe for building lifted generators. Applications to a convex-potential underdamped Langevin family and a boundary-driven quantum model are developed, providing rigorous infinite- and finite-dimensional verifications of the structural hypotheses behind the lower bound, together with numerical evidence for the $\sqrt{\lambda(Q_O)}$ mechanism.

1. Introduction

1.1. **Motivation and main question.** Many systems of interest—forced diffusions [24, 34], boundary-driven quantum systems [28], and actively controlled open-system settings such as active reservoir engineering [25]—operate away from detailed balance. Their generators contain a non-reversible component of strength η and converge to a non-equilibrium steady state (NESS). For reversible systems ($\eta = 0$), lifting to a larger phase space can accelerate mixing [11, 16, 6, 31]: at optimal intermediate friction, the rate improves from the diffusive scale to its square root (quadratic speedup). More recently, non-reversible lifting mechanisms beyond the reversible-target setting have also been studied from a different viewpoint in Markov-process theory [15]. In this paper we ask:

> For a system that is η away from equilibrium, under what conditions does the kinetic acceleration (lifting) persist, and how does the convergence rate degrade as a function of η?

Concretely, we consider generators on a Hilbert space $\mathcal{H}$ of the form

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}, \tag{1}$$

where $\gamma > 0$ is a friction (scale-separation) parameter, $\mathcal{R} \leq 0$ is self-adjoint with nontrivial kernel $\mathcal{H}_S := \ker(\mathcal{R})$, $\mathcal{V}_{\text{skew}}$ is skew-adjoint, and $\eta\mathcal{V}_{\text{pert}}$ is a non-reversible perturbation whose strength $\eta \geq 0$ is *given by the physics*. In the singular limit $\gamma \to \infty$, adiabatic elimination [27, 39] reduces the dynamics to an effective slow generator

$$\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$$

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acting on $\mathcal{H}_S$, where P_S is the orthogonal projection onto $\mathcal{H}_S$ and $\mathcal{R}^+$ is the Moore–Penrose pseudoinverse. This overdamped limit gives a decay rate of order $\lambda(\mathcal{L}_O)/\gamma$, i.e., a diffusive bottleneck. We then ask: (i) whether the quadratic acceleration at finite friction survives the non-equilibrium perturbation, and (ii) how large η can be before the accelerated regime breaks down. Throughout the introduction, $\mathcal{H}_0$ denotes the invariant orthogonal complement of the stationary direction, i.e. the mean-zero subspace on which all decay statements are formulated.

1.2. **Main results.** The decomposition (1) isolates a self-adjoint *reversible backbone*

$$Q_O := -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S, \tag{2}$$

which captures the reversible skeleton of the slow dynamics. Let $\lambda(Q_O)$ denote its spectral gap on the mean-zero slow subspace $\mathcal{H}_{S,0} := \mathcal{H}_S \cap \mathcal{H}_0$.

Within the DMS hypocoercivity framework [13, 14], we construct a modified Lyapunov functional $\mathcal{E}_{\alpha,\varpi}(f) = \frac{1}{2}\|f\|^2 + \alpha\, \Re\langle \mathcal{A}_\varpi f, f\rangle$ with an explicit corrector operator $\mathcal{A}_\varpi$ adapted from [24, 7]. Our two main results are a constructive lower bound and a conditional overdamped ceiling.

In the formulas below, $\Lambda_{\text{hypo}}(\gamma)$ is the unperturbed rate profile, $B_{\text{pert}}(\gamma)$ the perturbative loss, and $\delta_*(\gamma)$ the Lyapunov-norm distortion.

Theorem A (Theorem 5.1). *Under the structural hypotheses stated precisely in Section 4—microscopic gap, centered coupling, flux regularity, perturbation bounds, and a kernel-compatible slow gap, together with the standard common-core/closure input in the unbounded setting—for every $\gamma > 0$ and every perturbation strength within the certified range $\eta \le \eta^*(\gamma)$, the semigroup decays exponentially on the invariant mean-zero subspace $\mathcal{H}_0$:*

$$\|e^{t\mathcal{L}} f\| \le C_*(\gamma)\, e^{-\lambda_\eta t}\|f\|, \qquad \forall\, f \in \mathcal{H}_0,\ t \ge 0,$$

The rate is fully explicit and satisfies

$$\lambda_\eta \ge \frac{\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)}{1 + \delta_*(\gamma)}.$$

Moreover, the Lyapunov distortion is uniformly controlled: $\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\| \le 2^{-1/2}$, hence $C_*(\gamma) \le 1 + \sqrt{2}$, and therefore

$$\lambda_\eta \ge (2 - \sqrt{2})(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)).$$

The gap-adapted choice $\varpi_* := \lambda(Q_O)$ is the one that restores $\sqrt{\lambda(Q_O)}$ scaling; the certified threshold

$$\eta^*(\gamma) := \min\left\{\eta_{\text{sym}}(\gamma),\ \frac{\Lambda_{\text{hypo}}(\gamma)}{2B_{\text{pert}}(\gamma)}\right\}, \qquad \eta_{\text{sym}}(\gamma) := \begin{cases} \infty, & a_2 = 0, \\ 2\gamma/a_2, & a_2 > 0, \end{cases}$$

is completely computable once the structural constants of the model are known.

Theorem B (Theorem 5.6). *Under Assumption 5.5—that is, the structural hypotheses plus contraction of the lifted semigroup in the working Hilbert norm, eigenfunction attainment, and corrector regularity—any admissible L^2 exponential rate ν with prefactor C obeys the overdamped ceiling*

$$\nu \le (1 + \log C)\left[\frac{(1+\kappa_{\text{ell}})\lambda(Q_O)}{\gamma} + \eta(b_1{+}b_2) + \eta(a_1{+}a_2)\frac{\sqrt{\lambda(Q_O)}}{\gamma}\right].$$

In particular, for $\gamma \to \infty$ and fixed η, the rate is at most $\mathcal{O}(\gamma^{-1}) + \mathcal{O}(\eta)$: the diffusive bottleneck persists, so acceleration is necessarily a finite-friction phenomenon.

Theorem A is fully constructive, whereas Theorem B is a conditional complement requiring norm-level contraction of the lifted semigroup. At equilibrium ($\eta = 0$) this contraction hypothesis is automatic; for $\eta \neq 0$ it becomes an additional L^2 assumption. In the unbounded Langevin example, the extra analytic input is standard: the core-to-semigroup passage is the usual DMS closure step

[14, §2], and the needed confining Kramers–Fokker–Planck regularity is classical [22, Introduction and §§1–2]; see also [21].

Theorems A and B should be read together. Theorem A gives a model-wise lower bound under Assumptions 4.1–4.2; Corollary 5.2 extracts the $\sqrt{\lambda(Q_O)}$ law for families whose structural constants stay bounded as $\lambda(Q_O) \downarrow 0$; and Theorem B shows that, under its additional ceiling hypotheses, this exponent cannot be improved. Thus acceleration is an intermediate-friction effect: within the certified regime $\eta \le \eta^*(\gamma)$ the lower bound deteriorates linearly in η, but at $\gamma \asymp \sqrt{\lambda(Q_O)}$ the best possible exponent is still of order $\sqrt{\lambda(Q_O)}$. The same square-root ceiling was established independently by Eberle–Lörler [16, 17] and Brigati–Lörler–Wang [6].

1.3. **Kinetic acceleration and non-equilibrium stability.** For uniformly controlled families, lifting gives a *quadratic speedup*: the decay scale improves from $\lambda(Q_O)$ to $\sqrt{\lambda(Q_O)}$ at intermediate friction $\gamma_{\mathrm{opt}} \asymp \sqrt{\lambda(Q_O)}$. The same slow–fast argument quantifies how non-equilibrium forcing erodes this gain. In particular, the certified lower bound degrades linearly in η, and a key cancellation makes $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$, so the certified threshold still satisfies $\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}$.

Section 6.1 turns this mechanism into a construction: given a target slow generator $\mathcal{L}_O$, we build a lifted generator whose reversible backbone is Q_O. This gives a concrete *algorithmic lift* and also clarifies the steady-state bias. In general $\|P_S h_\eta - g_\eta\| = \mathcal{O}(\eta/\gamma)$, but for designed lifts one may enforce $P_F \mathcal{V} P_F = 0$ and remove the bias exactly (Proposition 6.1).

1.4. **Related work.** The L^2 hypocoercive method based on modified Lyapunov functionals was developed by Dolbeault–Mouhot–Schmeiser [13, 14], building on Villani's H^1 approach [40]; its adaptation to Langevin dynamics using resolvent-type correctors is due to Iacobucci–Olla–Stoltz [24], Roussel–Stoltz [37], and Cao–Lu–Wang [7]. In a quantum direction, Fang–Lu–Tong [18] derive hypocoercive mixing-time estimates separating Hamiltonian and dissipative effects; our boundary-driven example differs in that $\mathcal{L}_O$ is already non-self-adjoint.

The lifting perspective—embedding slow dynamics into an enlarged space—originates with Chen–Lovász–Pak [11]; fundamental limits were identified by Apers–Ticozzi–Sarlette [3]. Extensions to continuous time are due to Eberle–Lörler [16, 17] and Brigati–Lörler–Wang [6]; the quantum analogue was developed by Li–Lu [31]; complementary flow-Poincaré methods appear in [15]. Our framework differs in being an operator-theoretic slow–fast perturbation theory centered on the self-adjoint backbone Q_O, tailored to NESS.

In the non-equilibrium setting, Dietert [12] develops L^2-hypocoercivity for kinetic equations with non-equilibrium forcing, and Monmarché [33] establishes geometric ergodicity for kinetic Langevin with NESS via a Lyapunov technique. We differ in providing an abstract framework with explicit η-dependence and a certified stability threshold. For boundary-driven quantum systems, see Carlen–Huse–Lebowitz [9] and the Zeno-limit analysis in [8].

Among hypocoercivity approaches, the variational method of [2] yields sharp regularity for kinetic Fokker–Planck; explicit matrix analyses apply in structured settings [1, 29]; and operator-theoretic precedents include [20, 4]. We adopt the DMS corrector framework for its natural extension to NESS. Our key novelty is the gap-adapted shift $\varpi_* = \lambda(Q_O)$, which upgrades the classical fixed-shift bound from $\mathcal{O}(\lambda^{5/2})$ to $\mathcal{O}(\sqrt{\lambda})$.

1.5. **Outline.** Section 2 gives a self-contained motivating example based on equilibrium underdamped Langevin, previewing the gap-adapted shift mechanism in the model most closely connected with our later nonequilibrium example. Section 3 sets up the abstract framework and adiabatic elimination. Section 4 formulates the structural assumptions and introduces the DMS corrector. Section 5 proves the main decay-rate bounds. Section 6 presents the convex-potential Langevin family, the reverse adiabatic elimination recipe, and a boundary-driven toy model. Section 7 discusses extensions and open problems.

2. Motivating example: equilibrium Langevin and the gap-adapted DMS corrector

This section gives the equilibrium underdamped Langevin warm-up behind the whole paper. It is the model underlying our later nonequilibrium example, and it isolates the same gap-adapted mechanism with much less bookkeeping: once the DMS corrector is shifted by the overdamped gap m, the slow coercive block becomes order one and the proof reduces to an explicit 2×2 matrix estimate.

For simplicity we normalize the inverse temperature to 1 in this warm-up. Let

$$d\rho_\infty(x,v) = d\mu(x)\, d\kappa(v), \qquad d\mu(x) = Z^{-1} e^{-U(x)}\, dx, \qquad d\kappa(v) = (2\pi)^{-d/2} e^{-|v|^2/2}\, dv,$$

and assume that μ satisfies the Poincaré inequality

$$\|g\|^2_{L^2(\mu)} \le \frac{1}{m} \|\nabla_x g\|^2_{L^2(\mu)}, \qquad g \in H^1(\mu), \ \int g\, d\mu = 0, \tag{3}$$

for some $m > 0$. Consider on $L^2_0(\rho_\infty)$ the equilibrium Langevin generator

$$\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma \mathcal{R}, \qquad \mathcal{V}_{\text{skew}} = v \cdot \nabla_x - \nabla_x U \cdot \nabla_v, \qquad \mathcal{R} = -v \cdot \nabla_v + \Delta_v. \tag{4}$$

The slow projection is the velocity average

$$P_S f(x) := \int_{\mathbb{R}^d} f(x,v)\, d\kappa(v), \qquad P_F := \mathrm{id} - P_S,$$

so the slow space consists of functions of x alone. The associated positive overdamped operator is

$$\mathcal{L}_{\text{ovd}} := -Q_O = -\Delta_x + \nabla_x U \cdot \nabla_x, \tag{5}$$

acting on $L^2_0(\mu)$. By (3), $\mathcal{L}_{\text{ovd}} \ge m$ on $L^2_0(\mu)$, equivalently $\lambda(Q_O) = m$.

We do not reprove here the standard regularity input for this model. Every identity below is first written on $C_c^\infty(\mathbb{R}^{2d}) \cap L^2_0(\rho_\infty)$ and then extended by the standard DMS closure argument [14, §2]; the confining kinetic Fokker–Planck regularization needed for that passage is classical [22, Introduction and §§1–2].

The gap-adapted corrector is

$$\mathcal{A}_m := (m + \mathcal{L}_{\text{ovd}})^{-1} (\mathcal{V}_{\text{skew}} P_S)^*, \tag{6}$$

and the modified Lyapunov functional is

$$\mathsf{E}_\varepsilon(f) := \frac{1}{2}\|f\|^2 - \varepsilon\, \Re\langle \mathcal{A}_m f, f\rangle. \tag{7}$$

The three bounds below are the equilibrium Langevin counterpart of the abstract corrector estimates proved later.

Lemma 2.1 (Corrector estimates for the Langevin warm-up)**.** *For every $\phi \in L^2_0(\rho_\infty)$, the corrector $\mathcal{A}_m$ satisfies*

$$\|\mathcal{A}_m \phi\| \le \frac{1}{2\sqrt{m}} \|\phi\|, \tag{8}$$

$$\|\mathcal{V}_{\text{skew}} \mathcal{A}_m \phi\| \le \|\phi\|, \tag{9}$$

$$\|\mathcal{A}_m \mathcal{V}_{\text{skew}} P_F \phi\| \le \sqrt{2}\, \|\phi\|. \tag{10}$$

Consequently, if $\varepsilon \le \sqrt{m}$, then

$$\frac{1 - \varepsilon/\sqrt{m}}{2} \|\phi\|^2 \le \mathsf{E}_\varepsilon(\phi) \le \frac{1 + \varepsilon/\sqrt{m}}{2} \|\phi\|^2. \tag{11}$$

Proof. Let $u = \mathcal{A}_m\phi$ and $w = (\mathcal{V}_{\rm skew}P_S)^*\phi$, so $u = (m+\mathcal{L}_{\rm ovd})^{-1}w$ and $w \in L^2_0(\mu)$. Since u depends only on x,

$$\|\mathcal{V}_{\rm skew}u\|^2 = \|\nabla_x u\|^2_{L^2(\mu)} = \langle \mathcal{L}_{\rm ovd}u, u\rangle_{L^2(\mu)}.$$

Taking the $L^2(\mu)$ inner product of $(m+\mathcal{L}_{\rm ovd})u = w$ with u and applying Cauchy–Schwarz yields

$$m\|u\|^2 + \|\mathcal{L}^{1/2}_{\rm ovd}u\|^2 \le \|\phi\|\,\|\mathcal{L}^{1/2}_{\rm ovd}u\|.$$

This gives both (8) and (9). For (10), set $B := \mathcal{A}_m\mathcal{V}_{\rm skew}P_F$. If $g \in \mathrm{Ran}(P_S)$ and $h = (m+\mathcal{L}_{\rm ovd})^{-1}g$, then an integration by parts shows

$$B^*g = -\sum_{i,j}(v_iv_j - \delta_{ij})\,\partial_{ij}h.$$

The Gaussian fourth-moment identity gives $\|B^*g\|^2 = 2\|\nabla^2_x h\|^2_{L^2(\mu)}$, and the convex Bochner estimate implies $\|\nabla^2_x h\|_{L^2(\mu)} \le \|\mathcal{L}_{\rm ovd}h\|_{L^2(\mu)} \le \|g\|_{L^2(\mu)}$. Hence $\|B^*g\| \le \sqrt{2}\|g\|$, which is (10). The norm equivalence follows from (8) and Cauchy–Schwarz. □

Write $f_s = P_Sf$ and $f_f = P_Ff$. The Gaussian Poincaré inequality in v gives $\|f_f\|^2 \le \|\nabla_v f\|^2$. Therefore, if we set

$$X := \|\nabla_v f\|, \qquad Y := \|f_s\|, \qquad Z := \frac{\gamma}{2\sqrt{m}} + \sqrt{2},$$

then differentiating E_ε along $\partial_t f = \mathcal{L}f$ and using Lemma 2.1 yields

$$\frac{d}{dt}\mathsf{E}_\varepsilon(f) \le -\begin{pmatrix} X \\ Y\end{pmatrix}^T \begin{pmatrix} \gamma-\varepsilon & -\dfrac{\varepsilon Z}{2} \\ -\dfrac{\varepsilon Z}{2} & \dfrac{\varepsilon}{2}\end{pmatrix}\begin{pmatrix} X \\ Y\end{pmatrix}. \tag{12}$$

Indeed, the microscopic dissipation gives $-\gamma X^2$; the macroscopic term is

$$-\varepsilon\Big\langle \frac{\mathcal{L}_{\rm ovd}}{m+\mathcal{L}_{\rm ovd}}f_s, f_s\Big\rangle \le -\frac{\varepsilon}{2}Y^2$$

because $\mathcal{L}_{\rm ovd} \ge m$ on $L^2_0(\mu)$; the fast–fast remainder contributes at most εX^2 by (9); the dual regularity estimate (10) contributes $\varepsilon\sqrt{2}\,XY$; and the correction involving $\gamma\mathcal{R}$ contributes $\varepsilon\gamma(2\sqrt{m})^{-1}XY$ by (8).

Proposition 2.2. *For the gap-adapted corrector $\mathcal{A}_m$, the DMS lower-rate profile satisfies, for every $\gamma > 0$,*

$$\Lambda^{\rm Lan}_{\rm hypo}(\gamma) \ge \frac{\gamma}{4\big((\gamma/(2\sqrt{m})+\sqrt{2})^2+2\big)}. \tag{13}$$

The optimum is attained at $\gamma_{\rm opt} = 4\sqrt{m}$, yielding

$$\Lambda^{\rm Lan}_{\rm hypo}(\gamma_{\rm opt}) \ge \frac{2-\sqrt{2}}{8}\sqrt{m}. \tag{14}$$

Since $m = \lambda(Q_O)$, this is exactly the kinetic scale $\mathcal{O}(\sqrt{\lambda(Q_O)})$.

Proof. Let $M_{\gamma,\varepsilon}$ denote the dissipation matrix in (12). Its determinant is

$$\det M_{\gamma,\varepsilon} = \frac{\gamma\varepsilon}{2} - \frac{\varepsilon^2(Z^2+2)}{4},$$

so the natural choice is

$$\varepsilon_* = \frac{\gamma}{Z^2+2}.$$

For a positive 2×2 matrix, $\lambda_{\min}(M) \geq \det(M)/\operatorname{tr}(M)$, and here $\operatorname{tr}(M_{\gamma,\varepsilon_*}) = \gamma - \varepsilon_*/2 \leq \gamma$. Hence

$$\lambda_{\min}(M_{\gamma,\varepsilon_*}) \geq \frac{\gamma}{4(Z^2+2)}.$$

This gives the stated lower bound. Writing $x = \gamma/(2\sqrt{m})$, the right-hand side becomes

$$\sqrt{m}\,\frac{x}{2\big((x+\sqrt{2})^2+2\big)},$$

and a direct derivative computation shows that it is maximized at $x = 2$, i.e. at $\gamma_{\text{opt}} = 4\sqrt{m}$. Substituting $x = 2$ yields (14). □

For comparison, the classical fixed-shift choice $\varpi = 1$ makes the macroscopic coercive block of order m rather than order one, and the same determinant–trace estimate gives the familiar $m^{5/2}$ loss. Thus the equilibrium Langevin warm-up already contains the precise mechanism behind the later abstract theory: the structural constants are $\lambda_R = M_R = 1$, $\Theta = \frac{1}{2}$, $k_{\text{reg}} = 1$, $K_{\text{geom}} = \sqrt{2}$, and the correct gap-adapted shift is exactly $\varpi_* = m = \lambda(Q_O)$. The rest of the paper asks how much of this equilibrium $\sqrt{m}$ acceleration survives after the nonequilibrium perturbation $\eta\mathcal{V}_{\text{pert}}$ is switched on.

3. Preliminaries

We set up the abstract Hilbert-space framework, the analytic toolbox for unbounded operators, and the adiabatic-elimination reduction used throughout.

3.1. **Hilbert-space setup.** Let $\mathcal{H}$ be a separable complex Hilbert space with inner product $\langle\cdot,\cdot\rangle$ (conjugate-linear in the first entry) and norm $\|\cdot\| = \langle\cdot,\cdot\rangle^{1/2}$. The generator $\mathcal{L}$ (possibly unbounded) generates a strongly continuous semigroup $(\mathcal{P}_t)_{t\geq 0}$ on $\mathcal{H}$, with adjoint $\mathcal{L}^\dagger$.

Concrete realizations. In the noncommutative (quantum) setting, $\mathcal{H} = L^2(\sigma)$ for a faithful state σ on a finite-dimensional space $\mathfrak{h}$, equipped with the KMS inner product $\langle X, Y\rangle_\sigma := \operatorname{tr}(\sigma^{1/2}X^\dagger\sigma^{1/2}Y)$. The generator is then a GKSL Lindbladian $\mathcal{L}(X) = i[H,X] + \sum_\alpha \mathcal{D}^\dagger[L_\alpha]X$. In the classical (PDE) setting, $\mathcal{H} = L^2(\mu)$ for a reference measure μ, where μ is typically the invariant measure of the unperturbed dynamics (e.g. a Gibbs measure).

Convention on operator domains. Every unbounded operator that appears in this work—$\mathcal{R}$, $\mathcal{V}_{\text{skew}}$, $\mathcal{V}_{\text{pert}}$, Q_O—is assumed to be *closed and densely defined* on $\mathcal{H}$ (or on the appropriate slow/fast subspace). An operator T being *closed* means that its graph $\{(f, Tf) : f \in \operatorname{dom}(T)\}$ is a closed subspace of $\mathcal{H} \oplus \mathcal{H}$; this is automatic for generators of strongly continuous semigroups and for self-adjoint operators. In the finite-dimensional setting that covers the quantum spins example of Section 6.3, all operators are bounded and $\operatorname{dom}(T) = \mathcal{H}$; the domain considerations below are then vacuous.

3.2. **Spectral gaps.** Let Π denote the projection onto the stationary direction and $\mathcal{H}_0 := \ker(\Pi)$ the mean-zero subspace. Write $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \leq 0$ for the symmetric part of the generator.

Definition 3.1 (Gap parameters)**.** On $\mathcal{H}_0$ we define the *coercive gap* $\lambda_S := \inf_{f\in\mathcal{H}_0\setminus\{0\}}(-\langle f, \mathcal{S}f\rangle/\|f\|^2)$ and the *spectral gap* $\lambda(\mathcal{L}) := \inf\{\Re\mu : \mu \in \operatorname{Spec}(-\mathcal{L}|_{\mathcal{H}_0})\}$.

When $\mathcal{L}$ is self-adjoint (detailed balance), $\lambda(\mathcal{L}) = \lambda_S$. In the non-reversible regime, it may happen that $\lambda_S = 0$ while $\lambda(\mathcal{L}) > 0$: the skew part $\frac{1}{2}(\mathcal{L} - \mathcal{L}^\dagger)$ couples degenerate directions to dissipative ones, producing exponential relaxation by hypocoercivity [40, 13]. The standard hypocoercive estimate asserts $\|\mathcal{P}_t f\| \leq Ce^{-\lambda t}\|f\|$ for $f \in \mathcal{H}_0$; when $C > 1$ is necessary, the relaxation is *strictly hypocoercive*.

Lemma 3.2 (Coercivity characterization)**.** *Assume* $\ker(\mathcal{L}) = \mathbb{C}\mathbf{1}$. *Then:*

(i) If $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ *and* $-\mathcal{S}|_{\mathcal{H}_0}$ *has a positive spectral gap (automatic in finite dimensions), then the semigroup is coercive* ($C = 1$)*. Conversely, coercivity implies* $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$*.*

(ii) If $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$*, any exponential decay estimate requires* $C > 1$*.*

3.3. **Slow–fast decomposition and adiabatic elimination.** Consider the generator $\mathcal{L}_\gamma = \gamma\mathcal{R} + \mathcal{V}$ with self-adjoint dissipator $\mathcal{R} \leq 0$ and interaction $\mathcal{V}$. Define the *slow* and *fast* subspaces

$$\mathcal{H}_S := \ker(\mathcal{R}), \qquad \mathcal{H}_F := \mathcal{H}_S^\perp,$$

with orthogonal projections P_S, $P_F := \mathrm{id} - P_S$. We assume $\mathcal{R}$ has spectral gap $\lambda_R > 0$ on $\mathcal{H}_F$. Any $f \in \mathcal{H}$ decomposes as $f = f_S + f_F$ with $f_S := P_S f$ (slow/macroscopic) and $f_F := P_F f$ (fast/microscopic). In the quantitative theory below, the stationary direction is generated by a vector $\mathbf{1} \in \mathcal{H}_S$, so Π is the orthogonal projection onto $\mathbb{C}\mathbf{1}$. Because $\mathcal{H}_F \perp \mathcal{H}_S$, this implies

$$\mathcal{H}_F \subset \mathcal{H}_0, \qquad \mathcal{H}_0 = (\mathcal{H}_S \cap \mathcal{H}_0) \oplus \mathcal{H}_F.$$

In particular, if $f \in \mathcal{H}_0$, then both $f_S := P_S f$ and $f_F := P_F f$ remain in $\mathcal{H}_0$.

Definition 3.3 (Centered coupling)**.** The interaction satisfies $P_S \mathcal{V} P_S = 0$.

This centering condition ensures that the effective slow dynamics is second-order in γ^{-1}—generated by slow–fast–slow excursions—rather than driven by an $\mathcal{O}(1)$ tangential drift [34, 8].

Definition 3.4 (Effective generator)**.** Let $\mathcal{R}^+$ denote the Moore–Penrose pseudoinverse of $\mathcal{R}$, bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$. Under the centered coupling condition, we define

$$\mathcal{L}_O := -P_S \mathcal{V} \mathcal{R}^+ \mathcal{V} P_S. \tag{15}$$

In the context of lifting, $\mathcal{L}_O$ is the effective "target" dynamics one seeks to accelerate. On the macroscopic timescale $\tau = t/\gamma$, the full evolution converges to $e^{\tau\mathcal{L}_O}$ as $\gamma \to \infty$ (quantitative Zeno-limit error bounds are given in [8]). From the rate perspective, Theorem 5.6 shows that the decay is bounded by a diffusive γ^{-1} ceiling in this overdamped regime, motivating the finite-friction analysis of Section 5.

When the interaction is decomposed as $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ (skew-adjoint part plus perturbation), we impose the stronger componentwise conditions $P_S \mathcal{V}_{\mathrm{skew}} P_S = 0$ and $P_S \mathcal{V}_{\mathrm{pert}} P_S = 0$. For a single fixed value of η, this is stronger than merely assuming $P_S \mathcal{V} P_S = 0$ for the full interaction; we adopt it because the quantitative estimates below are intended to hold uniformly over an admissible range of η, and because it isolates the reversible backbone Q_O from the η-dependent perturbation. This componentwise assumption implies the centered coupling condition of Definition 3.3 for every admissible η.

3.4. **Analytic toolbox for the slow–fast operators.** In this subsection we collect the functional-analytic constructions underlying the slow–fast framework. In finite dimensions every step is trivial; the statements below are needed only for the unbounded (PDE/diffusion) setting, where operator domains and closures require care.

Lemma 3.5 (Pseudoinverse via spectral calculus)**.** *Let* $\mathcal{R}$ *be a non-positive self-adjoint operator on* $\mathcal{H}$ *with* $\ker(\mathcal{R}) = \mathcal{H}_S$ *and spectral gap* $\lambda_R > 0$ *on* $\mathcal{H}_F = \mathcal{H}_S^\perp$*. Let* $\{E_\lambda\}_{\lambda \geq 0}$ *be the spectral family of* $-\mathcal{R}|_{\mathcal{H}_F}$*, so that* $-\mathcal{R}|_{\mathcal{H}_F} = \int_{\lambda_R}^\infty \lambda \, dE_\lambda$*. Then the pseudoinverse* $\mathcal{R}^+ := -(-\mathcal{R}|_{\mathcal{H}_F})^{-1} P_F$ *is a bounded self-adjoint operator on* $\mathcal{H}$ *satisfying:*

(i) $\mathcal{R}^+$ *maps* $\mathcal{H}_F$ *into* $\mathrm{dom}(\mathcal{R}) \cap \mathcal{H}_F$ *and annihilates* $\mathcal{H}_S$*;*

(ii) $\mathcal{R}\mathcal{R}^+ = P_F$ *and* $\mathcal{R}^+\mathcal{R} = P_F$ *on* $\mathrm{dom}(\mathcal{R})$*;*

(iii) $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$ *and* $-\mathcal{R}^+ \geq 0$ *on* $\mathcal{H}_F$*;*

(iv) the square root satisfies $\|(-\mathcal{R}^+)^{1/2}\| \leq \lambda_R^{-1/2}$*.*

Proof. Since the spectrum of $-\mathcal{R}|_{\mathcal{H}_F}$ is contained in $[\lambda_R, \infty)$, the inverse $(-\mathcal{R}|_{\mathcal{H}_F})^{-1} = \int_{\lambda_R}^{\infty} \lambda^{-1}\, dE_\lambda$ is a bounded non-negative self-adjoint operator on $\mathcal{H}_F$ with norm at most λ_R^{-1}. The remaining properties follow from the spectral theorem and the identities $E_\lambda \cdot \mathrm{dom}(\mathcal{R}) = \mathrm{dom}(\mathcal{R}) \cap E_\lambda \mathcal{H}$ for each spectral projection. □

Lemma 3.6 (Friedrichs construction of the slow operator Q_O)**.** *Under the standing hypotheses on $\mathcal{R}$ and $\mathcal{V}_{\mathrm{skew}}$ (including the centered coupling condition $P_S \mathcal{V}_{\mathrm{skew}} P_S = 0$), and given a dense subspace $\mathcal{C} \subset \mathrm{dom}(\mathcal{V}_{\mathrm{skew}}) \cap \mathcal{H}_S$ that is invariant under $\mathcal{R}^+$ and $P_S \mathcal{V}_{\mathrm{skew}}$, define the quadratic form*

$$q_O(\psi) := \langle \mathcal{V}_{\mathrm{skew}}\psi,\, (-\mathcal{R}^+)\, \mathcal{V}_{\mathrm{skew}}\psi \rangle, \qquad \psi \in \mathcal{C}. \tag{16}$$

Then:

(i) q_O is a non-negative closed symmetric form on $\mathcal{H}_S$ (the closure being taken in the $\|\cdot\| + q_O(\cdot)^{1/2}$ graph topology).

(ii) The Friedrichs extension of q_O is a non-positive self-adjoint operator Q_O on $\mathcal{H}_S$ satisfying

$$\langle \psi, (-Q_O)\psi \rangle = q_O(\psi), \qquad \forall \psi \in \mathrm{dom}((-Q_O)^{1/2}). \tag{17}$$

(iii) On $\mathcal{C}$, this form identity reads $\langle \psi, (-Q_O)\psi \rangle = \langle \mathcal{V}_{\mathrm{skew}}\psi,\, (-\mathcal{R}^+)\, \mathcal{V}_{\mathrm{skew}}\psi \rangle$, recovering the operator expression (19)*.*

(iv) For every $\varpi > 0$, the resolvent $(\varpi - Q_O)^{-1}$ is a bounded self-adjoint operator on $\mathcal{H}_S$ with $\|(\varpi - Q_O)^{-1}\| \le \varpi^{-1}$. If $\lambda(Q_O) > 0$ (Assumption 4.2), then additionally $\|(\varpi - Q_O)^{-1}\| \le (\varpi + \lambda(Q_O))^{-1}$ on $\mathcal{H}_{S,0}$.

Proof. Symmetry follows from the skew-adjointness of $\mathcal{V}_{\mathrm{skew}}$:

$$q_O(\psi) = \langle \mathcal{V}_{\mathrm{skew}}\psi,\, (-\mathcal{R}^+)\, \mathcal{V}_{\mathrm{skew}}\psi \rangle \ge 0,$$

since $-\mathcal{R}^+ \ge 0$ on $\mathcal{H}_F$. The centering condition $P_S \mathcal{V}_{\mathrm{skew}} P_S = 0$ ensures that $\mathcal{V}_{\mathrm{skew}}\psi \in \mathcal{H}_F$ for $\psi \in \mathcal{H}_S$, so the form is well defined. Closedness is standard (the form norm $\|\psi\|^2 + q_O(\psi)$ controls the graph norm [26, Thm. VI.2.1]). The Friedrichs extension theorem [36, Thm. X.23] then provides the self-adjoint operator, and the resolvent bound is an immediate consequence of $\sigma(-Q_O) \subset [0, \infty)$ on $\mathcal{H}_S$. On $\mathcal{H}_{S,0}$, the spectrum of $-Q_O$ is contained in $[\lambda(Q_O), \infty)$, giving the sharper resolvent bound. □

4. The non-equilibrium lifting framework

We now state the structural assumptions used for quantitative hypocoercivity at finite friction. The goal is to isolate the ingredients that transfer dissipation from fast modes to slow modes through the DMS corrector [13, 14, 24]: fast spectral gap, centering, flux regularity, and perturbation bounds. Standing convention on the stationary direction. Throughout Sections 4–6, we fix a unit vector $\mathbf{1} \in \mathcal{H}_S$ such that $\mathcal{L}\mathbf{1} = 0$, and denote by Π the orthogonal projection onto $\mathbb{C}\mathbf{1}$. We assume $\Pi e^{t\mathcal{L}} = e^{t\mathcal{L}}\Pi$ for all $t \ge 0$, so the mean-zero subspace $\mathcal{H}_0 := \ker(\Pi)$ is invariant under the semigroup. Because $\mathbf{1} \in \mathcal{H}_S$, both P_S and P_F preserve $\mathcal{H}_0$, and every $f \in \mathcal{H}_0$ decomposes as $f = f_S + f_F$ with $f_S \in \mathcal{H}_S \cap \mathcal{H}_0$ and $f_F \in \mathcal{H}_F$.

4.1. Structural assumptions.

Let $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ be an orthogonal slow–fast decomposition. We study generators depending on friction $\gamma > 0$ and perturbation strength $\eta \ge 0$:

$$\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}. \tag{18}$$

The transport term is decomposed as $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$, and P_S, P_F denote the orthogonal projections onto $\mathcal{H}_S, \mathcal{H}_F$.

Before stating the structural assumptions, we record the slow operator that drives the DMS corrector. The reversible contribution to the effective diffusion is

$$Q_O := -P_S \mathcal{V}_{\mathrm{skew}} \mathcal{R}^+ \mathcal{V}_{\mathrm{skew}} P_S. \tag{19}$$

Whenever this composition is well defined, Q_O acts on $\mathcal{H}_S$. From the skew-adjointness of $\mathcal{V}_{\text{skew}}$ and the negativity of $\mathcal{R}^+$, it follows that for any $f \in \mathcal{H}_S$,

$$\langle f, Q_O f\rangle = -\langle f, P_S\mathcal{V}_{\text{skew}}\mathcal{R}^+\mathcal{V}_{\text{skew}}P_S f\rangle = -\langle \mathcal{V}_{\text{skew}}^\dagger f, \mathcal{R}^+\mathcal{V}_{\text{skew}} f\rangle = \langle \mathcal{V}_{\text{skew}} f, \mathcal{R}^+\mathcal{V}_{\text{skew}} f\rangle \le 0. \tag{20}$$

Thus Q_O is self-adjoint and dissipative on its natural domain. In finite dimensions this is automatic; in unbounded settings Assumption 4.3 below supplies the common-core bookkeeping needed for the same composition.

Assumption 4.1 (Structural conditions)**.** The generator has the form $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$. We assume:

(1) **Microscopic Spectral Gap.** $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$, with kernel $\ker(\mathcal{R}) = \mathcal{H}_S$, and there exists $\lambda_R > 0$ such that

$$-\langle \mathcal{R}f, f\rangle \ge \lambda_R\|f\|^2, \qquad \forall f \in \mathcal{H}_F. \tag{21}$$

(2) **Centered Coupling.**

$$P_S\mathcal{V}_{\text{skew}}P_S = 0, \qquad P_S\mathcal{V}_{\text{pert}}P_S = 0. \tag{22}$$

(3) **Fast-to-slow Regularity.** For $\psi \in \text{dom}(Q_O) \subset \mathcal{H}_S$, define its *microscopic flux* induced by transport as

$$g_\psi := P_F\mathcal{V}_{\text{skew}}P_S\psi \in \mathcal{H}_F. \tag{23}$$

There exist constants $M_R, \kappa_{\text{ell}} > 0$ such that

$$\|g_\psi\|^2 \le M_R\,\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle, \qquad \forall\psi \in \text{dom}(Q_O) \subset \mathcal{H}_S. \tag{24}$$

$$\|P_F\mathcal{V}_{\text{skew}}P_F\,(-\mathcal{R}^+)\,g_\psi\| \le \kappa_{\text{ell}}\|P_S\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi\| = \kappa_{\text{ell}}\|(-Q_O)\psi\|. \tag{25}$$

(4) **Structural Decomposition of the Perturbation.**

$$\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}, \tag{26}$$

where

$$\mathcal{K}_{\text{pert}} := \tfrac{1}{2}(\mathcal{V}_{\text{pert}} - \mathcal{V}_{\text{pert}}^*), \qquad \mathcal{S}_{\text{pert}} := \tfrac{1}{2}(\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}^*),$$

with the adjoint taken in $\mathcal{H}$. Assume:

(a) **Skew Perturbation Control:** The operator $\mathcal{K}_{\text{pert}}$ is skew-symmetric, i.e., for all $f \in \text{dom}(\mathcal{V}_{\text{pert}})$:

$$\Re\langle \mathcal{K}_{\text{pert}} f, f\rangle = 0. \tag{27}$$

There exist constants $a_1, b_1 \ge 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{K}_{\text{pert}} f\| \le a_1\,\|(-\mathcal{R})^{1/2} f\| + b_1\,\|f\|. \tag{28}$$

(b) **Symmetric Perturbation Control:** $\mathcal{S}_{\text{pert}}$ is symmetric on $\text{dom}((-\mathcal{R})^{1/2})$, and there exist constants $a_2, b_2 \ge 0$ such that for all $f \in \text{dom}((-\mathcal{R})^{1/2})$,

$$\|\mathcal{S}_{\text{pert}} f\| \le a_2\,\|(-\mathcal{R})^{1/2} f\| + b_2\,\|f\|. \tag{29}$$

The pair (λ_R, M_R) measures fast dissipation and flux conditioning, κ_{ell} controls the geometric transport remainder, and (a_i, b_i) quantify the perturbation in the energy norm; Section 5 combines them into the rate $\Lambda_{\text{hypo}}(\gamma)$ and the perturbative loss $B_{\text{pert}}(\gamma)$.

The operator Q_O from (19) is the reversible contribution to the effective diffusion. Assumption 4.2 now isolates the kernel-compatible spectral gap needed for the slow coercive term in the DMS functional.

Assumption 4.2 (Kernel-compatible slow gap)**.** We assume that the stationary direction is the whole kernel of the reversible backbone on the slow space:

$$\ker(Q_O) = \mathbb{C}\mathbf{1} \subset \mathcal{H}_S.$$

Equivalently, on $\mathcal{H}_{S,0} := \mathcal{H}_S \cap \mathcal{H}_0$,

$$-\langle f, Q_O f\rangle \geq \lambda(Q_O)\|f\|^2, \qquad \forall f \in \mathcal{H}_{S,0}. \tag{30}$$

Hence $(\varpi - Q_O)^{-1}$ is bounded on $\mathcal{H}_S$ for every $\varpi > 0$, and in the unbounded setting we additionally assume that it maps $\mathcal{H}_S$ into $\mathrm{dom}(Q_O)$.

Assumption 4.3 (Analytic Core in Unbounded Settings)**.** This assumption is vacuous in finite dimensions. In unbounded settings, fix a dense subspace

$$\mathcal{C} \subset \mathrm{dom}(\mathcal{L}) \cap \mathrm{dom}((-\mathcal{R})^{1/2})$$

satisfying:

(i) $\mathcal{C} \subset \mathrm{dom}(\mathcal{R}) \cap \mathrm{dom}(\mathcal{V}_{\mathrm{skew}}) \cap \mathrm{dom}(\mathcal{V}_{\mathrm{pert}})$.

(ii) $\mathcal{C}$ is invariant under the structural operators:

$$\mathcal{R}(\mathcal{C}) \subset \mathcal{C}, \quad \mathcal{V}_{\mathrm{skew}}(\mathcal{C}) \subset \mathcal{C}, \quad P_S(\mathcal{C}) \subset \mathcal{C}, \quad P_F(\mathcal{C}) \subset \mathcal{C}. \tag{31}$$

Moreover, $\mathcal{C} \cap \mathcal{H}_S$ is a core for Q_O.

(iii) The pseudoinverse–transport composition maps the slow core back into the fast core:

$$\mathcal{R}^+\mathcal{V}_{\mathrm{skew}}(\mathcal{C} \cap \mathcal{H}_S) \ \subset \ \mathcal{C} \cap \mathcal{H}_F.$$

(iv) $\mathcal{C}$ is a core for $\mathcal{R}$ and for $\mathcal{V}_{\mathrm{skew}}$.

(v) The operator compositions defining Q_O, $\mathcal{A}_\varpi$, and the bilinear forms used in the Lyapunov and singular-value estimates are well defined on $\mathcal{C}$ and extend continuously to the relevant energy spaces. For every $\varpi > 0$, $(\varpi - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\mathrm{dom}(Q_O)$.

In finite dimensions Assumption 4.3 is automatic; in the Langevin example it is supplied by the standard Schwartz-core theory quoted later. Assumption 4.2 is needed because Theorem 5.1 is a decay statement on all of $\mathcal{H}_0$: extra zero modes of Q_O would leave uncontrolled directions. Finally, since $\mathcal{L}_O - Q_O$ is perturbative in η, any regime in which the slow spectrum is stable under this perturbation has $\lambda(\mathcal{L}_O)$ and $\lambda(Q_O)$ of the same order.

4.2. **The DMS corrector.** To close the hypocoercive estimate, we introduce a bounded corrector that solves a shifted slow-space Poisson problem. Following the Dolbeault–Mouhot–Schmeiser (DMS) approach [13, 14], extended to L^2 settings by Iacobucci–Olla–Stoltz [24] and Cao–Lu–Wang [7], this operator captures the response of the slow component to fast transport excursions and provides the cross term that restores coercivity at the Lyapunov level. This is the abstract counterpart of the concrete corrector from Section 2: in the equilibrium Langevin warm-up the gap-adapted choice is $\varpi_* = m = \lambda(Q_O)$, and in the abstract theory the same rule becomes $\varpi_* = \lambda(Q_O)$.

Lemma 4.4 (Resolvent of the slow backbone)**.** *Let Q_O be as in* (19) *and Assumption 4.2. Then for every shift parameter $\varpi > 0$,*

(i) the operator $(\varpi - Q_O)$ is strictly positive on $\mathcal{H}_S$ with spectrum contained in $[\varpi, \infty)$;

(ii) the resolvent $(\varpi - Q_O)^{-1}$ is a bounded, self-adjoint, positive operator on $\mathcal{H}_S$ with $\|(\varpi - Q_O)^{-1}\| \leq \varpi^{-1}$;

(iii) on the mean-zero subspace $\mathcal{H}_{S,0}$, the sharper bound $\|(\varpi - Q_O)^{-1}|_{\mathcal{H}_{S,0}}\| \leq (\varpi + \lambda(Q_O))^{-1}$ holds;

(iv) $(\varpi - Q_O)^{-1}$ maps $\mathcal{H}_S$ into $\mathrm{dom}(Q_O)$ (by Assumption 4.3(v) in the unbounded setting; automatic in finite dimensions).

Proof. Since $-Q_O \geq 0$ is self-adjoint with $\ker(Q_O) = \mathbb{C}\mathbf{1}$ (Assumption 4.2), the spectral theorem gives $\sigma(-Q_O|_{\mathcal{H}_{S,0}}) \subset [\lambda(Q_O), \infty)$ and $\sigma(-Q_O|_{\ker Q_O}) = \{0\}$. Therefore $\sigma(\varpi - Q_O) \subset [\varpi, \varpi + \|Q_O\|]$ in finite dimensions, or $\sigma(\varpi - Q_O) \subset [\varpi, \infty)$ in general. Items (i)–(iii) are direct consequences; (iv) is part of Assumption 4.3(v). □

Definition 4.5 (DMS Corrector)**.** Let $\varpi > 0$ be a shift parameter, to be optimized in Section 5. We define the **corrector** $\mathcal{A}_\varpi : \mathcal{H} \to \mathcal{H}_S$ by

$$\mathcal{A}_\varpi := (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F. \tag{32}$$

The key obstruction to a direct energy estimate is that dissipation is degenerate on $\mathcal{H}_S$: for any $\phi \in \mathcal{H}_S$ with $\phi \neq 0$, one has $\Re\langle \mathcal{L}\phi, \phi\rangle = 0$, so the plain energy $\frac{1}{2}\|f\|^2$ cannot produce a spectral gap. The remedy is to augment it with a cross term involving the corrector $\mathcal{A}_\varpi$; differentiating $\Re\langle \mathcal{A}_\varpi f, f\rangle$ along the flow generates a slow coercive contribution tied to $-Q_O$, together with controlled remainders from the fast and perturbative components.

The following lemma establishes the critical boundedness properties of $\mathcal{A}_\varpi$, including the control over non-equilibrium perturbations. The reader may compare with the equilibrium Langevin analogues in Lemma 2.1.

Lemma 4.6 (Structural Estimates for the DMS Corrector)**.** *Under Assumption 4.1, together with Assumption 4.3 in the unbounded setting, the operator $\mathcal{A}_\varpi$ satisfies the following structural estimates, with constants depending solely on the structural parameters $(M_R, \lambda_R, \kappa_{\text{ell}}, a_1, b_1, a_2, b_2)$ and* not *on ϖ:*

(1) ***Corrector Boundedness:***

$$\|\mathcal{A}_\varpi\| \leq k_{\mathcal{A}} \varpi^{-1/2}, \quad \textit{with } k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}. \tag{33}$$

(2) ***Regularity Estimates (Forward and Dual):*** *The corrector acts as a bounded regularizer for the skew-symmetric transport in both directions:*

$$\|\mathcal{V}_{\text{skew}} \mathcal{A}_\varpi\| \leq k_{\text{reg}}, \quad \textit{with } k_{\text{reg}} := \sqrt{M_R/\lambda_R}, \tag{34a}$$

$$\|\mathcal{A}_\varpi \mathcal{V}_{\text{skew}} P_F\| \leq k_{\text{dual}}, \quad \textit{with } k_{\text{dual}} := \kappa_{\text{ell}}. \tag{34b}$$

The dual estimate (34b) *is a direct consequence of the Macroscopic Flux Regularity (Assumption 4.1.3).*

(3) ***Perturbation Control:*** *The interaction with the non-equilibrium perturbation is controlled in both directions:*

$$\|\mathcal{V}_{\text{pert}}^\dagger \mathcal{A}_\varpi\| \leq k_P(\varpi), \tag{35a}$$

$$\|\mathcal{A}_\varpi \mathcal{V}_{\text{pert}}\| \leq \tilde{k}_P(\varpi). \tag{35b}$$

with control constants

$$k_P(\varpi) := (b_1 + b_2) k_{\mathcal{A}}\, \varpi^{-1/2}, \tag{36}$$

$$\tilde{k}_P(\varpi) := \Big((b_1 + b_2) k_{\mathcal{A}} + \tfrac{a_1 + a_2}{2} \Big) \varpi^{-1/2}. \tag{37}$$

Moreover, the centered perturbation coupling in Assumption 4.1.2 implies

$$P_S \mathcal{V}_{\text{pert}} P_S = 0 \qquad \Longrightarrow \qquad P_S \mathcal{V}_{\text{pert}}^\dagger P_S = 0, \tag{38}$$

and therefore

$$\mathcal{V}_{\text{pert}}\, \mathcal{H}_S \subset \mathcal{H}_F, \qquad \mathcal{V}_{\text{pert}}^\dagger\, \mathcal{H}_S \subset \mathcal{H}_F. \tag{39}$$

Proof. See Appendix B. □

4.3. **The modified Lyapunov functional.** We define a modified squared norm that explicitly incorporates the correlation between slow and fast degrees of freedom mediated by the corrector $\mathcal{A}_\varpi$.

Definition 4.7 (Modified Lyapunov functional [13, 14])**.** For any $f \in \mathcal{H}$ and a parameter $\alpha \in (0,1)$, we define the functional:

$$\mathcal{E}_{\alpha,\varpi}(f) := \frac{1}{2}\|f\|^2 + \alpha\Re\langle \mathcal{A}_\varpi f, f\rangle. \tag{40}$$

This functional defines a norm equivalent to the underlying Hilbert norm, provided the weight α is sufficiently small relative to the bounded solver $\mathcal{A}_\varpi$.

Proposition 4.8 (Norm equivalence)**.** *If the parameter α satisfies $\alpha\|\mathcal{A}_\varpi\| \le 1/2$, then for all $f \in \mathcal{H}$:*

$$\frac{1}{2}(1 - 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2 \le \mathcal{E}_{\alpha,\varpi}(f) \le \frac{1}{2}(1 + 2\alpha\|\mathcal{A}_\varpi\|)\|f\|^2. \tag{41}$$

Proof. For every $f \in \mathcal{H}$, Cauchy–Schwarz gives

$$\left|\Re\langle \mathcal{A}_\varpi f, f\rangle\right| \le \|\mathcal{A}_\varpi f\|\,\|f\| \le \|\mathcal{A}_\varpi\|\,\|f\|^2.$$

Substituting this bound into the definition

$$\mathcal{E}_{\alpha,\varpi}(f) = \tfrac{1}{2}\|f\|^2 + \alpha\Re\langle \mathcal{A}_\varpi f, f\rangle$$

immediately yields

$$\tfrac{1}{2}\|f\|^2 - \alpha\|\mathcal{A}_\varpi\|\,\|f\|^2 \le \mathcal{E}_{\alpha,\varpi}(f) \le \tfrac{1}{2}\|f\|^2 + \alpha\|\mathcal{A}_\varpi\|\,\|f\|^2,$$

which is exactly the claimed norm equivalence. □

5. Convergence bounds for the lifted semigroup

We now derive quantitative decay bounds for the semigroup on the mean-zero subspace. The strategy has two parts: a constructive lower bound from a 2×2 dissipation matrix, and a complementary overdamped upper ceiling from singular values. Together they identify the finite-friction acceleration window.

5.1. **Lower bound via modified hypocoercivity.** We derive an explicit lower bound by optimizing the Lyapunov parameters (α, ϖ); the full matrix estimates are in Appendix C. We package the structural input from Assumption 4.1 into derived quantities: a coupling scale $Z(\gamma)$, an optimal Lyapunov weight $\alpha_*(\gamma)$, an unperturbed decay profile $\Lambda_{\text{hypo}}(\gamma)$, and a perturbative loss rate $B_{\text{pert}}(\gamma)$.
Notation for the rate functions. Define the structural scalars

$$\Theta := \frac{\sqrt{M_R}}{2}, \qquad K_{\text{geom}} := \kappa_{\text{ell}}, \qquad b_{2,\text{eff}} := b_2 + \frac{a_2}{2}(1 + \lambda_R), \tag{42}$$

where κ_{ell} is the flux regularity constant from (25); here Θ measures slow–fast flux amplitude and K_{geom} collects the cross-term transport geometry. We note the distinction from the forward regularity constant $k_{\text{reg}} = \sqrt{M_R/\lambda_R}$ of Lemma 4.6: k_{reg} enters the trace of $\mathbf{M}$ (diagonal block M_{11}), whereas $K_{\text{geom}} = \kappa_{\text{ell}}$ enters the coupling scale $Z(\gamma)$ (off-diagonal block M_{12}).

For each friction $\gamma > 0$, define the *coupling scale*

$$Z(\gamma) := \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}, \tag{43}$$

which measures effective slow–fast interaction strength as a function of γ, and set:

$$\alpha_*(\gamma) := \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}, \tag{44}$$

$$\Lambda_{\text{hypo}}(\gamma) := \frac{\frac{1}{4}\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\text{reg}}}, \tag{45}$$

$$B_{\text{pert}}(\gamma) := b_{2,\text{eff}} + \alpha_*(\gamma)\, C_{\text{sing}}\, \lambda(Q_O)^{-1/2}, \tag{46}$$

where $C_{\text{sing}} := 2(b_1 + b_2)\, k_{\mathcal{A}} + (a_1 + a_2)$ is the perturbation interaction constant (see Appendix C for the derivation), with $k_{\mathcal{A}}$ the resolvent norm bound from Lemma 4.6. Here $\alpha_*(\gamma)$ is the optimal Lyapunov mixing weight, $\Lambda_{\text{hypo}}(\gamma)$ is the lower-rate profile, and $B_{\text{pert}}(\gamma)$ quantifies perturbation destabilization. Although B_{pert} contains an apparent $\lambda(Q_O)^{-1/2}$ factor, it is cancelled at optimal friction because $\alpha_*(\gamma_{\text{opt}}) \asymp \sqrt{\lambda(Q_O)}$ (Corollary 5.2).

We also isolate the explicit admissibility restriction coming from the symmetric perturbation in the basic energy estimate:

$$\eta_{\text{sym}}(\gamma) := \begin{cases} +\infty, & a_2 = 0, \\ 2\gamma/a_2, & a_2 > 0. \end{cases} \tag{47}$$

Equivalently, $\eta \le \eta_{\text{sym}}(\gamma)$ is exactly the condition ensuring that the coefficient of $\|(-\mathcal{R})^{1/2} f\|^2$ remains nonnegative after the symmetric perturbation is absorbed into the dissipative part; see Appendix C.

With the gap-adapted choice $\varpi_* := \lambda(Q_O)$, define the associated Lyapunov-distortion parameter and norm-equivalence prefactor by

$$\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|, \tag{48}$$

$$C_*(\gamma) := \sqrt{\frac{1 + \delta_*(\gamma)}{1 - \delta_*(\gamma)}}. \tag{49}$$

Main result. The structural properties established in Appendix C yield an explicit norm equivalence with distortion parameter $\delta_*(\gamma) < 1$ for all $\gamma > 0$. In fact the appendix proves the uniform bound $\delta_*(\gamma) \le 2^{-1/2}$; equivalently,

$$\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\| = \frac{\delta_*(\gamma)}{2} \le 2^{-3/2} < \frac{1}{2},$$

so Proposition 4.8 applies to the optimized Lyapunov functional, and hence $C_*(\gamma) \le 1 + \sqrt{2}$.

Theorem 5.1 (Explicit quantitative hypocoercivity)**.** *Suppose Assumptions 4.1 and 4.2 hold, and in the unbounded setting also Assumption 4.3. Set the explicit certified threshold*

$$\eta^*(\gamma) := \min\left\{\eta_{\text{sym}}(\gamma),\ \frac{\Lambda_{\text{hypo}}(\gamma)}{2B_{\text{pert}}(\gamma)}\right\}.$$

For any $\gamma > 0$, if $\eta \le \eta^*(\gamma)$, then $e^{t\mathcal{L}}$ is exponentially stable on the invariant mean-zero subspace $\mathcal{H}_0 = \ker(\Pi)$:

$$\|e^{t\mathcal{L}} f\| \le C_*(\gamma)\, e^{-\lambda_\eta t}\|f\| \le (1 + \sqrt{2})\, e^{-\lambda_\eta t}\|f\|, \quad \forall t \ge 0,\ \forall f \in \mathcal{H}_0. \tag{50}$$

The global decay rate λ_η satisfies the strict lower bound:

$$\lambda_\eta \ge \frac{\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)}{1 + \delta_*(\gamma)} \ge (2 - \sqrt{2})\,(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)). \tag{51}$$

Proof outline. The full computation is in Appendix C; here we record the logical structure of the argument and where each constant enters. The Lyapunov derivative is first computed on the analytic core and then extended to all of $\mathcal{H}_0$ by density, using the boundedness of $\mathcal{A}_{\varpi_*}$.

Step 1: Lyapunov derivative. With the resolvent shift fixed at $\varpi_* = \lambda(Q_O)$, expanding $\frac{d}{dt}\mathcal{E}_{\alpha,\varpi_*}(f_t) = \Re\langle \mathcal{L}f, f\rangle + \alpha\,\Re[\langle \mathcal{A}_{\varpi_*}\mathcal{L}f, f\rangle + \langle \mathcal{A}_{\varpi_*}f, \mathcal{L}f\rangle]$ and grouping terms by their P_S/P_F component yields the quadratic bound $\frac{d}{dt}\mathcal{E}_{\alpha,\varpi_*} \le -\mathbf{v}^T\mathbf{M}(\alpha)\mathbf{v}$ for the vector $\mathbf{v} = (\|(-\mathcal{R})^{1/2}f_F\|, \|f_S\|)^T$ and a 2×2 dissipation matrix

$$\mathbf{M}(\alpha) = \begin{pmatrix} \gamma - \alpha\, k_{\mathrm{reg}} & -\frac{1}{2}\alpha\, Z(\gamma)/\sqrt{\lambda_R} \\ -\frac{1}{2}\alpha\, Z(\gamma)/\sqrt{\lambda_R} & \frac{\alpha}{2}\cdot\frac{\lambda(Q_O)}{\varpi_*+\lambda(Q_O)} \end{pmatrix}, \tag{52}$$

where we have used: (a) the basic energy estimate $\Re\langle \mathcal{R}f_F, f_F\rangle \le -\lambda_R\|f_F\|^2$ for M_{11}; (b) the corrector bound $\|\mathcal{V}_{\mathrm{skew}}\mathcal{A}_\varpi\| \le k_{\mathrm{reg}}$ and coupling estimate from the cross-term for M_{12}; (c) the slow coercive identity $\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\mathrm{skew}}f_F\rangle = \langle (\varpi_* - Q_O)^{-1}(-Q_O)f_S, f_S\rangle \ge \frac{\lambda(Q_O)}{\varpi_*+\lambda(Q_O)}\|f_S\|^2$ for M_{22}. The "term (c)" analogue $\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\mathrm{skew}}f_S\rangle$ vanishes because $P_S\mathcal{V}_{\mathrm{skew}}P_S = 0$ (Assumption 4.1.2), so $\mathcal{V}_{\mathrm{skew}}f_S \in \mathcal{H}_F$ and the resulting expression is absorbed into the off-diagonal coupling M_{12}. Here (52) is written for the unperturbed case $\eta = 0$; perturbative corrections to M_{11} and M_{12} are proportional to η and are estimated in Appendix C.

Step 2: Matrix optimization. The entries of $\mathbf{M}(\alpha)$ come from three ingredients: microscopic dissipation through λ_R, slow coercivity through the spectral gap $\lambda(Q_O)$, and the transport/perturbation remainders through the constants Θ, K_{geom}, and $b_{2,\mathrm{eff}}$. Maximising the determinant of $\mathbf{M}(\alpha)$ yields the optimal weight $\alpha_*(\gamma)$ from (44) and the rate $\Lambda_{\mathrm{hypo}}(\gamma)$.

Step 3: Distortion control. The distortion parameter $\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|$ satisfies

$$\delta_*(\gamma) \le \frac{\sqrt{\lambda_R}}{\sqrt{M_R}\big(K_{\mathrm{geom}} + \sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}}\big)} \le 2^{-1/2} \qquad \forall\, \gamma > 0, \tag{53}$$

by the solver bound $\|\mathcal{A}_{\varpi_*}\| \le (2\sqrt{\lambda_R})^{-1}\lambda(Q_O)^{-1/2}$, the formula for α_* in (44), and the elementary extremum $\sup_{x>0} x\big[(x+K_{\mathrm{geom}})^2 + 2k_{\mathrm{reg}}\big]^{-1} = 1/\big(2(K_{\mathrm{geom}} + \sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}})\big)$. This gives the explicit norm equivalence encoded by $C_*(\gamma)$, together with the uniform simplification $C_*(\gamma) \le 1+\sqrt{2}$ and the lower bound in (51). □

Corollary 5.2 (Kinetic acceleration and singularity cancellation)**.** *Assume the setting of Theorem 5.1. Assume furthermore that the structural constants Θ, K_{geom}, k_{reg}, λ_R, $b_{2,\mathrm{eff}}$, and C_{sing} are independent of $\lambda(Q_O)$ (in particular, they remain $\mathcal{O}(1)$ as $\lambda(Q_O)\downarrow 0$), and that $b_{2,\mathrm{eff}} > 0$.*

Then the unperturbed rate function $\Lambda_{\mathrm{hypo}}(\gamma)$ defined in (45) attains its global maximum at

$$\gamma_{\mathrm{opt}} = \frac{\sqrt{\lambda(Q_O)}}{\Theta}\sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}} \asymp \sqrt{\lambda(Q_O)}. \tag{54}$$

At this friction scale, the maximal lower bound exhibits kinetic scaling:

$$\max_{\gamma>0} \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{\lambda(Q_O)}. \tag{55}$$

Moreover, the mixing weight in (46) satisfies

$$\alpha_*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}, \qquad \alpha_*(\gamma_{\mathrm{opt}})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1), \tag{56}$$

so the solver singularity $\lambda(Q_O)^{-1/2}$ in $B_{\mathrm{pert}}(\gamma)$ is cancelled at the optimal friction. Consequently, $B_{\mathrm{pert}}(\gamma_{\mathrm{opt}}) = \mathcal{O}(1)$ and the certified threshold satisfies

$$\eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{57}$$

Proof. Write

$$\Lambda_{\mathrm{hypo}}(\gamma) = \frac{\lambda_R}{4}\,\frac{\gamma}{\big(\gamma\Theta\lambda(Q_O)^{-1/2} + K_{\mathrm{geom}}\big)^2 + 2k_{\mathrm{reg}}}. \tag{58}$$

Set $x := \gamma\Theta\lambda(Q_O)^{-1/2}$ and $C_0 := 2k_{\rm reg}$. Then $\Lambda_{\rm hypo}(\gamma) = \frac{\lambda_R}{4\Theta}\sqrt{\lambda(Q_O)}\,\frac{x}{(x+K_{\rm geom})^2+C_0}$. A direct derivative computation shows that the function $x \mapsto \frac{x}{(x+K_{\rm geom})^2+C_0}$ is maximized at $x = \sqrt{K_{\rm geom}^2 + C_0}$, giving the stated $\gamma_{\rm opt}$ and the scaling $\max_\gamma \Lambda_{\rm hypo}(\gamma) \asymp \sqrt{\lambda(Q_O)}$.

For the perturbation term, by (46), $B_{\rm pert}(\gamma) = b_{2,\rm eff} + \alpha_*(\gamma)\, C_{\rm sing}\, \lambda(Q_O)^{-1/2}$. At $\gamma = \gamma_{\rm opt}$ we have $x = \mathcal{O}(1)$ and $Z(\gamma_{\rm opt}) = x + K_{\rm geom} = \mathcal{O}(1)$, hence $\alpha_*(\gamma_{\rm opt}) = \frac{\gamma_{\rm opt}\lambda_R}{Z(\gamma_{\rm opt})^2+2k_{\rm reg}} = \mathcal{O}(\sqrt{\lambda(Q_O)})$, which implies $\alpha_*(\gamma_{\rm opt})\,\lambda(Q_O)^{-1/2} = \mathcal{O}(1)$ and therefore $B_{\rm pert}(\gamma_{\rm opt}) = \mathcal{O}(1)$. Finally, the matrix-controlled threshold satisfies $\Lambda_{\rm hypo}(\gamma_{\rm opt})/(2B_{\rm pert}(\gamma_{\rm opt})) \asymp \sqrt{\lambda(Q_O)}$. If $a_2 = 0$, then $\eta_{\rm sym}(\gamma_{\rm opt}) = +\infty$ and therefore $\eta^*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}$. If $a_2 > 0$, then $\eta_{\rm sym}(\gamma_{\rm opt}) = 2\gamma_{\rm opt}/a_2 \asymp \sqrt{\lambda(Q_O)}$. In either case, $\eta^*(\gamma_{\rm opt}) = \min\{\eta_{\rm sym}(\gamma_{\rm opt}), \Lambda_{\rm hypo}(\gamma_{\rm opt})/(2B_{\rm pert}(\gamma_{\rm opt}))\} \asymp \sqrt{\lambda(Q_O)}$. □

The threshold $\eta^*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}$ says that forcing and kinetic relaxation are measured on the same scale. This is still perturbative, as in earlier approaches [24, 37, 16, 17], but here the threshold is explicit once the structural constants are known. For the equilibrium Langevin warm-up of Section 2, the general formulas recover Proposition 2.2 exactly.

Remark 5.3 (Why the gap-adapted shift matters: root-cause spectral analysis). The $\sqrt{\lambda(Q_O)}$ scaling comes from a single spectral multiplier in the $(2,2)$-entry of the dissipation matrix:

$$M_{22}^0 \;=\; \frac{\alpha}{2}\cdot\frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)}. \tag{59}$$

If ϖ is fixed while $\lambda := \lambda(Q_O) \downarrow 0$, then $\lambda/(\varpi+\lambda) \asymp \lambda$, so the slow coercive block is too small and the determinant–trace bound gives the familiar $\lambda^{5/2}$ loss at $\gamma \asymp \sqrt{\lambda}$. If instead $\varpi = \lambda(Q_O)$, then the multiplier is exactly $1/2$, the slow block is order one, and the same argument gives $\Lambda_{\rm hypo} \asymp \sqrt{\lambda}$. More generally, any shift $\varpi = c\,\lambda(Q_O)$ with fixed $c > 0$ gives the same exponent; $c = 1$ is simply the cleanest choice for the explicit bound (45).

Define the (rigorous) L^2 relaxation time as in [31] by

$$t_{\rm rel}(\mathcal{L}) := \inf\Big\{t \ge 0\,;\ \|e^{t\mathcal{L}}f\| \le e^{-1}\,\|f\| \text{ for all } f \in \mathcal{H}_0\Big\}. \tag{60}$$

The following holds:

Corollary 5.4 (Acceleration in the L^2 relaxation time)**.** *Assume the setting of Theorem 5.1. Then for $\eta \le \eta^*(\gamma)$, the exponential decay estimate from Theorem 5.1 implies*

$$t_{\rm rel}(\mathcal{L}) \le \frac{1+\log C_*(\gamma)}{\lambda_\eta} \le \frac{1+\log(1+\sqrt{2})}{\lambda_\eta}. \tag{61}$$

In particular, at the optimal friction $\gamma_{\rm opt} \asymp \sqrt{\lambda(Q_O)}$ (Corollary 5.2) and for perturbations within the certified threshold, one has

$$t_{\rm rel}(\mathcal{L}) = \mathcal{O}\big(\lambda(Q_O)^{-1/2}\big). \tag{62}$$

Proof. By Theorem 5.1, for all $f \in \mathcal{H}_0$ and $t \ge 0$ one has $\|e^{t\mathcal{L}}f\| \le C_*(\gamma)\,e^{-\lambda_\eta t}\,\|f\|$. Choose $t = (1+\log C_*(\gamma))/\lambda_\eta$ so that $C_*(\gamma)\,e^{-\lambda_\eta t} = e^{-1}$. □

5.2. **Overdamped diffusive ceiling.** To establish that the high-friction γ^{-1} behavior is intrinsic and not an artifact of the lower-bound construction, we derive a universal upper bound on any admissible L^2 exponential rate in terms of the singular-value gap.

Define the *singular-value gap*

$$s(\mathcal{L}) := \inf_{f\in\mathcal{H}_0\cap{\rm dom}(\mathcal{L})\setminus\{0\}} \frac{\|\mathcal{L}f\|}{\|f\|}, \tag{63}$$

that is, the smallest singular value of $\mathcal{L}|_{\mathcal{H}_0}$ in finite dimensions, remaining meaningful as an infimum over the operator domain in unbounded settings. Quantities of this type are continuous-time analogues of *multiplicative reversibilizations* and the *pseudo spectral gap* used in non-reversible Markov chain theory; see, e.g., [35]. In the discrete-time finite-state setting, a closely related point of view is developed by Chatterjee, who defines a nonreversible "spectral gap" via singular values of the generator $I - P$ [10].

Assumption 5.5 (Overdamped ceiling hypotheses)**.** In addition to Assumptions 4.1, 4.2 (and 4.3 in the unbounded setting), we assume:

(i) $\ker(\mathcal{L}) \cap \mathcal{H}_0 = \{0\}$;
(ii) $\lambda(Q_O)$ is attained by a normalized eigenfunction $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0$ (automatic in finite dimensions and many compact-resolvent settings);
(iii) the first-order corrector $\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi$ belongs to $\mathrm{dom}(\mathcal{L}) \cap \mathrm{dom}((-\mathcal{R})^{1/2})$ with $P_F\mathcal{V}_{\mathrm{skew}}\psi$ well defined;
(iv) $e^{t\mathcal{L}}$ is a contraction semigroup on $\mathcal{H}_0$.

Theorem 5.6 (Overdamped diffusive ceiling via singular value gap)**.** *Assume Assumption 5.5. Let $C \geq 1$ and $\nu > 0$ be such that*

$$\|e^{t\mathcal{L}} f\| \leq C\, e^{-\nu t}\, \|f\|, \qquad \forall t \geq 0,\ \forall f \in \mathcal{H}_0. \tag{64}$$

Then for every $\gamma > 0$,

$$\nu \leq (1 + \log C)\, s(\mathcal{L}) \leq (1 + \log C) \left[\frac{(1 + \kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\,(b_1 + b_2) + \eta\,(a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma} \right], \tag{65}$$

where κ_{ell} is the flux-regularity constant from Assumption 4.1.3 and a_1, a_2, b_1, b_2 are from Assumption 4.1.4. In particular, for $\eta \leq c/\gamma$ one obtains the diffusive overdamped scaling $\nu = \mathcal{O}(\gamma^{-1})$ as $\gamma \to \infty$.

Proof outline. The proof is given in Appendix D; we indicate the two main steps. The key point is that one can test the singular value gap on a *first-order corrector* $f_\gamma := \phi - \gamma^{-1}\psi$ built from a slow eigenfunction $Q_O\phi = -\lambda(Q_O)\phi$ and $\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi$. The additional domain hypothesis ensures that $f_\gamma \in \mathrm{dom}(\mathcal{L})$ and that all terms below are well defined. The flux regularity bound (25) controls the only nontrivial fast remainder term $P_F\mathcal{V}_{\mathrm{skew}}\psi$ without ever invoking $\|\mathcal{V}_{\mathrm{skew}}\|$. The perturbation contribution is controlled by the $(-\mathcal{R})^{1/2}$-energy bounds in Assumption 4.1.4 and the explicit identity $\|(-\mathcal{R})^{1/2}\psi\|^2 = \lambda(Q_O)$. This yields the bound on $s(\mathcal{L})$. The first inequality $\nu \leq (1+\log C)\, s(\mathcal{L})$ then follows from Lemma D.1, which combines the contraction bound $\|e^{t\mathcal{L}} f\| \leq \|f\|$ on $[0, (\log C)/\nu]$ with the assumed exponential decay on $[(\log C)/\nu, \infty)$. □

Theorem 5.6 shows that in the overdamped regime the best achievable L^2 exponential decay rate (with any finite prefactor C) is bounded above by $(1 + \log C)$ times the macroscopic diffusion scale $\lambda(Q_O)/\gamma$, up to explicit perturbative contributions. Combined with Theorem 5.1 and, for the explicit $\sqrt{\lambda(Q_O)}$ scaling statement, Corollary 5.2, this shows that the accelerated regime must occur at *intermediate* friction $\gamma \asymp \sqrt{\lambda(Q_O)}$, rather than in the large-friction limit.

At equilibrium ($\eta = 0$), the contraction hypothesis (Assumption 5.5(iv)) holds unconditionally: $\mathrm{Re}\langle \mathcal{L}f, f\rangle = \gamma\langle \mathcal{R}f, f\rangle \leq 0$ by skew-adjointness of $\mathcal{V}_{\mathrm{skew}}$, so $e^{t\mathcal{L}}$ is a contraction semigroup on $\mathcal{H}_0$. Together with Theorem 5.1, this gives a matching characterization at equilibrium: the exponent 1/2 in the $\sqrt{\lambda(Q_O)}$ scaling is optimal. For $\eta \neq 0$, the generator is generically not dissipative in $L^2(\mu_0)$; however, recent works establish L^2 convergence in stationary-state-adapted norms where contraction is built into the framework; see, for instance, Monmarché [33, Assumption 1 and Thm. 2] and Dietert [12]. Reformulating the ceiling in such a norm would make the hypothesis automatic, at the cost of modifying the structural constants; this gap is one of reference-measure choice rather than a fundamental obstruction (see Open Problem 3). For designed lifts in finite dimensions,

the remaining ceiling hypotheses are automatic; in PDE models they reduce to the same kind of first-order corrector regularity that already appears in the Langevin example below.

Corollary 5.7 (Diffusive ceiling for the L^2 relaxation time)**.** *Assume the setting of Theorem 5.6. Define $t_{\mathrm{rel}}(\mathcal{L})$ as in Corollary 5.4. Then the singular-value gap satisfies the explicit ceiling*

$$s(\mathcal{L}) \le \frac{(1+\kappa_{\mathrm{ell}})\,\lambda(Q_O)}{\gamma} + \eta\,(b_1+b_2) + \eta\,(a_1+a_2)\,\frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{66}$$

Moreover, the relaxation time is bounded below in terms of the singular-value gap (see Lemma D.2):

$$t_{\mathrm{rel}}(\mathcal{L}) \ge \frac{1}{2\,s(\mathcal{L})}. \tag{67}$$

Consequently, if $\eta \le c/\gamma$, then $s(\mathcal{L}) = \mathcal{O}(\gamma^{-1})$ as $\gamma \to \infty$ and hence

$$t_{\mathrm{rel}}(\mathcal{L}) = \Omega(\gamma) \qquad (\gamma \to \infty). \tag{68}$$

In particular, combining this ceiling with Corollary 5.4 shows that, at optimal friction, the relaxation-time upper bound is of kinetic order $t_{\mathrm{rel}}(\mathcal{L}) = \mathcal{O}(\lambda(Q_O)^{-1/2})$.

6. Examples

This section explains how to build lifts from a prescribed slow generator and then applies the construction to two examples: a convex-potential Langevin family and a finite-dimensional boundary-driven model. In both examples Theorem 5.1 applies rigorously; the overdamped ceiling remains conditional away from equilibrium because the extra contraction hypothesis is not verified in the present Hilbert norms.

6.1. **Reverse adiabatic elimination: from effective dynamics to lifted dynamics.** Given a non-equilibrium effective generator $\mathcal{L}_O$ on the slow space $\mathcal{H}_S$, we seek a lifted generator $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ on $\mathcal{H} = \mathcal{H}_S \oplus \mathcal{H}_F$ whose adiabatic elimination (Definition 3.4) is exactly $\mathcal{L}_O$. We call this *reverse adiabatic elimination*: identify the reversible backbone Q_O, choose a fast space and skew coupling that reproduce it, and then add the perturbation so that the effective generator is exactly $\mathcal{L}_O$. In finite dimensions this is algebraic; in unbounded settings one must also verify the relevant domain properties.

Perturbation expansion of $\mathcal{L}_O$. By Definition 3.4, $\mathcal{L}_O = -P_S\mathcal{V}\mathcal{R}^+\mathcal{V}P_S$ with $\mathcal{V} = \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$. Expanding in powers of η gives

$$\mathcal{L}_O = \underbrace{-P_S\mathcal{V}_{\mathrm{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{skew}}\,P_S}_{Q_O}\ \underbrace{-\,\eta\,P_S\big(\mathcal{V}_{\mathrm{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{pert}}\ +\ \mathcal{V}_{\mathrm{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{skew}}\big)P_S}_{\text{cross terms } \mathcal{O}(\eta)}\ \underbrace{-\,\eta^2\,P_S\mathcal{V}_{\mathrm{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\mathrm{pert}}\,P_S}_{\text{pure perturbation } \mathcal{O}(\eta^2)}. \tag{69}$$

The first term is the reversible backbone Q_O; the other two are the $\mathcal{O}(\eta)$ and $\mathcal{O}(\eta^2)$ corrections. In the boundary-driven example they are written explicitly in (95).

Step 1: Identify Q_O. Set $\eta = 0$ in the slow model. The result is Q_O, whose spectral gap $\lambda(Q_O)$ is the basic scale in Theorem 5.1.

Step 2: Choose fast space and coupling to match Q_O. The adiabatic-elimination identity implies

$$Q_O = T^*\,(\mathcal{R}_F)^{-1}\,T, \qquad T := P_F\mathcal{V}_{\mathrm{skew}}P_S, \tag{70}$$

where $\mathcal{R}_F := \mathcal{R}|_{\mathcal{H}_F} < 0$. So, after choosing $\mathcal{H}_F$ and $\mathcal{R}_F$, we pick $T : \mathcal{H}_S \to \mathcal{H}_F$ to satisfy $T^*\mathcal{R}_F^{-1}T = Q_O$. A universal choice is $\mathcal{H}_F = \mathcal{H}_S$, $\mathcal{R}_F = -\lambda_R\,\mathrm{id}$, $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. Then $T^*\mathcal{R}_F^{-1}T = Q_O$, and we set $\mathcal{R} := 0 \oplus \mathcal{R}_F$ and $\mathcal{V}_{\mathrm{skew}} := \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$.

This universal choice always matches the prescribed Q_O and is the version used in the algorithmic-lift comparisons later in the paper.

Step 3: Determine $\mathcal{V}_{\text{pert}}$ from consistency. With $\mathcal{R}$ and $\mathcal{V}_{\text{skew}}$ fixed, we choose $\mathcal{V}_{\text{pert}}$ so that adiabatic elimination exactly reproduces $\mathcal{L}_O$ whenever the resulting consistency problem is solvable in the chosen class of operators. This is equivalent to the consistency relation

$$\mathcal{L}_O - Q_O \;=\; -\eta\, P_S\big(\mathcal{V}_{\text{skew}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{skew}}\big)P_S \;-\; \eta^2\, P_S\mathcal{V}_{\text{pert}}\,\mathcal{R}^+\,\mathcal{V}_{\text{pert}}\,P_S, \tag{71}$$

with the centering constraint $P_S\mathcal{V}_{\text{pert}}P_S = 0$. In the Langevin example, $\mathcal{V}_{\text{pert}}P_S = 0$, so the formula simplifies strongly. In the boundary-driven example, all terms are present, and we write them explicitly in (95). In finite dimensions this is an algebraic problem, usually with non-unique solutions; in infinite dimensions solvability is model-dependent. The resulting lifted generator $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ automatically satisfies centering. To apply Theorem 5.1, one must in addition verify the remaining structural bounds and the kernel-compatible gap assumption $\ker(Q_O) = \mathbb{C}\mathbf{1}$. The key structural advantage is that, for a broad class of designed lifts, the projection bias can be removed exactly.

Proposition 6.1 (Unbiasedness of designed lifts)**.** *Consider a lift $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}}$ constructed via the reverse adiabatic elimination recipe above, with $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$ (Step 2) and $\mathcal{V}_{\text{pert}}$ satisfying the centering condition $P_S\mathcal{V}_{\text{pert}}P_S = 0$ and the consistency condition* (71)*. If additionally $P_F\mathcal{V}_{\text{pert}}P_F = 0$ (the perturbation has no fast-to-fast component), then:*

(i) $P_F\mathcal{V}P_F = 0$;
(ii) the Schur complement (72) *satisfies $\gamma\, S(\gamma) = \mathcal{L}_O$ for all $\gamma > 0$;*
(iii) whenever the lifted and effective stationary states are well defined, the lifted NESS satisfies $P_S h_\eta = g_\eta$ for all $\gamma > 0$, i.e., the projection bias $\Delta_{\text{bias}}(\gamma) \equiv 0$.

The condition $P_F\mathcal{V}_{\text{pert}}P_F = 0$ is a design choice: (71) *constrains only the off-diagonal blocks, so the fast–fast block may be set to zero without changing the effective dynamics.*

Proof. (i) Since $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$ is block-off-diagonal, $P_F\mathcal{V}_{\text{skew}}P_F = 0$. Combined with $P_F\mathcal{V}_{\text{pert}}P_F = 0$, we get $P_F\mathcal{V}P_F = P_F(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})P_F = 0$. (ii) With $P_F\mathcal{V}P_F = 0$, the Schur complement (72) reduces to $S(\gamma) = -P_S\mathcal{V}(\gamma\mathcal{R}_F)^{-1}\mathcal{V}\,P_S = \gamma^{-1}\mathcal{L}_O$. (iii) Since $\gamma S(\gamma) = \mathcal{L}_O$, the NESS equation $S(\gamma)u = 0$ becomes $\mathcal{L}_O u = 0$, so $u = g_\eta$. □

Proposition 6.1 gives the design rule we use later: if $P_F\mathcal{V}_{\text{pert}}P_F = 0$, then the projected lifted steady state equals the overdamped one for all γ. This does not remove non-equilibrium effects, because $\mathcal{L}_O$ still depends on the off-diagonal perturbation blocks.

When $\eta \neq 0$ and $P_F\mathcal{V}P_F \neq 0$, the NESS $h_\eta \in \ker(\mathcal{L})$ acquires a fast component $P_F h_\eta = \mathcal{O}(\eta/\gamma)$, and the slow marginal $u = P_S h_\eta$ satisfies the Schur complement equation

$$S(\gamma)\,u = 0, \qquad S(\gamma) := -P_S\mathcal{V}\big(\gamma\mathcal{R}_F + P_F\mathcal{V}P_F\big)^{-1}\mathcal{V}\,P_S. \tag{72}$$

Expanding in a Neumann series gives

$$\gamma\, S(\gamma) = \mathcal{L}_O + \gamma^{-1}\, P_S\mathcal{V}\,\mathcal{R}^+\,(P_F\mathcal{V}P_F)\,\mathcal{R}^+\,\mathcal{V}\,P_S + \mathcal{O}(\gamma^{-2}). \tag{73}$$

Thus $P_S h_\eta \to g_\eta$ as $\gamma \to \infty$, and under the usual isolated-eigenvalue perturbation hypotheses one gets $\Delta_{\text{bias}}(\gamma) := \|P_S h_\eta - g_\eta\| = \mathcal{O}(\eta/\gamma)$. When $P_F\mathcal{V}P_F = 0$, as in Proposition 6.1, the expansion collapses to $\gamma S(\gamma) = \mathcal{L}_O$ and the bias vanishes identically.

Since the lifted dynamics converges to h_η, the approximation error for the overdamped NESS g_η satisfies the triangle inequality

$$\|P_S e^{t\mathcal{L}} f - g_\eta\|_{\mathcal{H}_S} \;\le\; \underbrace{C_*(\gamma)\, e^{-\lambda_\eta t}\,\|f - h_\eta\|}_{\text{convergence error}} \;+\; \underbrace{\|P_S h_\eta - g_\eta\|_{\mathcal{H}_S}}_{\text{bias } \Delta_{\text{bias}}(\gamma)}. \tag{74}$$

For lifts satisfying the conditions of Proposition 6.1, $\Delta_{\text{bias}}(\gamma) = 0$ and the accelerated mixing time is

$$T_{\text{mix}}(\varepsilon) \;=\; \mathcal{O}\Big(\frac{\log(1/\varepsilon)}{\sqrt{\lambda(Q_O)}}\Big), \qquad \forall\, \varepsilon > 0. \tag{75}$$

Thus, in the unbiased case, the construction has three steps: factor $Q_O = T^*\mathcal{R}_F^{-1}T$, choose $\mathcal{V}_{\text{skew}} = \left(\begin{smallmatrix} 0 & -T^* \\ T & 0 \end{smallmatrix}\right)$ and solve the consistency relation (71) with $P_F\mathcal{V}_{\text{pert}}P_F = 0$, then run the lifted dynamics at $\gamma_{\text{opt}} \asymp \sqrt{\lambda(Q_O)}$ until the right-hand side of (74) falls below the target accuracy. If $P_F\mathcal{V}P_F \neq 0$, the same estimate suggests a two-phase variant: first run the lifted dynamics down to the bias floor $\Delta_{\text{bias}}(\gamma_{\text{opt}}) = \mathcal{O}(\eta/\sqrt{\lambda(Q_O)})$, then switch to $\mathcal{L}_O$ to remove the residual error. This construction is in the spirit of previous acceleration methods in Markov-chain and MCMC settings [38, 29, 23, 31], cast here in the continuous-time operator language of Assumption 4.1; for the universal algorithmic lift one has $M_R = \lambda_R$ and $\kappa_{\text{ell}} = 0$, while for the convex-potential Langevin family one has $\lambda_R = M_R = 1$ and a family-uniform Bochner bound on κ_{ell}.

6.2. **Example 1: Nonequilibrium Langevin family with convex potential.** We study the underdamped Langevin family on $\mathbb{R}^{2d}$ with a uniformly convex confining potential and a bounded non-gradient perturbation. This is our main infinite-dimensional example, and it verifies the $\sqrt{\lambda(Q_O)}$ mechanism uniformly as $m \downarrow 0$. Throughout this example we fix $d \geq 2$, $\beta > 0$, $m \in (0, 1]$, and take $U \in C^\infty(\mathbb{R}^d)$ to satisfy

$$\nabla^2 U(q) \geq m \quad \text{for all } q \in \mathbb{R}^d, \tag{76}$$

with all derivatives of positive order growing at most polynomially at infinity. These regularity assumptions are used only to place the model within the standard confining Kramers–Fokker–Planck framework; they do not affect the explicit constants below. We also assume $F \in C_b^\infty(\mathbb{R}^d; \mathbb{R}^d)$ is bounded and not of gradient form, so $\mathcal{L}_O$ is genuinely nonequilibrium for every $\eta \neq 0$.

Setup. Let

$$d\mu(q, p) = Z^{-1} \exp\Big(-\beta\big(\tfrac{1}{2}|p|^2 + U(q)\big)\Big)\, dq\, dp = d\mu_q(q)\, d\mu_p(p), \tag{77}$$

where $\mu_q(dq) \propto e^{-\beta U(q)}\, dq$, and work in $\mathcal{H} = L^2(\mu)$ with slow projection

$$P_S f = \Pi_p f := \int_{\mathbb{R}^d} f(\cdot, p)\, d\mu_p(p). \tag{78}$$

We use the Schwartz core

$$\mathcal{C}_{\text{Lan}} := \mathcal{S}(\mathbb{R}^d_q \times \mathbb{R}^d_p). \tag{79}$$

Generator decomposition. The Langevin generator is written as

$$\mathcal{L} = \mathcal{V}_{\text{skew}} + \gamma\mathcal{R} + \eta\mathcal{V}_{\text{pert}}, \tag{80}$$

with

$$\mathcal{V}_{\text{skew}} := p \cdot \nabla_q - \nabla_q U \cdot \nabla_p, \qquad \mathcal{R} := -p \cdot \nabla_p + \beta^{-1}\Delta_p, \qquad \mathcal{V}_{\text{pert}} f := F(q) \cdot \nabla_p f. \tag{81}$$

Effective generator $\mathcal{L}_O$ (explicit formula). Since $\mathcal{R}(p_i) = -p_i$ and $\int p_i\, d\mu_p = 0$, Definition 3.4 gives, for $\phi = \phi(q)$,

$$\mathcal{L}_O\phi = \beta^{-1}\Delta_q\phi - (\nabla_q U - \eta F(q)) \cdot \nabla_q\phi, \tag{82}$$

and therefore

$$Q_O\phi = \beta^{-1}\Delta_q\phi - \nabla_q U \cdot \nabla_q\phi. \tag{83}$$

The slow reference measure is $\mu_q(dq) \propto e^{-\beta U(q)}\, dq$. The Bakry–Émery criterion [5], applied to the uniform convexity $\nabla^2 U \geq m > 0$, yields the Poincaré inequality

$$-\langle \phi, Q_O \phi \rangle_{L^2(\mu_q)} = \beta^{-1} \|\nabla_q \phi\|^2_{L^2(\mu_q)} \geq m \, \|\phi - \textstyle\int \phi \, d\mu_q\|^2_{L^2(\mu_q)}. \tag{84}$$

Hence

$$\ker(Q_O) = \mathbb{C}\mathbf{1}, \qquad \lambda(Q_O) = m. \tag{85}$$

Since F is not a gradient field, the drift $\nabla_q U - \eta F(q)$ is not of gradient form for $\eta \neq 0$, so the effective generator $\mathcal{L}_O$ is genuinely nonequilibrium.

Non-self-adjointness on the slow space. In $L^2(\mu_q)$, integration by parts gives

$$(F \cdot \nabla_q)^\dagger = -F \cdot \nabla_q - (\nabla \cdot F) + \beta \, \nabla_q U \cdot F. \tag{86}$$

Hence $\mathcal{L}_O$ is generically non-self-adjoint as soon as $\eta \neq 0$. For the concrete force used in the numerics below,

$$F(q) = (-\tanh q_2, \tanh q_1), \tag{87}$$

one has $\nabla \cdot F = 0$ but $\nabla_q U \cdot F \not\equiv 0$ for any non-radial U, so $\mathcal{L}_O$ is explicitly non-self-adjoint for every $\eta \neq 0$.

Verification of Assumptions 4.1 and 4.2. The following proposition collects the verification of all structural hypotheses, with structural constants that are uniform in m and depend only on $(\beta, d, \|F\|_{L^\infty})$.

Proposition 6.2 (Verification for the convex-potential family)**.** *For the family* (81) *with* $U \in C^\infty(\mathbb{R}^d)$ *satisfying* $\nabla^2 U \geq m > 0$ *and with all derivatives of positive order growing at most polynomially, and bounded non-gradient force* F*, Assumptions 4.1, 4.2, and 4.3 hold in* $L^2(\mu)$*. Moreover, one may choose structural constants satisfying*

$$\lambda_R = 1, \qquad M_R = 1, \qquad \kappa_{\mathrm{ell}} = \sqrt{2}, \tag{88}$$

and the perturbation constants (a_1, b_1, a_2, b_2) *depend only on* $(\beta, d, \|F\|_{L^\infty})$ *and are independent of* m*. The extra polynomial-growth assumptions on the derivatives of* U *enter only in the qualitative verification of Assumption 4.3; they do not affect these quantitative constants.*

Corollary 6.3 (Rigorous kinetic scaling for the family)**.** *For the convex-potential Langevin family above, the hypotheses of Corollary 5.2 are satisfied uniformly as* $m \downarrow 0$*. Consequently,*

$$\gamma_{\mathrm{opt}} \asymp \sqrt{m}, \qquad \max_{\gamma > 0} \Lambda_{\mathrm{hypo}}(\gamma) \asymp \sqrt{m}, \qquad \eta^*(\gamma_{\mathrm{opt}}) \asymp \sqrt{m}. \tag{89}$$

Thus this family gives a fully rigorous classical nonequilibrium verifier of the $\sqrt{\lambda(Q_O)}$ *acceleration mechanism within the present theorem.*

The proof of Proposition 6.2 is given in Appendix E. Thus Example 6.2 satisfies the hypotheses of Theorem 5.1 and Corollary 5.2. We do not verify Assumption 5.5(iv) for $\eta \neq 0$ in the present $L^2(\mu)$ norm, so Theorem 5.6 remains conditional here away from equilibrium.

Numerical evidence on the truncated model. We verify the predicted scaling on the quadratic instance $U(q) = \frac{m}{2}|q|^2$ in $d = 2$, with $F(q) = (-\tanh q_2, \tanh q_1)$ and $\eta = c\sqrt{m}$, $c \in \{0, \frac{1}{2}, 1\}$. The generator is discretized in a tensor-product Hermite–Galerkin basis of dimension 256. These computations are illustrative and are not used in the proofs.

Figure 1 shows the three-regime structure and the predicted $\sqrt{\lambda(Q_O)}$ scaling: the observed log–log slopes on the truncated matrices are 0.51, 0.53, and 0.52 for $c = 0$, $\frac{1}{2}$, 1. At $m = 0.08$, $c = \frac{1}{2}$, the truncated lifted gap $\lambda(\mathcal{L}) \approx 0.157$ gives about a 1.9× speedup over $\lambda(\mathcal{L}_O) \approx 0.080$, with mixing times $t_{\mathrm{mix}} \approx 60$ versus 115.

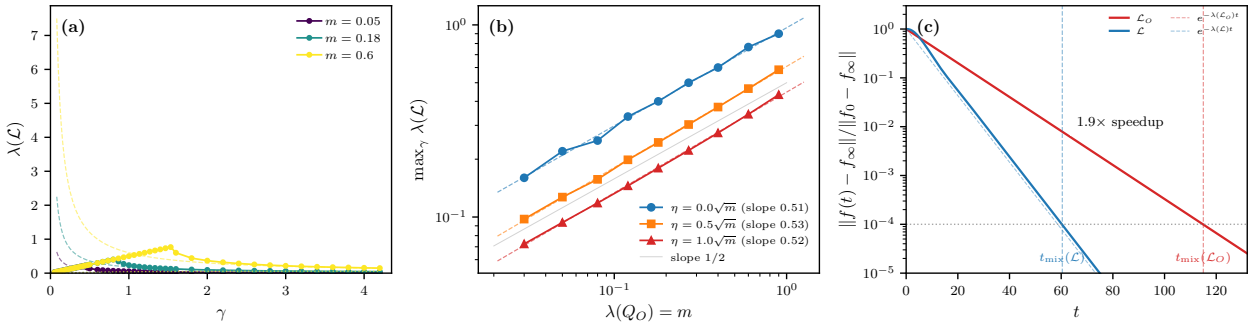

FIGURE 1. Numerical experiment on the truncated Hermite–Galerkin model for Example 6.2 ($\beta = 1$, $m \in [0.03, 0.90]$, 45-point γ-grid). **(a)** Spectral gap versus γ at $\eta = 0$; dashed: overdamped envelope m/γ; dotted: gap-adapted DMS lower bound. **(b)** Peak gap versus m on log–log scale for $c \in \{0, \frac{1}{2}, 1\}$; fitted slopes ≈ 0.51–0.53. **(c)** Semigroup decay at $m = 0.08$, $c = \frac{1}{2}$, $\gamma^* = 0.5$; vertical markers: ε-mixing times ($\varepsilon = 10^{-4}$).

Algorithmic lift via generic square-root factorization. Here we use exactly the same three-step construction as in Example 6.3, now on the truncated slow Hermite space underlying Figure 1.

Step 1 (compute slow operators). On the 16-dimensional slow space we assemble the Galerkin matrices Q_O and $\mathcal{L}_O$ from the physical Langevin model.

Step 2 (build a generic lifted fast block). Set $\mathcal{H}_{F,\mathrm{alg}} = \mathcal{H}_S$, $\mathcal{R}_{F,\mathrm{alg}} = -\lambda_R \,\mathrm{id}$ with $\lambda_R = 1$, and $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. Then $T^*\mathcal{R}_{F,\mathrm{alg}}^{-1}T = Q_O$, so this is the universal square-root factorization of the truncated slow operator. It yields a 32×32 algorithmic lift, compared with the 256×256 physical lift.

Step 3 (match the non-equilibrium part). For Langevin, the perturbation satisfies $\mathcal{V}_{\mathrm{pert}}P_S = 0$, so we impose the same triangular structure in the designed lift by choosing $P_F\mathcal{V}_{\mathrm{pert}}P_S = 0$ and $P_F\mathcal{V}_{\mathrm{pert}}P_F = 0$. On the truncated mean-zero slow space, the consistency relation (71) reduces exactly to

$$\mathcal{L}_O - Q_O = \frac{\eta}{\lambda_R}\,(P_S\mathcal{V}_{\mathrm{pert}}P_F)\,T. \tag{90}$$

Since $\mathcal{L}_O - Q_O$ annihilates constants, one exact solution on that truncated space is

$$P_S\mathcal{V}_{\mathrm{pert}}P_F = \frac{\lambda_R}{\eta}(\mathcal{L}_O - Q_O)T^+, \tag{91}$$

with T^+ the Moore–Penrose pseudoinverse of the truncated operator T. On the matched grid used below the relative residual stays below 2×10^{-16}, which is a machine-precision check of the finite-dimensional construction rather than a new analytic input. Because $P_F\mathcal{V}_{\mathrm{pert}}P_F = 0$, Proposition 6.1 applies to the designed lift.

Figure 2 and Table 1 compare the physical (256×256) and algorithmic (32×32) lifts on the matched grid $m \in \{0.03, 0.05, 0.08, 0.12\}$, $c \in \{\frac{1}{4}, \frac{1}{2}\}$. The two constructions track each other closely: fitted log–log slopes are 0.50–0.51 for both. At $m = 0.08$, $c = \frac{1}{4}$, the mixing times ($\varepsilon = 10^{-4}$) are $t_{\mathrm{mix}} \approx 47.5$ (physical), 47.6 (algorithmic), and 115.2 (overdamped), confirming that the algorithmic lift reproduces the kinetic speedup while compressing the fast space from dimension 240 to 16.

6.3. **Example 2: A Boundary-Driven Two-Qubit Model.** We now study a two-qubit boundary-driven model. This is the finite-dimensional example: the reduced slow dynamics is already nonequilibrium for $\eta \neq 0$, the reverse-adiabatic-elimination algebra is explicit, and all operators are bounded because $\dim \mathcal{H} = 16$.

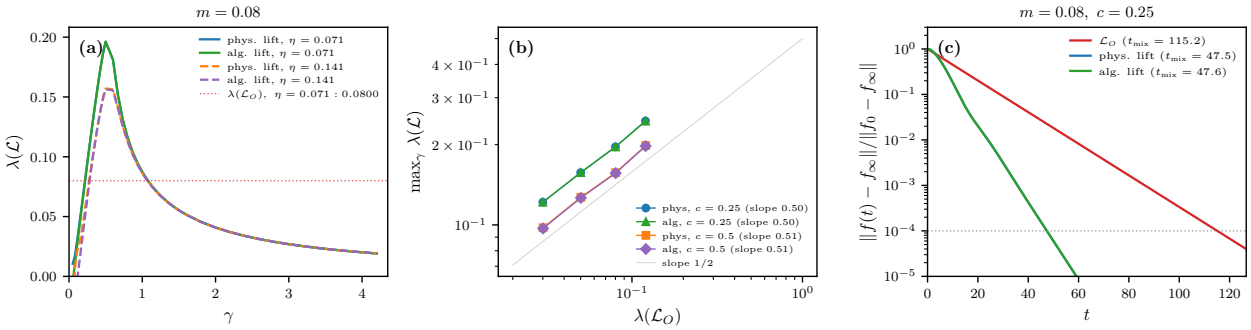

FIGURE 2. Comparison of the *physical* (256×256) and *algorithmic* (32×32, square-root factorization) lifted generators for the truncated model from Example 6.2. **(a)** Spectral gap versus γ at $m = 0.08$ for both lifts with $c = \frac{1}{4}$ and $c = \frac{1}{2}$ (that is, $\eta = c\sqrt{m}$); the overdamped reference is $\lambda(\mathcal{L}_O)$. **(b)** Log–log plot of peak spectral gap versus $\lambda(\mathcal{L}_O)$ on the matched grid $m \in \{0.03, 0.05, 0.08, 0.12\}$ for both lifts. **(c)** Convergence comparison at $m = 0.08$, $c = \frac{1}{4}$, $\varepsilon = 10^{-4}$: algorithmic lift, physical lift, and overdamped $\mathcal{L}_O$.

TABLE 1. Numerically computed peak spectral gaps $\max_\gamma \lambda(\mathcal{L})$ for the physical and algorithmic lifts on the matched grid. The final column reports the physical-lift speedup $\max_\gamma \lambda(\mathcal{L}_{\rm phys})/\lambda(\mathcal{L}_O)$.

m	c	$\lambda(\mathcal{L}_O)$	phys. peak	alg. peak	phys. speedup
0.03	$\frac{1}{4}$	0.03001	0.12175	0.12163	4.06×
0.03	$\frac{1}{2}$	0.03010	0.09757	0.09701	3.24×
0.05	$\frac{1}{4}$	0.05001	0.15718	0.15703	3.14×
0.05	$\frac{1}{2}$	0.05016	0.12696	0.12629	2.53×
0.08	$\frac{1}{4}$	0.08002	0.19628	0.19613	2.45×
0.08	$\frac{1}{2}$	0.08024	0.15707	0.15640	1.96×
0.12	$\frac{1}{4}$	0.12002	0.24546	0.24519	2.05×
0.12	$\frac{1}{2}$	0.12032	0.19855	0.19765	1.65×

Setup. Both fast and slow subsystems are qubits: $\mathfrak{h}_F \cong \mathfrak{h}_S \cong \mathbb{C}^2$. We work in $\mathcal{H} = L^2(\sigma)$ with $\sigma = \frac{1}{2}\mathbf{1}_F \otimes \frac{1}{2}\mathbf{1}_S$. The slow subspace is $\mathcal{H}_S = \mathbf{1}_F \otimes \mathcal{B}(\mathfrak{h}_S)$, with projection $P_S(X) = \mathbf{1}_F \otimes \mathrm{tr}_F(\frac{1}{2}\mathbf{1}_F \cdot X)$.

Generator decomposition. The decomposition is $\mathcal{L} = \mathcal{V}_{\rm skew} + \gamma\mathcal{R} + \eta\mathcal{V}_{\rm pert}$. Here $\mathcal{R} = \mathcal{R}_F \otimes \mathrm{id}_S$ with $\mathcal{R}_F X = \Gamma \sum_{j\in\{x,y,z\}} \mathcal{D}^\dagger[\sigma_j]X$, so $\lambda_R = 4\Gamma$. The transport part uses a *two-channel* coupling,

$$\mathcal{V}_{\rm skew} X = -i[g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z), X], \tag{92}$$

and the perturbation is $\mathcal{V}_{\rm pert} = \sigma_z \otimes \mathcal{G}_S$, $\mathcal{G}_S(X) = \mathcal{D}^\dagger[\sigma_-]X$. Because $\mathrm{tr}(\frac{1}{2}\mathbf{1}_F \cdot \sigma_z) = 0$, both terms map $\mathcal{H}_S$ into $\mathcal{H}_F$, so centering is satisfied.

Effective generator $\mathcal{L}_O$ (explicit formula). Let $X = \mathbf{1}_F \otimes X_S \in \mathcal{H}_S$. Because all operators are bounded in this 16×16 model, the three-step computation below is an exact evaluation of Definition 3.4, not a formal heuristic.

Step 1 (first excursion to fast space). The two-channel coupling gives

$$(\mathcal{V}_{\rm skew} + \eta\mathcal{V}_{\rm pert})X = \sum_{a\in\{x,z\}} \sigma_a \otimes \Big(-ig\,[\sigma_a, X_S]\Big) + \sigma_z \otimes \eta\,\mathcal{G}_S(X_S). \tag{93}$$

Step 2 (Poisson equation on the fast ancilla). Since $\mathcal{R}_F(\sigma_a) = -\kappa\,\sigma_a$ with $\kappa := 4\Gamma$ for each $a \in \{x, y, z\}$, we have $\mathcal{R}^+(\sigma_a \otimes Y) = -(1/\kappa)\sigma_a \otimes Y$. Therefore

$$\mathcal{R}^+(\mathcal{V}_{\text{skew}} + \eta\mathcal{V}_{\text{pert}})X = -\frac{1}{\kappa}\sum_{a\in\{x,z\}} \sigma_a \otimes \Big(- ig\,[\sigma_a, X_S]\Big) - \frac{1}{\kappa}\sigma_z \otimes \eta\,\mathcal{G}_S(X_S). \tag{94}$$

Step 3 (return map and projection). Using $\mathcal{V}_{\text{skew}}(\sigma_a \otimes Y) = -ig\sum_{b\in\{x,z\}} \sigma_a\sigma_b \otimes [\sigma_b, Y]$, the cross-channel products $\sigma_x\sigma_z = -i\sigma_y$ and $\sigma_z\sigma_x = i\sigma_y$ produce σ_y factors, which vanish under P_S (since $\text{tr}(\sigma_y) = 0$). Only same-channel terms survive. Projecting gives

$$\begin{aligned}\mathcal{L}_O(\mathbf{1}_F \otimes X_S) = -\frac{1}{\kappa}\,\mathbf{1}_F \otimes \Big(g^2 &\sum_{a\in\{x,z\}} [\sigma_a, [\sigma_a, X_S]] \\ &+ i\eta g\,\big[\sigma_z, \mathcal{G}_S(X_S)\big] \\ &+ i\eta g\,\mathcal{G}_S([\sigma_z, X_S]) \\ &- \eta^2\,\mathcal{G}_S^2(X_S)\Big).\end{aligned} \tag{95}$$

The reference operator is the $\eta = 0$ part,

$$Q_O(\mathbf{1}_F \otimes X_S) = -\frac{g^2}{\kappa}\,\mathbf{1}_F \otimes \sum_{a\in\{x,z\}} [\sigma_a, [\sigma_a, X_S]]. \tag{96}$$

Non-self-adjointness on the slow space. Since the slow marginal of σ is proportional to the identity, the $L^2(\sigma)$ inner product on $\mathcal{H}_S$ reduces to the Hilbert–Schmidt inner product, and one checks that $(\mathcal{D}^\dagger[\sigma_-])^\dagger = \mathcal{D}[\sigma_-] \neq \mathcal{D}^\dagger[\sigma_-]$. Thus $\mathcal{G}_S$ is not self-adjoint on $\mathcal{H}_S$. Consequently, for the chosen two-channel coupling and every $\eta \neq 0$, the $\mathcal{O}(\eta)$ cross terms in $\mathcal{L}_O - Q_O$ involve the mixed operators $[\sigma_z, \mathcal{G}_S(\cdot)] + \mathcal{G}_S([\sigma_z, \cdot])$, which are not self-adjoint, so $\mathcal{L}_O$ is non-self-adjoint on $\mathcal{H}_S$. In particular, the effective slow dynamics is already a NESS-type evolution before any finite-friction acceleration is invoked; we are not merely lifting an equilibrium reduced model.

This can be seen explicitly on the Pauli basis. Using

$$\mathcal{G}_S(\mathbf{1}) = 0, \qquad \mathcal{G}_S(\sigma_x) = -\tfrac{1}{2}\sigma_x, \qquad \mathcal{G}_S(\sigma_y) = -\tfrac{1}{2}\sigma_y, \qquad \mathcal{G}_S(\sigma_z) = -(\mathbf{1} + \sigma_z), \tag{97}$$

the effective generator (95) acts as

$$\begin{aligned}\mathcal{L}_O(\mathbf{1}) &= 0, \\ \mathcal{L}_O(\sigma_x) &= -\frac{4g^2 - \eta^2/4}{\kappa}\,\sigma_x - \frac{2\eta g}{\kappa}\,\sigma_y, \\ \mathcal{L}_O(\sigma_y) &= \frac{2\eta g}{\kappa}\,\sigma_x - \frac{8g^2 - \eta^2/4}{\kappa}\,\sigma_y, \\ \mathcal{L}_O(\sigma_z) &= \frac{\eta^2}{\kappa}\,\mathbf{1} - \frac{4g^2 - \eta^2}{\kappa}\,\sigma_z.\end{aligned} \tag{98}$$

The σ_x–σ_y coupling and the $\sigma_z \to \mathbf{1}$ drift vanish when $\eta = 0$; for $\eta \neq 0$ they provide an explicit algebraic witness that the reduced dynamics itself is already away from detailed balance.

Verification of Assumptions 4.1 and 4.2. The following proposition collects the structural verification; all operators are bounded on the finite-dimensional space $\mathcal{H}$, so the analysis reduces to linear algebra.

Proposition 6.4 (Verification for the boundary-driven model)**.** *For the boundary-driven two-qubit family above with the two-channel coupling $H_{\text{int}} = g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z)$, Assumptions 4.1 and 4.2 hold. Moreover,*

$$\lambda_R = 4\Gamma, \qquad M_R = \|\mathcal{R}\|, \qquad \ker(Q_O) = \mathbb{C}\mathbf{1}, \qquad \lambda(Q_O) = g^2/\Gamma, \tag{99}$$

and the perturbation constants (a_1, b_1, a_2, b_2) *satisfy* $a_1 = a_2 = 0$, $b_j = \|\mathcal{K}_{\text{pert}}\|$ *or* $\|\mathcal{S}_{\text{pert}}\|$.

The proof is given in Appendix F. Thus the boundary-driven toy satisfies the hypotheses of Theorem 5.1; the ceiling theorem remains conditional here because the extra contraction hypothesis is not automatic for $\eta \neq 0$ in the Hilbert–Schmidt/KMS norm used above.

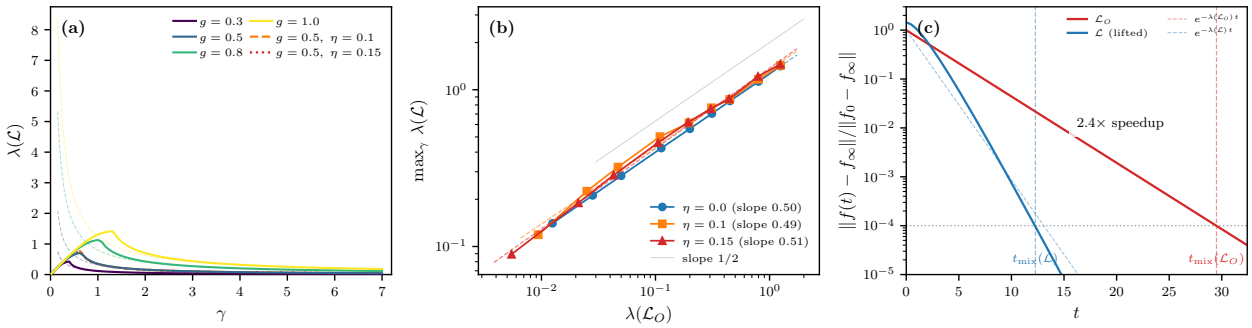

FIGURE 3. Numerical verification for Example 6.3. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ at $\eta = 0$ for several values of g; dashed curves show the overdamped envelope $\lambda(\mathcal{L}_O)/\gamma$. **(b)** Log–log plot of $\max_\gamma \lambda(\mathcal{L})$ versus $\lambda(\mathcal{L}_O)$. In this finite-dimensional setting the fitted slopes track the half-law closely, so the panel serves as a clean numerical benchmark for the mechanism. **(c)** Decay of the projected slow component for the lifted dynamics at γ^* and for $\mathcal{L}_O$ ($g = 0.5$, $\eta = 0$); dashed lines show the exponential envelopes. Vertical markers indicate the ε-level crossing times ($\varepsilon = 10^{-4}$). Parameters: $\Gamma = 0.8$, $\kappa = 3.2$, $g \in \{0.1, \ldots, 1.0\}$, 53-point γ-grid over $[0.02, 7.0]$.

TABLE 2. Kernel compatibility and spectral gap for Q_O with the two-channel coupling $H_{\text{int}} = g(\sigma_x \otimes \sigma_x + \sigma_z \otimes \sigma_z)$. For all $g > 0$: $\dim\ker(Q_O) = 1$ ($= \mathbb{C}\mathbf{1}$) and $\lambda(Q_O) = g^2/\Gamma$.

g	$\dim\ker(Q_O)$	$\lambda(Q_O)$
0.3	1	0.1125
0.5	1	0.3125
1.0	1	1.2500

Numerical evidence on the exact finite-dimensional model. We assemble the exact 16×16 generator and scan spectral gaps over $g \in \{0.1, \ldots, 1.0\}$, $\Gamma = 0.8$, $\gamma \in [0.02, 7.0]$, and $\eta \in \{0, 0.10, 0.15\}$. Figure 3 again shows the three-regime structure and the $\sqrt{\lambda(Q_O)}$ scaling: the fitted log–log slopes are 0.50, 0.49, 0.51 for $\eta = 0$, 0.10, 0.15. At $g = 0.5$, $\eta = 0$, the lifted gap yields about a 2.4× speedup ($t_{\text{mix}} \approx 12.3$ versus 29.5, $\varepsilon = 10^{-4}$).

Algorithmic lift via generic square-root factorization. Here we use exactly the same three-step construction as above, now in the boundary-driven toy model.

Step 1 (compute slow operators). Compute $\mathcal{L}_O$ and Q_O from the physical model.

Step 2 (build a generic lifted fast block). Set $\mathcal{H}_{F,\text{alg}} = \mathcal{H}_S$, $\mathcal{R}_{F,\text{alg}} = -\lambda_R\,\text{id}$, and $T = \sqrt{\lambda_R}(-Q_O)^{1/2}$. Then $T^*\mathcal{R}_{F,\text{alg}}^{-1}T = Q_O$, so this again matches the universal square-root factorization. The resulting lifted generator is 8×8 (the physical model is 16×16).

Step 3 (match the non-equilibrium part). Solve the consistency relation (71) numerically for the perturbation block on the finite slow space. This linear algebra problem has multiple solutions; we

choose the one with smallest matrix norm. The final residual is below 10^{-13}, which serves only as a numerical diagnostic for the designed lift.

Figure 4 and Table 3 compare the physical (16×16) and algorithmic (8×8) lifts on the matched grid $g \in \{0.3, 0.5, 0.8, 1.0\}$, $\eta \in \{0.10, 0.15\}$. The two constructions track each other closely in both peak gaps and convergence curves; stationary states agree to machine precision.

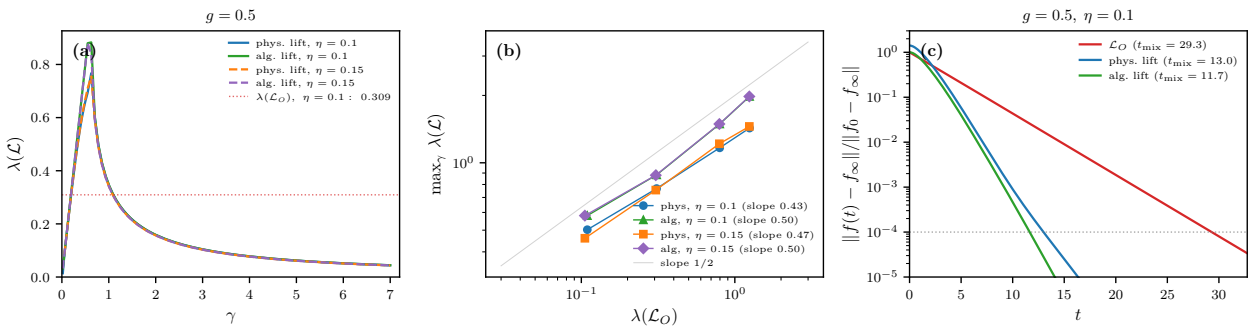

FIGURE 4. Comparison of the *physical* (16×16) and *algorithmic* (8×8, square-root factorization) lifted generators for the finite-dimensional model of Example 6.3. **(a)** Spectral gap $\lambda(\mathcal{L})$ vs. γ at $g = 0.5$ for both lifts ($\eta = 0.10$ and $\eta = 0.15$); the overdamped reference is $\lambda(\mathcal{L}_O)$. **(b)** Log–log plot of peak spectral gap vs. $\lambda(\mathcal{L}_O)$ for both lifts. **(c)** Convergence comparison ($g = 0.5$, $\eta = 0.10$, $\varepsilon = 10^{-4}$): algorithmic lift, physical lift, and overdamped $\mathcal{L}_O$.

TABLE 3. Numerically computed peak spectral gaps $\max_\gamma \lambda(\mathcal{L})$ for the physical and algorithmic lifts (53-point γ-grid). The "speedup" column is $\max_\gamma \lambda(\mathcal{L}_{\rm phys})/\lambda(\mathcal{L}_O)$.

g	η	$\lambda(\mathcal{L}_O)$	phys. peak	alg. peak	phys. speedup
0.3	0.10	0.1094	0.5024	0.5806	4.59×
0.3	0.15	0.1055	0.4600	0.5806	4.36×
0.5	0.10	0.3094	0.7673	0.8831	2.48×
0.5	0.15	0.3055	0.7552	0.8800	2.47×
0.8	0.10	0.7969	1.1663	1.4906	1.46×
0.8	0.15	0.7930	1.2163	1.4906	1.53×
1.0	0.10	1.2469	1.4291	1.9789	1.15×
1.0	0.15	1.2430	1.4534	1.9789	1.17×

For the *algorithmic* lifts in both examples, $P_F \mathcal{V}_{\rm pert} P_F = 0$ by construction, so Proposition 6.1 predicts zero projection bias. Numerically, $\|P_S h_\eta^{\rm alg} - g_\eta\|/\|g_\eta\| \lesssim 10^{-16}$ at all tested parameter values; the physical lifts give the same result on the adopted truncations.

7. Conclusion and open problems

We have established explicit model-wise bounds for the interplay between friction, reversible backbone gap, and non-equilibrium forcing in lifted generators, recovering the kinetic $\sqrt{\lambda(Q_O)}$ scaling for uniformly controlled families. We close with several open directions.

(1) Sharp stability thresholds and non-convex potentials. The threshold η^* involves structural constants that may not be sharp; numerically, acceleration persists well beyond η^*. The present convex-potential Langevin family (Section 6.2) establishes family-uniform

structural constants for all U with $\nabla^2 U \geq m > 0$, but extending the result to non-convex potentials (where the Bochner inequality fails) remains open. For models where forcing enhances the effective gap (e.g. optimally rotated diffusions [23]), one expects $\lambda(\mathcal{L}_O) > \lambda(Q_O)$, and a refined analysis could yield

$$\max_{\gamma>0} \lambda(\mathcal{L}_\gamma) \asymp \sqrt{\lambda(\mathcal{L}_O)} \qquad \text{(conjectured, when } \lambda(\mathcal{L}_O) > \lambda(Q_O)\text{)}. \tag{100}$$

(2) Optimal factorization. The reverse adiabatic elimination recipe lifts any target Q_O, but the structural constants depend on the auxiliary space $(\mathcal{H}_F, \mathcal{R}_F, T)$. Characterizing the factorization that maximises η^* in infinite-dimensional settings, and extending the framework to multilevel decompositions $\mathcal{H}_{S_1} \subset \mathcal{H}_{S_2} \subset \cdots$, remain open.

(3) Beyond the perturbative regime. Extending the analysis to large η may require space-time divergence methods [16, 17] or probabilistic contractivity arguments [33]. A related problem is to formulate the overdamped ceiling directly in a stationary-state-adapted norm, making the contraction hypothesis automatic.

(4) Quantum many-body NESS. The two-qubit toy model of Section 6.3 shows that the framework applies to boundary-driven Lindbladians [19, 32] when suitable noncommutative Poincaré inequalities hold; compare [30]. For many-body systems (e.g., boundary-driven XXZ chains), scaling the structural constants with system size and establishing meaningful $\lambda(Q_O)$ lower bounds remain open.

References

[1] Franz Achleitner, Anton Arnold, and Eric A. Carlen, *On multi-dimensional hypocoercive BGK models*, Kinet. Relat. Models **11** (2018), no. 4, 953–1009.

[2] Dallas Albritton, Scott Armstrong, Jean-Christophe Mourrat, and Matthew Novack, *Variational methods for the kinetic Fokker-Planck equation*, Anal. PDE **17** (2024), no. 6, 1953–2010.

[3] Simon Apers, Francesco Ticozzi, and Alain Sarlette, *Lifting Markov chains to mix faster: limits and opportunities*, arXiv:1705.08253, 2017.

[4] Anton Arnold, Irene M. Gamba, Maria Pia Gualdani, Stéphane Mischler, Clément Mouhot, and Christof Sparber, *The Wigner–Fokker–Planck equation: stationary states and large-time behavior*, arXiv:1010.2791, 2012.

[5] Dominique Bakry and Michel Émery, *Diffusions hypercontractives*, Séminaire de probabilités, XIX, 1983/84, Lecture Notes in Math., vol. 1123, Springer, Berlin, 1985, pp. 177–206.

[6] Giovanni Brigati, Francis Lörler, and Lihan Wang, *Hypocoercivity meets lifts*, arXiv:2412.10890, 2025.

[7] Yu Cao, Jianfeng Lu, and Lihan Wang, *On explicit L^2-convergence rate estimate for underdamped Langevin dynamics*, Arch. Ration. Mech. Anal. **247** (2023), no. 5, 90.

[8] Eric A. Carlen, David A. Huse, and Joel L. Lebowitz, *Boundary-driven quantum systems near the Zeno limit: steady states and long-time behavior*, arXiv:2512.12825, 2025.

[9] Eric A. Carlen, David A. Huse, and Joel L. Lebowitz, *Stationary states of boundary driven quantum systems: some exact results*, Phys. Rev. A **111** (2025), no. 1, 012210.

[10] Sourav Chatterjee, *Spectral gap of nonreversible Markov chains*, arXiv:2310.10876, 2023.

[11] Fang Chen, László Lovász, and Igor Pak, *Lifting Markov chains to speed up mixing*, Proceedings of the Thirty-First Annual ACM Symposium on Theory of Computing (New York), STOC '99, ACM, 1999, pp. 275–281.

[12] Helge Dietert, *L^2-hypocoercivity for non-equilibrium kinetic equations*, arXiv:2310.13456, 2023.

[13] Jean Dolbeault, Clément Mouhot, and Christian Schmeiser, *Hypocoercivity for kinetic equations with linear relaxation terms*, arXiv:0810.3493, 2009.

[14] Jean Dolbeault, Clément Mouhot, and Christian Schmeiser, *Hypocoercivity for linear kinetic equations conserving mass*, Trans. Amer. Math. Soc. **367** (2015), no. 6, 3807–3828.

[15] Andreas Eberle, Arnaud Guillin, Leo Hahn, Francis Lörler, and Manon Michel, *Convergence of non-reversible Markov processes via lifting and flow Poincaré inequality*, arXiv:2503.04238, 2025.

[16] Andreas Eberle and Francis Lörler, *Non-reversible lifts of reversible diffusion processes and relaxation times*, arXiv:2402.05041, 2024.

[17] Andreas Eberle and Francis Lörler, *Space-time divergence lemmas and optimal non-reversible lifts of diffusions on Riemannian manifolds with boundary*, arXiv:2412.16710, 2024.

[18] Di Fang, Jianfeng Lu, and Yu Tong, *Mixing time of open quantum systems via hypocoercivity*, Phys. Rev. Lett. **134** (2025), 140405.

[19] Vittorio Gorini, Andrzej Kossakowski, and E. C. G. Sudarshan, *Completely positive dynamical semigroups of N-level systems*, J. Math. Phys. **17** (1976), no. 5, 821–825. MR 0413882
[20] Martin Grothaus and Patrik Stilgenbauer, *Hypocoercivity for Kolmogorov backward evolution equations and applications*, J. Funct. Anal. **267** (2014), no. 10, 3515–3556.
[21] Bernard Helffer and Francis Nier, *Hypoelliptic estimates and spectral theory for Fokker–Planck operators and Witten Laplacians*, Lecture Notes in Mathematics, vol. 1862, Springer, 2005.
[22] Frédéric Hérau, *Short and long time behavior of the Fokker–Planck equation in a confining potential and applications*, J. Funct. Anal. **244** (2007), no. 1, 95–118.
[23] Chii-Ruey Hwang, Sheng-Yao Hwang-Ma, and Shuenn-Jyi Sheu, *Accelerating diffusions*, Ann. Appl. Probab. **15** (2005), no. 2, 1433–1444.
[24] Alessandra Iacobucci, Stefano Olla, and Gabriel Stoltz, *Convergence rates for nonequilibrium Langevin dynamics*, arXiv:1702.03685, 2017.
[25] Marcelo Janovitch, Matteo Brunelli, and Patrick P. Potts, *Active quantum reservoir engineering: using a qubit to manipulate its environment*, arXiv:2505.16898, 2025.
[26] Tosio Kato, *Perturbation theory for linear operators*, reprint of the 1980 edition ed., Classics in Mathematics, Springer-Verlag, Berlin, 1995.
[27] E. M. Kessler, *Generalized Schrieffer-Wolff formalism for dissipative systems*, Phys. Rev. A **86** (2012), no. 1, 012126.
[28] Gabriel T. Landi, Dario Poletti, and Gernot Schaller, *Nonequilibrium boundary-driven quantum systems: Models, methods, and properties*, Rev. Mod. Phys. **94** (2022), 045006.
[29] Tony Lelièvre, Francis Nier, and Grigorios A. Pavliotis, *Optimal non-reversible linear drift for the convergence to equilibrium of a diffusion*, J. Stat. Phys. **152** (2013), no. 2, 237–274.
[30] Bowen Li and Jianfeng Lu, *Quantum space-time Poincaré inequality for Lindblad dynamics*, arXiv:2406.09115, 2024.
[31] Bowen Li and Jianfeng Lu, *Speeding up quantum Markov processes through lifting*, arXiv:2505.12187, 2025.
[32] G. Lindblad, *On the generators of quantum dynamical semigroups*, Comm. Math. Phys. **48** (1976), no. 2, 119–130. MR 0413878
[33] Pierre Monmarché, *L^2 geometric ergodicity for the kinetic Langevin process with non-equilibrium steady states*, arXiv:2501.18004, 2025.
[34] Pierre Monmarché and Mouad Ramil, *Overdamped limit at stationarity for non-equilibrium Langevin diffusions*, arXiv:2110.01238, 2022.
[35] Daniel Paulin, *Concentration inequalities for Markov chains by Marton couplings and spectral methods*, Electron. J. Probab. **20** (2015), 1–32.
[36] Michael Reed and Barry Simon, *Methods of modern mathematical physics, vol. II: Fourier analysis, self-adjointness*, Academic Press, New York, 1975.
[37] Julien Roussel and Gabriel Stoltz, *Spectral methods for Langevin dynamics and associated error estimates*, ESAIM Math. Model. Numer. Anal. **52** (2018), no. 3, 1051–1083.
[38] Mario Szegedy, *Quantum speed-up of Markov chain based algorithms*, 45th Annual IEEE Symposium on Foundations of Computer Science, IEEE, 2004, pp. 32–41.
[39] Masaaki Tokieda and Angela Riva, *Time-convolutionless master equation applied to adiabatic elimination*, Phys. Rev. A **111** (2025), no. 5, 052206.
[40] Cédric Villani, *Hypocoercivity*, Mem. Amer. Math. Soc., vol. 202, American Mathematical Society, Providence, RI, 2009. MR 2562709

Appendix A. Proofs for Preliminaries

This appendix collects the proofs from Section 3, carried out at Hilbert-space level with $\mathcal{H} = \mathbb{C}\mathbf{1} \oplus \mathcal{H}_0$.

A.1. **Proof of Lemma 3.2.** We first clarify the geometric relationship between the fixed point space $\mathcal{F}(\mathcal{L}) = \mathbb{C}\mathbf{1}$ and the kernel of the symmetric part $\mathcal{S}$.

Proposition A.1 (Kernel Inclusion)**.** *For any dissipative generator with symmetric part $\mathcal{S} := \frac{1}{2}(\mathcal{L} + \mathcal{L}^\dagger) \leq 0$, one has*

$$\ker(\mathcal{L}) \subseteq \ker(\mathcal{S}). \tag{101}$$

Proof. Let $X \in \ker(\mathcal{L})$. Then $\mathcal{L}(X) = 0$. Consequently,

$$\langle X, \mathcal{S}(X)\rangle = \Re\langle X, \mathcal{L}(X)\rangle = 0. \tag{102}$$

Since $\mathcal{S}$ is self-adjoint and non-positive, the operator $-\mathcal{S} \geq 0$ admits a square root. The identity above is equivalent to

$$\|(-\mathcal{S})^{1/2}X\|^2 = \langle X, (-\mathcal{S})X\rangle = 0.$$

Hence $(-\mathcal{S})^{1/2}X = 0$, and therefore $\mathcal{S}X = 0$. Thus $X \in \ker(\mathcal{S})$. □

Proof of Lemma 3.2. We prove the coercivity characterization and show that, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$, then no exponential decay estimate can hold with prefactor $C = 1$.

(⇐): Assume $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$ and $-\mathcal{S}|_{\mathcal{H}_0}$ has a positive spectral gap $\lambda_S > 0$.

$$\langle X, \mathcal{S}X\rangle \leq -\lambda_S \|X\|^2, \qquad \forall X \in \mathcal{H}_0. \tag{103}$$

For any $X_0 \in \mathrm{dom}(\mathcal{L}) \cap \mathcal{H}_0$, set $X_t = \mathcal{P}_t X_0$. Then $t \mapsto X_t$ is differentiable in $\mathcal{H}_0$, and the evolution of the norm satisfies:

$$\frac{1}{2}\frac{d}{dt}\|X_t\|^2 = \Re\langle X_t, \mathcal{L}X_t\rangle = \langle X_t, \mathcal{S}X_t\rangle \leq -\lambda_S \|X_t\|^2. \tag{104}$$

Integrating this differential inequality yields $\|X_t\| \leq e^{-\lambda_S t}\|X_0\|$ for all such X_0. By density of $\mathrm{dom}(\mathcal{L}) \cap \mathcal{H}_0$ in $\mathcal{H}_0$ and continuity of $\mathcal{P}_t$, the same estimate extends to every $X_0 \in \mathcal{H}_0$. This is an exponential decay with prefactor $C = 1$ and rate λ_S. Thus, the system is coercive, not strictly hypocoercive.

(⇒): Assume the system is coercive ($C = 1$). By definition, there exists $\nu > 0$ such that $\|\mathcal{P}_t X\| \leq e^{-\nu t}\|X\|$ for all $X \in \mathcal{H}_0$. For $X \in \mathrm{dom}(\mathcal{L}) \cap \mathcal{H}_0$, divide the squared estimate by $2t$ and let $t \downarrow 0$:

$$\Re\langle X, \mathcal{L}X\rangle \leq -\nu\|X\|^2, \quad \forall X \in \mathrm{dom}(\mathcal{L}) \cap \mathcal{H}_0. \tag{105}$$

Substituting $\Re\langle X, \mathcal{L}X\rangle = \langle X, \mathcal{S}X\rangle$, we obtain:

$$\langle X, \mathcal{S}X\rangle \leq -\nu\|X\|^2. \tag{106}$$

Now, suppose there exists a non-trivial vector $Y \in \ker(\mathcal{S}) \cap \mathcal{H}_0$. Then $\langle Y, \mathcal{S}Y\rangle = 0$. However, the inequality above requires $\langle Y, \mathcal{S}Y\rangle \leq -\nu\|Y\|^2 < 0$, a contradiction. Therefore, $\ker(\mathcal{S}) \cap \mathcal{H}_0 = \{0\}$, which implies $\ker(\mathcal{S}) = \mathbb{C}\mathbf{1} = \ker(\mathcal{L})$.

Finally, if $\ker(\mathcal{S}) \cap \mathcal{H}_0 \neq \{0\}$ then the above argument shows that coercivity ($C = 1$ with some positive rate) is impossible. In that case, any exponential decay estimate (when it holds) must necessarily come with some prefactor $C > 1$. □

Appendix B. Proof of Lemma 4.6

This appendix establishes the structural bounds of Lemma 4.6.

Proof. In the unbounded setting, all identities below are first established for f in the common core $\mathcal{C}$ from Assumption 4.3; the resulting operator bounds then extend to the stated energy spaces by density and continuity. Recall the definition of the reference effective dynamics $Q_O = -P_S\mathcal{V}_{\mathrm{skew}}\mathcal{R}^+\mathcal{V}_{\mathrm{skew}}P_S$. By Assumption 4.1.1, the pseudoinverse is bounded on $\mathcal{H}_F$ with $\|\mathcal{R}^+\| \leq \lambda_R^{-1}$, hence

$$\langle g, -\mathcal{R}^+ g\rangle \leq \frac{1}{\lambda_R}\|g\|^2, \quad \forall g \in \mathcal{H}_F. \tag{107}$$

The complementary (lower) control needed below is supplied by the flux conditioning assumption (24), which applies to fluxes of the form $g = g_\psi = P_F\mathcal{V}_{\mathrm{skew}}P_S\psi$.

B.1. **Proof of Properties 1 and 2 (Boundedness and Regularity). Solver boundedness and forward regularity.** Let $f \in \mathcal{H}$ and define $u := \mathcal{A}_\varpi f \in \mathcal{H}_S$. The defining equation is $(\varpi - Q_O)u = P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f$. Taking the inner product with u and using the skew-adjointness $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$:

$$\varpi\|u\|^2 + \langle u, -Q_O u\rangle = \langle u, P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F f\rangle = -\langle \mathcal{V}_{\text{skew}}u, (-\mathcal{R}^+)P_F f\rangle. \tag{108}$$

Let $v := \mathcal{V}_{\text{skew}}u$. Since $P_S\mathcal{V}_{\text{skew}}P_S = 0$ (Centered Coupling), we strictly have $v \in \mathcal{H}_F$. Using the definition of Q_O and the pseudoinverse bound (107):

$$\langle u, -Q_O u\rangle = \langle v, -\mathcal{R}^+ v\rangle = \|(-\mathcal{R}^+)^{1/2}v\|^2. \tag{109}$$

Substituting this into the energy identity and applying Cauchy–Schwarz yields:

$$\varpi\|u\|^2 + \|(-\mathcal{R}^+)^{1/2}v\|^2 \le \|(-\mathcal{R}^+)^{1/2}v\|\,\|(-\mathcal{R}^+)^{1/2}P_F f\|. \tag{110}$$

Writing $a := \|(-\mathcal{R}^+)^{1/2}v\|$ and $b := \|(-\mathcal{R}^+)^{1/2}P_F f\|$, we have $\varpi\|u\|^2 + a^2 \le ab$. Completing the square,

$$a^2 - ab = \left(a - \tfrac{1}{2}b\right)^2 - \tfrac{1}{4}b^2 \ge -\tfrac{1}{4}b^2, \tag{111}$$

so $\varpi\|u\|^2 \le \frac{1}{4}b^2$, hence

$$\|u\| \le \frac{1}{2}\,\varpi^{-1/2}\,\|(-\mathcal{R}^+)^{1/2}P_F f\| \le \frac{1}{2\sqrt{\lambda_R}}\,\varpi^{-1/2}\,\|f\|. \tag{112}$$

The first inequality is the spectral bound $\varpi^{-1/2}\cdot\frac{1}{2}$, which can equivalently be derived from the spectral calculus: writing $Q_O = -\int_0^\infty \lambda\, dE_\lambda$ on $\mathcal{H}_{S,0}$, one has $\|(\varpi - Q_O)^{-1}(-Q_O)^{1/2}\| = \sup_{\lambda\ge 0}\frac{\sqrt{\lambda}}{\varpi+\lambda}$, and the calculus identity $\sup_{\lambda\ge 0}\sqrt{\lambda}/(\varpi+\lambda) = 1/(2\sqrt{\varpi})$ (attained at $\lambda = \varpi$) gives the stated bound. The second inequality uses $\|(-\mathcal{R}^+)^{1/2}P_F f\| \le \lambda_R^{-1/2}\|f\|$. Moreover, $a \le b$. By construction of the resolvent, $u \in \operatorname{dom}(Q_O)$; in the unbounded setting this is exactly the mapping property required in Assumption 4.3. Then the vector $v = \mathcal{V}_{\text{skew}}u$ is a slow-induced flux of the form g_ψ with $\psi \in \operatorname{dom}(Q_O)$; we may therefore apply the flux conditioning bound (24) to v to obtain

$$\|v\|^2 \le M_R\,\langle v, -\mathcal{R}^+ v\rangle = M_R a^2 \le M_R b^2 \le \frac{M_R}{\lambda_R}\,\|f\|^2, \tag{113}$$

so $\|v\| \le \sqrt{M_R/\lambda_R}\,\|f\|$. This proves $\|\mathcal{A}_\varpi\| \le k_{\mathcal{A}}\varpi^{-1/2}$ and $\|\mathcal{V}_{\text{skew}}\mathcal{A}_\varpi\| \le k_{\text{reg}}$, with

$$k_{\mathcal{A}} := \frac{1}{2\sqrt{\lambda_R}}, \qquad k_{\text{reg}} := \sqrt{M_R/\lambda_R}. \tag{114}$$

Dual regularity. We consider the operator $\mathcal{T} := \mathcal{A}_\varpi\mathcal{V}_{\text{skew}}P_F$. Its norm equals that of its adjoint $\mathcal{T}^\dagger$. We compute the adjoint of the solver:

$$\mathcal{A}_\varpi^\dagger = \left((\varpi - Q_O)^{-1}P_S\mathcal{V}_{\text{skew}}(-\mathcal{R}^+)P_F\right)^\dagger = -P_F(-\mathcal{R}^+)\mathcal{V}_{\text{skew}}P_S(\varpi - Q_O)^{-1}. \tag{115}$$

(Using $Q_O^\dagger = Q_O$ and $\mathcal{V}_{\text{skew}}^\dagger = -\mathcal{V}_{\text{skew}}$). Therefore:

$$\mathcal{T}^\dagger = P_F\mathcal{V}_{\text{skew}}^\dagger\mathcal{A}_\varpi^\dagger = P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)\mathcal{V}_{\text{skew}}P_S(\varpi - Q_O)^{-1}. \tag{116}$$

Let $g \in \mathcal{H}_S$ and $\psi := (\varpi - Q_O)^{-1}g$. Define the macroscopic flux

$$g_\psi := P_F\mathcal{V}_{\text{skew}}P_S\psi \in \mathcal{H}_F. \tag{117}$$

Then $\mathcal{T}^\dagger g = P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi$. Invoking the **Flux Regularity** condition from Assumption 4.1.3:

$$\|P_F\mathcal{V}_{\text{skew}}P_F(-\mathcal{R}^+)g_\psi\| \le \kappa_{\text{ell}}\|(-Q_O)\psi\|. \tag{118}$$

Substituting ψ back:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\rm ell}\|(-Q_O)(\varpi - Q_O)^{-1} g\|. \tag{119}$$

Since Q_O is self-adjoint and non-positive, the operator $(-Q_O)(\varpi - Q_O)^{-1}$ is a spectral contraction: by functional calculus, $\|(-Q_O)(\varpi - Q_O)^{-1}\| = \sup_{\lambda\ge 0} \lambda/(\varpi+\lambda) \le 1$ (the supremum is ≤ 1 for every $\varpi > 0$, with equality in the limit $\lambda \to \infty$). Thus:

$$\|\mathcal{T}^\dagger g\| \le \kappa_{\rm ell}\|g\|. \tag{120}$$

This proves $\|\mathcal{A}_\varpi \mathcal{V}_{\rm skew} P_F\| \le \kappa_{\rm ell}$.

B.2. **Proof of Property 3 (Perturbation Control).** We use the structural decomposition $\mathcal{V}_{\rm pert} = \mathcal{K}_{\rm pert} + \mathcal{S}_{\rm pert}$ from Assumption 4.1.4. We also recall the centered mapping property (38), which will be used later to sharpen the perturbation bookkeeping in the dissipation matrix. Recall the bound for the skew part (Assumption 4.1.4(a)):

$$\|\mathcal{K}_{\rm pert} f\| \le a_1 \|(-\mathcal{R})^{1/2} f\| + b_1 \|f\|, \qquad f \in {\rm dom}((-\mathcal{R})^{1/2}). \tag{121}$$

For the symmetric part, we use the energy bound (Assumption 4.1.4(b)):

$$\|\mathcal{S}_{\rm pert} f\| \le a_2 \|(-\mathcal{R})^{1/2} f\| + b_2 \|f\|, \qquad f \in {\rm dom}((-\mathcal{R})^{1/2}). \tag{122}$$

Reverse estimate ($\mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi$). Let $f \in \mathcal{H}$ and $u := \mathcal{A}_\varpi f$. The adjoint is $\mathcal{V}_{\rm pert}^\dagger = -\mathcal{K}_{\rm pert} + \mathcal{S}_{\rm pert}$. Applying the triangle inequality and the bounds above:

$$\|\mathcal{V}_{\rm pert}^\dagger u\| \le \|\mathcal{K}_{\rm pert} u\| + \|\mathcal{S}_{\rm pert} u\| \tag{123}$$

$$\le b_1 \|u\| + b_2 \|u\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} u\|. \tag{124}$$

Since $u \in \mathcal{H}_S = \ker(\mathcal{R})$ (by definition of $\mathcal{A}_\varpi$), one has $(-\mathcal{R})^{1/2} u = 0$, hence

$$\|\mathcal{V}_{\rm pert}^\dagger u\| \le (b_1 + b_2) \|u\|. \tag{125}$$

Substituting the Solver Bound ($\|u\| \le k_{\mathcal{A}} \varpi^{-1/2} \|f\|$):

$$\|\mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f\| \le (b_1 + b_2) k_{\mathcal{A}} \varpi^{-1/2} \|f\|. \tag{126}$$

This establishes (35a) with $k_P(\varpi) := (b_1 + b_2) k_{\mathcal{A}} \varpi^{-1/2}$.

Forward estimate ($\mathcal{A}_\varpi \mathcal{V}_{\rm pert}$). We aim to bound the operator norm of $\mathcal{W}_{\rm fwd} := \mathcal{A}_\varpi \mathcal{V}_{\rm pert}$. By adjoints on Hilbert spaces, $\|\mathcal{W}_{\rm fwd}\| = \|\mathcal{W}_{\rm fwd}^\dagger\| = \|\mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi^\dagger\|$. Let $g \in \mathcal{H}_S$ be arbitrary with $\|g\| = 1$ and set $w := \mathcal{A}_\varpi^\dagger g$. Using the bounds on $\mathcal{K}_{\rm pert}$ and the energy bound on $\mathcal{S}_{\rm pert}$,

$$\|\mathcal{V}_{\rm pert}^\dagger w\| \le \|\mathcal{K}_{\rm pert} w\| + \|\mathcal{S}_{\rm pert} w\|$$

$$\le (b_1 + b_2) \|w\| + (a_1 + a_2) \|(-\mathcal{R})^{1/2} w\|. \tag{127}$$

We compute the norms on the right-hand side.

(1) **Solver term ($\|w\|$):**

$$\|w\| = \|\mathcal{A}_\varpi^\dagger g\| \le \|\mathcal{A}_\varpi\| \, \|g\| \le k_{\mathcal{A}} \, \varpi^{-1/2}. \tag{128}$$

(2) **Energy term ($\|(-\mathcal{R})^{1/2} w\|$):** Using $w = \mathcal{A}_\varpi^\dagger g$ and the functional calculus for the non-positive self-adjoint operator Q_O,

$$\|(-\mathcal{R})^{1/2} w\| = \|(-\mathcal{R})^{1/2} \mathcal{A}_\varpi^\dagger g\| = \|(-\mathcal{R})^{-1/2} P_F \mathcal{V}_{\rm skew} P_S (\varpi - Q_O)^{-1} g\|. \tag{129}$$

Setting $\psi := (\varpi - Q_O)^{-1} g \in \mathcal{H}_S$, we note the identity (for such slow-induced fluxes)

$$g_\psi := P_F \mathcal{V}_{\rm skew} P_S \psi, \tag{130}$$

$$\|(-\mathcal{R})^{-1/2} g_\psi\|^2 = \langle g_\psi, -\mathcal{R}^+ g_\psi \rangle = -\langle \psi, Q_O \psi \rangle = \|(-Q_O)^{1/2} \psi\|^2, \tag{131}$$

hence

$$\|(-\mathcal{R})^{1/2}w\| = \|(-Q_O)^{1/2}(\varpi - Q_O)^{-1}g\| \le \frac{1}{2}\,\varpi^{-1/2}\,\|g\|. \tag{132}$$

(The last step follows from the scalar bound $\sup_{x\ge 0}\frac{\sqrt{x}}{\varpi+x} = \frac{1}{2\sqrt{\varpi}}$.)

Substituting these estimates into (127) yields

$$\|\mathcal{V}_{\mathrm{pert}}^{\dagger}\mathcal{A}_{\varpi}^{\dagger}g\| \le \Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\,\varpi^{-1/2}\,\|g\|. \tag{133}$$

Since g is arbitrary, this establishes (35b) with $\tilde{k}_P(\varpi) := \big((b_1+b_2)k_{\mathcal{A}} + \frac{a_1+a_2}{2}\big)\varpi^{-1/2}$.

□

Appendix C. Proof of Theorem 5.1

This appendix derives the dissipation matrix governing the hypocoercive convergence. The computation is based on the modified Lyapunov functional with resolvent shift $\varpi > 0$ and mixing weight $\alpha > 0$:

$$\mathcal{E}_{\alpha,\varpi}(f) := \frac{1}{2}\|f\|^2 + \alpha\Re\langle\mathcal{A}_{\varpi}f, f\rangle, \tag{134}$$

where $\mathcal{A}_{\varpi} = (\varpi - Q_O)^{-1}P_S\mathcal{V}_{\mathrm{skew}}(-\mathcal{R}^+)P_F$ is the DMS corrector associated with the reference effective dynamics Q_O.

C.1. **Preliminaries.** This appendix uses the structural estimates of Lemma 4.6 (solver bound, regularity, and perturbation control) and the notation of equations (42)–(46). We set $\varpi := \varpi_* = \lambda(Q_O)$ throughout and use the effective perturbation coefficient

$$b_{2,\mathrm{eff}} := b_2 + \frac{a_2}{2}(1+\lambda_R), \tag{135}$$

as defined in (42). We also record the energy-admissibility threshold

$$\eta_{\mathrm{sym}}(\gamma) := \begin{cases} +\infty, & a_2 = 0,\\ 2\gamma/a_2, & a_2 > 0,\end{cases} \tag{136}$$

which is exactly the range for which the coefficient of $\|(-\mathcal{R})^{1/2}f\|^2$ in the basic energy estimate below remains nonnegative.

C.2. **Derivation of Differential Estimates. Standing regularity for the Lyapunov computation.** In the bounded-operator setting, for any $f_0 \in \mathcal{H}$ the solution $f_t = e^{t\mathcal{L}}f_0$ is continuously differentiable and satisfies $\partial_t f_t = \mathcal{L}f_t$. In unbounded settings, the algebraic identities below are first derived for classical trajectories $t \mapsto f_t$ taking values in the common core $\mathcal{C}$ and satisfying $\partial_t f_t = \mathcal{L}f_t$ there; this is the only place where model-dependent regularity enters. Once the differential inequality has been established on that regular class, the resulting semigroup estimate extends to arbitrary $f_0 \in \mathcal{H}_0$ by density together with the norm equivalence from Proposition 4.8 and continuity of the semigroup.

Lemma C.1 (Energy identity for the fast dissipator)**.** *Assume $\mathcal{R}$ is self-adjoint and non-positive on $\mathcal{H}$. Then for all $f \in \mathrm{dom}(\mathcal{R})$,*

$$\Re\langle\mathcal{R}f, f\rangle = \langle\mathcal{R}f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2. \tag{137}$$

In particular, if $f = f_S + f_F$ with $f_S \in \ker(\mathcal{R})$ and $f_F \in \mathcal{H}_F$, then

$$\Re\langle\mathcal{R}f, f\rangle = \Re\langle\mathcal{R}f_F, f_F\rangle. \tag{138}$$

Proof. Since $\mathcal{R}$ is self-adjoint and non-positive, $-\mathcal{R} \ge 0$ admits a functional calculus square root $(-\mathcal{R})^{1/2}$ and $\langle\mathcal{R}f, f\rangle = -\langle(-\mathcal{R})f, f\rangle = -\|(-\mathcal{R})^{1/2}f\|^2$. Moreover $\mathcal{R}f_S = 0$ and $\mathcal{R}f = \mathcal{R}f_F$. □

Let f_t denote such a classical solution. By differentiating (40) along the flow and using that $\mathcal{A}_\varpi$ is time-independent, we obtain

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) = \Re\langle \partial_t f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \partial_t f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_t, \partial_t f_t\rangle \tag{139}$$

$$= \Re\langle \mathcal{L} f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L} f_t, f_t\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_t, \mathcal{L} f_t\rangle. \tag{140}$$

We now split the right-hand side into the "plain" dissipation coming from the generator itself and the hypocoercive correction generated by the cross term:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) = \mathcal{D}_I(f_t) + \mathcal{D}_{\mathrm{mod}}(f_t), \tag{141}$$

where we define

$$\mathcal{D}_I(f) := \Re\langle \mathcal{L} f, f\rangle, \tag{142}$$

$$\mathcal{D}_{\mathrm{mod}}(f) := \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{L} f\rangle. \tag{143}$$

Expanding $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ in $\mathcal{D}_I$ then yields the decomposition displayed below.

Natural dissipation ($\mathcal{D}_I$).

$$\mathcal{D}_I = \gamma\Re\langle \mathcal{R} f, f\rangle + \Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle + \eta\Re\langle \mathcal{V}_{\mathrm{pert}} f, f\rangle. \tag{144}$$

Using Assumption 4.1.1 ($\mathcal{R}$ coercivity) and Assumption 4.1.4 (Perturbation decomposition $\mathcal{V}_{\mathrm{pert}} = \mathcal{K}_{\mathrm{pert}} + \mathcal{S}_{\mathrm{pert}}$): the fast dissipator contributes $\gamma\Re\langle \mathcal{R} f, f\rangle \le -\gamma\lambda_R \|f_F\|^2$, while $\Re\langle \mathcal{V}_{\mathrm{skew}} f, f\rangle = 0$ and $\Re\langle \mathcal{K}_{\mathrm{pert}} f, f\rangle = 0$ by skew-adjointness. For the symmetric part, using (29) and Cauchy–Schwarz,

$$\eta\Re\langle \mathcal{S}_{\mathrm{pert}} f, f\rangle \le \eta\|\mathcal{S}_{\mathrm{pert}} f\|\,\|f\| \le \eta\Big(b_2\|f\|^2 + a_2\,\|(-\mathcal{R})^{1/2} f\|\,\|f\|\Big) \tag{145}$$

$$\le \eta\Big(\big(b_2 + \tfrac{a_2}{2}\big)\|f\|^2 + \tfrac{a_2}{2}\,\|(-\mathcal{R})^{1/2} f\|^2\Big). \tag{146}$$

Combining with the dissipator contribution, note that by Lemma C.1, $\Re\langle \mathcal{R} f, f\rangle = \Re\langle \mathcal{R} f_F, f_F\rangle = -\|(-\mathcal{R})^{1/2} f\|^2$. If $\eta \le \eta_{\mathrm{sym}}(\gamma)$, i.e. if $\gamma - \eta a_2/2 \ge 0$, then we may use the lower bound $\|(-\mathcal{R})^{1/2} f\|^2 \ge \lambda_R \|f_F\|^2$ without changing the inequality direction. Hence

$$\gamma\Re\langle \mathcal{R} f, f\rangle + \eta\Re\langle \mathcal{S}_{\mathrm{pert}} f, f\rangle \le -\big(\gamma - \tfrac{\eta a_2}{2}\big)\,\|(-\mathcal{R})^{1/2} f\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f\|^2 \tag{147}$$

$$\le -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2, \tag{148}$$

where we used $\|(-\mathcal{R})^{1/2} f\|^2 \ge \lambda_R \|f_F\|^2$, the decomposition $\|f\|^2 = \|f_S\|^2 + \|f_F\|^2$, and the fact that $\frac{a_2\lambda_R}{2} + (b_2 + \frac{a_2}{2}) = b_{2,\mathrm{eff}}$ by definition. The fast mode carries the full effective coefficient $b_{2,\mathrm{eff}}$, while the slow mode retains only the smaller $b_2 + \frac{a_2}{2}$ (since $f_S \in \ker(\mathcal{R})$ does not benefit from the λ_R-amplification).

Thus:

$$\mathcal{D}_I \le -\big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big)\,\|f_F\|^2 + \eta\big(b_2 + \tfrac{a_2}{2}\big)\,\|f_S\|^2. \tag{149}$$

Hypocoercive correction ($\mathcal{D}_{\mathrm{mod}}$). Recall

$$\mathcal{D}_{\mathrm{mod}}(f) = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{L} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{L} f\rangle. \tag{150}$$

Using $\mathcal{L} = \gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}$ and linearity, we regroup $\mathcal{D}_{\mathrm{mod}}$ according to which part of the generator appears:

$$\mathcal{D}_{\mathrm{mod}}(f) = T_{\mathcal{R}}(f) + T_{\mathcal{V}}(f) + T_{\mathcal{P}}(f), \tag{151}$$

where

$$T_{\mathcal{R}}(f) := \alpha\gamma\,\Re\langle \mathcal{A}_\varpi \mathcal{R} f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\varpi f, \mathcal{R} f\rangle, \tag{152}$$

$$T_{\mathcal{V}}(f) := \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{skew}} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\text{skew}} f\rangle, \tag{153}$$

$$T_{\mathcal{P}}(f) := \alpha\eta\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\text{pert}} f, f\rangle + \alpha\eta\,\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\text{pert}} f\rangle. \tag{154}$$

Interaction with the fast dissipator. We start from the regrouped definition

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\varpi \mathcal{R} f, f\rangle + \alpha\gamma\,\Re\langle \mathcal{A}_\varpi f, \mathcal{R} f\rangle. \tag{155}$$

Since $\mathcal{R} f = \mathcal{R} f_F$ and $\mathcal{A}_\varpi f \in \mathcal{H}_S = \ker(\mathcal{R})$, the second term vanishes by self-adjointness of $\mathcal{R}$:

$$\Re\langle \mathcal{A}_\varpi f, \mathcal{R} f\rangle = \Re\langle \mathcal{R}\mathcal{A}_\varpi f, f\rangle = 0. \tag{156}$$

Moreover $\mathcal{A}_\varpi \mathcal{R} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$, hence

$$T_{\mathcal{R}}(f) = \alpha\gamma\,\Re\langle \mathcal{A}_\varpi \mathcal{R} f_F, f_S\rangle. \tag{157}$$

Using $\mathcal{A}_\varpi = (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+) P_F$ and the pseudoinverse identity $\mathcal{R}^+\mathcal{R} = P_F$ on $\text{dom}(\mathcal{R})$, we obtain

$$\mathcal{A}_\varpi \mathcal{R} f_F = (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}}(-\mathcal{R}^+)\mathcal{R} f_F = -(\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} f_F. \tag{158}$$

Therefore, by skew-adjointness of $\mathcal{V}_{\text{skew}}$,

$$|T_{\mathcal{R}}| = \alpha\gamma\left|\Re\langle (\varpi - Q_O)^{-1} P_S \mathcal{V}_{\text{skew}} f_F, f_S\rangle\right| = \alpha\gamma\left|\Re\langle f_F, \mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\rangle\right| \tag{159}$$

$$\leq \alpha\gamma\,\|f_F\|\,\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\|. \tag{160}$$

To bound $\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\|$, set $\psi := (\varpi - Q_O)^{-1} f_S \in \text{dom}(Q_O) \subset \mathcal{H}_S$ and denote its induced microscopic flux

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi \in \mathcal{H}_F. \tag{161}$$

Since $\psi \in \mathcal{H}_S$, the centered-coupling assumption gives $P_S \mathcal{V}_{\text{skew}} \psi = 0$, hence $g_\psi = \mathcal{V}_{\text{skew}}\psi = \mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S$. Applying the flux conditioning bound (24) yields

$$\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\|^2 = \|g_\psi\|^2 \leq M_R\,\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle. \tag{162}$$

The "extra Q_O" comes from rewriting the right-hand side using the definition $Q_O := -P_S \mathcal{V}_{\text{skew}} \mathcal{R}^+ \mathcal{V}_{\text{skew}} P_S$: indeed, for $\psi \in \mathcal{H}_S$,

$$\langle g_\psi, -\mathcal{R}^+ g_\psi\rangle = \langle \mathcal{V}_{\text{skew}}\psi, (-\mathcal{R}^+)\mathcal{V}_{\text{skew}}\psi\rangle = \langle \psi, (-Q_O)\psi\rangle = \|(-Q_O)^{1/2}\psi\|^2. \tag{163}$$

Therefore

$$\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\| \leq \sqrt{M_R}\,\|(-Q_O)^{1/2}(\varpi - Q_O)^{-1} f_S\|. \tag{164}$$

Finally, by functional calculus for the self-adjoint operator $-Q_O \geq 0$,

$$\|(-Q_O)^{1/2}(\varpi - Q_O)^{-1}\| = \sup_{x\geq 0} \frac{\sqrt{x}}{\varpi + x} = \frac{1}{2\sqrt{\varpi}}, \tag{165}$$

which implies

$$\|\mathcal{V}_{\text{skew}}(\varpi - Q_O)^{-1} f_S\| \leq \frac{\sqrt{M_R}}{2}\,\varpi^{-1/2}\,\|f_S\|. \tag{166}$$

Hence

$$|T_{\mathcal{R}}| \leq \alpha\gamma\,\frac{\sqrt{M_R}}{2}\,\varpi^{-1/2}\,\|f_F\|\,\|f_S\|. \tag{167}$$

Interaction with transport. Starting from the regrouped definition

$$T_{\mathcal{V}}(f) = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\rm skew} f\rangle, \tag{168}$$

we decompose $f = f_S + f_F$ and use that $\mathcal{A}_\varpi$ maps into the slow space and annihilates the slow component: $\mathcal{A}_\varpi = \mathcal{A}_\varpi P_F$ (since P_F is on the right in (32)). Hence $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$. Expanding the *first* bracket gives

$$\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle = \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \tag{169}$$

$$+ \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_F\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_F\rangle. \tag{170}$$

The last term vanishes because $\mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F \in \mathcal{H}_S$ is orthogonal to $f_F \in \mathcal{H}_F$. Moreover, since $\mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S \in \mathcal{H}_S$, we have $\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_F\rangle = 0$. Therefore

$$\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f, f\rangle = \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle. \tag{171}$$

In the *second* bracket we use $\mathcal{A}_\varpi f = \mathcal{A}_\varpi f_F$ and expand $\mathcal{V}_{\rm skew} f$:

$$\Re\langle \mathcal{A}_\varpi f, \mathcal{V}_{\rm skew} f\rangle = \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle + \Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle. \tag{172}$$

Putting the pieces together, we obtain the explicit decomposition

$$T_{\mathcal{V}} = \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \tag{173}$$

$$+ \alpha\,\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle + \alpha\,\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle. \tag{174}$$

Main Coercivity. Since $f_S \in \mathcal{H}_S$ and $P_S \mathcal{V}_{\rm skew} P_S = 0$, we have $P_F \mathcal{V}_{\rm skew} f_S = \mathcal{V}_{\rm skew} f_S$. By the definition of $\mathcal{A}_\varpi$ and Q_O this yields

$$\alpha\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_S, f_S\rangle = \alpha\langle (\varpi - Q_O)^{-1} Q_O f_S, f_S\rangle = -\alpha\,\langle (\varpi - Q_O)^{-1}(-Q_O) f_S, f_S\rangle \tag{175}$$

$$\le -\alpha\,\frac{\lambda(Q_O)}{\varpi + \lambda(Q_O)}\,\|f_S\|^2. \tag{176}$$

Here we used that $f \in \mathcal{H}_0$ implies $f_S = P_S f \in \mathcal{H}_S \cap \mathcal{H}_0 = \mathcal{H}_{S,0}$, so Assumption 4.2 applies to f_S.
Cross Terms. Since $\mathcal{A}_\varpi f_F \in \mathcal{H}_S$ (by the range of the corrector) and $\mathcal{V}_{\rm skew} f_S \in \mathcal{H}_F$ (by centering, $P_S \mathcal{V}_{\rm skew} P_S = 0$), the third inner product vanishes: $\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_S\rangle = 0$. The only surviving cross term is bounded by the dual regularity estimate:

$$\alpha\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm skew} f_F, f_S\rangle \le \alpha\|\mathcal{A}_\varpi \mathcal{V}_{\rm skew} P_F\|\|f_F\|\|f_S\| \le \alpha\kappa_{\rm ell}\|f_F\|\|f_S\|. \tag{177}$$

Fast–fast remainder. Using $\mathcal{V}_{\rm skew}^\dagger = -\mathcal{V}_{\rm skew}$ and Cauchy–Schwarz,

$$\big|\alpha\Re\langle \mathcal{A}_\varpi f_F, \mathcal{V}_{\rm skew} f_F\rangle\big| = \alpha\big|\Re\langle -\mathcal{V}_{\rm skew}\mathcal{A}_\varpi f_F, f_F\rangle\big| \le \alpha\,\|\mathcal{V}_{\rm skew}\mathcal{A}_\varpi\|\,\|f_F\|^2 \le \alpha\,k_{\rm reg}\,\|f_F\|^2. \tag{178}$$

Interaction with perturbation.

$$T_{\mathcal{P}} = \alpha\eta\Re\langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f\rangle + \alpha\eta\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f\rangle. \tag{179}$$

We estimate each term using Lemma 4.6 and the centered mapping property (38).
Reverse term. Since $\mathcal{A}_\varpi f \in \mathcal{H}_S$ and $\mathcal{V}_{\rm pert}^\dagger \mathcal{H}_S \subset \mathcal{H}_F$, we have $\langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f\rangle = \langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f_F\rangle$. Therefore,

$$\big|\alpha\eta\Re\langle \mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f, f\rangle\big| \le \alpha\eta\,\|\mathcal{V}_{\rm pert}^\dagger \mathcal{A}_\varpi f\|\,\|f_F\| \le \alpha\eta\,k_P(\varpi)\,\|f\|\,\|f_F\| \tag{180}$$

$$\le \alpha\eta\,k_P(\varpi)\,\big(\|f_F\|^2 + \|f_F\|\,\|f_S\|\big). \tag{181}$$

Forward term. Since $\mathcal{A}_\varpi \mathcal{H} \subset \mathcal{H}_S$, we have $\langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f\rangle = \langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f_S\rangle$. Therefore,

$$\big|\alpha\eta\Re\langle \mathcal{A}_\varpi \mathcal{V}_{\rm pert} f, f\rangle\big| \le \alpha\eta\,\|\mathcal{A}_\varpi \mathcal{V}_{\rm pert} f\|\,\|f_S\| \le \alpha\eta\,\tilde{k}_P(\varpi)\,\|f\|\,\|f_S\| \tag{182}$$

$$\le \alpha\eta\,\tilde{k}_P(\varpi)\,\big(\|f_S\|^2 + \|f_F\|\,\|f_S\|\big). \tag{183}$$

C.3. **Dissipation Matrix Construction.** Summing all estimates and using Cauchy–Schwarz bounds of the form $|\langle X f_F, f_S\rangle| \le \|X\|\,\|f_F\|\,\|f_S\|$, we obtain a closed differential inequality of the form

$$\frac{d}{dt}\,\mathcal{E}_{\alpha,\varpi}(f_t) \le -\mathbf{v}(t)^T \mathbf{M}\,\mathbf{v}(t), \qquad \mathbf{v}(t) := \begin{pmatrix} \|P_F f_t\| \\ \|P_S f_t\| \end{pmatrix}. \tag{184}$$

Here $\mathbf{M}$ is a real symmetric 2×2 matrix whose entries depend only on the structural constants and the chosen parameters (α, ϖ). The entries of $\mathbf{M}$ are explicitly decomposed by powers of ϖ below.

Fast-Fast Block (M_{11}): From Natural Dissipation, the fast–fast transport remainder, and Reverse Perturbation.

$$\begin{aligned} M_{11} &= \big(\gamma\lambda_R - \eta b_{2,\mathrm{eff}}\big) - \alpha k_{\mathrm{reg}} - \alpha\eta k_P(\varpi) \\ &= \gamma\lambda_R - \eta b_{2,\mathrm{eff}} - \alpha k_{\mathrm{reg}} - \alpha\eta(b_1+b_2)k_{\mathcal{A}}\,\varpi^{-1/2}. \end{aligned} \tag{185}$$

Slow-Slow Block (M_{22}): From Main Coercivity, Natural Dissipation (gain), and both Perturbations.

$$\begin{aligned} M_{22} &= \alpha\frac{\lambda(Q_O)}{\varpi+\lambda(Q_O)} - \eta\big(b_2 + \tfrac{a_2}{2}\big) - \alpha\eta\,\tilde{k}_P(\varpi) \\ &= \alpha\frac{\lambda(Q_O)}{\varpi+\lambda(Q_O)} - \eta\big(b_2 + \tfrac{a_2}{2}\big) - \alpha\eta\left[\Big((b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\varpi^{-1/2}\right]. \end{aligned} \tag{186}$$

Cross Block (M_{12}): From Mixing, Transport Cross terms, and Forward Perturbation.

$$\begin{aligned} M_{12} &= -\frac{1}{2}\alpha\left[\gamma\frac{\sqrt{M_R}}{2}\,\varpi^{-1/2} + \kappa_{\mathrm{ell}} + \eta\big(k_P(\varpi) + \tilde{k}_P(\varpi)\big)\right] \\ &= -\frac{1}{2}\alpha\left[\kappa_{\mathrm{ell}} + \left(\gamma\frac{\sqrt{M_R}}{2} + \eta\Big(2(b_1+b_2)k_{\mathcal{A}} + \tfrac{a_1+a_2}{2}\Big)\right)\varpi^{-1/2}\right]. \end{aligned} \tag{187}$$

This explicit form allows for direct asymptotic analysis by choosing $\varpi \sim \lambda(Q_O)$, confirming the stability for small η.

C.4. **Optimization of the Decay Rate.** We work in the unperturbed regime $\eta = 0$ to identify the dominant hypocoercive rate. Let $\mathbf{M}_0(\alpha)$ denote the dissipation matrix at $\eta = 0$. Since $\mathbf{M}_0$ is 2×2 symmetric, its smallest eigenvalue satisfies the determinant–trace bound

$$\lambda_{\min}(\mathbf{M}_0) \ge \frac{\det(\mathbf{M}_0)}{\mathrm{Tr}(\mathbf{M}_0)}. \tag{188}$$

Choice of resolvent shift. The slow-coercivity contribution in M_{22}^0 is $\alpha\lambda(Q_O)/(\varpi+\lambda(Q_O))$, while the cross term $|M_{12}^0|$ contains $\gamma\Theta\varpi^{-1/2}$. A leading-order balance (maximising $\varpi(\varpi+\lambda(Q_O))^{-2}$ in the large-γ rate) motivates the gap-adapted choice

$$\varpi_* := \lambda(Q_O). \tag{189}$$

We do not claim here a separate global optimisation theorem over all admissible shifts; rather, ϖ_* is the canonical gap-scale choice that balances the slow coercive block against the cross term inside the determinant–trace estimate used in this paper. At this value, $\lambda(Q_O)/(\varpi_*+\lambda(Q_O)) = \frac{1}{2}$. Using the constants $\Theta, K_{\mathrm{geom}}, Z(\gamma)$ from (42)–(43), the unperturbed matrix entries at $\varpi = \varpi_*$ are:

$$M_{11}^0 = \gamma\lambda_R - \alpha k_{\mathrm{reg}}, \tag{190}$$

$$M_{22}^0 = \tfrac{1}{2}\,\alpha, \tag{191}$$

$$M_{12}^0 = -\tfrac{1}{2}\,\alpha\,Z(\gamma). \tag{192}$$

Optimal mixing weight. The determinant $D(\alpha) := M_{11}^0 M_{22}^0 - (M_{12}^0)^2$ is a concave quadratic in α:

$$D(\alpha) = \tfrac{1}{2}\gamma\lambda_R\,\alpha - \big(\tfrac{1}{2}k_{\rm reg} + \tfrac{1}{4}Z(\gamma)^2\big)\alpha^2. \tag{193}$$

It is maximised at

$$\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\rm reg}}, \tag{194}$$

in agreement with (44), giving

$$D(\alpha_*) = \frac{\frac{1}{4}(\gamma\lambda_R)^2}{Z(\gamma)^2 + 2k_{\rm reg}}. \tag{195}$$

Trace upper bound and rate estimate. The exact trace is $\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} = (\gamma\lambda_R - \alpha_* k_{\rm reg}) + \frac{1}{2}\alpha_* = \gamma\lambda_R - \alpha_*(k_{\rm reg} - \frac{1}{2})$. Since $k_{\rm reg} = \sqrt{M_R/\lambda_R} \geq 1$ (the flux conditioning bound gives $M_R \geq \lambda_R$), we have $k_{\rm reg} - \frac{1}{2} \geq \frac{1}{2} > 0$, hence

$$\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*} \leq \gamma\lambda_R. \tag{196}$$

Applying (188) yields

$$\lambda_{\min}(\mathbf{M}_0) \geq \frac{D(\alpha_*)}{\mathrm{Tr}(\mathbf{M}_0)|_{\alpha_*}} \geq \frac{\frac{1}{4}\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\rm reg}} =: \Lambda_{\rm hypo}(\gamma), \tag{197}$$

consistent with definition (45). This provides a rigorous lower bound on the unperturbed decay rate for all $\gamma > 0$.

Asymptotic scaling analysis. Equation (197) reveals the precise scaling behavior across regimes without relying on ad-hoc approximations.
Overdamped Limit ($\gamma \to \infty$). The denominator is dominated by the $\mathcal{O}(\gamma^2)$ term, giving

$$\lim_{\gamma\to\infty} \gamma\Lambda_{\rm hypo}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\lambda(Q_O)}{\Theta^2} = \lambda(Q_O)\,\frac{\lambda_R}{M_R}. \tag{198}$$

Thus, $\Lambda_{\rm hypo}(\gamma) = \lambda(Q_O)\,\frac{\lambda_R}{M_R}\,\gamma^{-1} + \mathcal{O}(\gamma^{-2})$. In particular, $\Lambda_{\rm hypo}(\gamma) \asymp \lambda(Q_O)\,\gamma^{-1}$.
Underdamped/Intermediate Regime. For small or order-one γ, the denominator is controlled by the γ-independent structural constants $K_{\rm geom}$ and $2k_{\rm reg}$. These arise from the slow–fast coupling through $\mathcal{V}_{\rm skew}$ and prevent the estimate from degenerating even though $\mathcal{R}$ does not dissipate on $\mathcal{H}_S$ directly. The transport mixes slow directions into fast dissipative directions, yielding $\Lambda_{\rm hypo}(\gamma) \asymp \gamma$ for small γ.

We now establish the stability condition by computing the operator norm of the perturbation matrix $\mathbf{E}(\gamma)$ as a function of γ, tracking the dependence on the macroscopic spectral gap $\varpi_* = \lambda(Q_O)$.

Explicit norm of the perturbation matrix. Recalling that $\mathbf{M} = \mathbf{M}_0 - \eta\mathbf{E}$, the perturbation matrix $\mathbf{E}$ has entries read off from the full $\mathbf{M}$ computation:

$$E_{11} = b_{2,\rm eff} + \alpha_*\,k_P(\varpi_*), \tag{199}$$

$$E_{22} = \big(b_2 + \tfrac{a_2}{2}\big) + \alpha_*\,\tilde{k}_P(\varpi_*), \tag{200}$$

$$E_{12} = \frac{1}{2}\,\alpha_*\,\big(k_P(\varpi_*) + \tilde{k}_P(\varpi_*)\big). \tag{201}$$

Here E_{11} contains the full effective coefficient $b_{2,\rm eff} = b_2 + \frac{a_2}{2}(1 + \lambda_R)$, while E_{22} contains only the slow-mode coefficient $b_2 + \frac{a_2}{2}$; the extra $\frac{a_2\lambda_R}{2}$ absorbed into $b_{2,\rm eff}$ arises in the fast block from the

bound $\|(-\mathcal{R})^{1/2}f\|^2 \geq \lambda_R\|f_F\|^2$ applied to the symmetric perturbation in $\mathcal{D}_I$ (see (149)). Since $\mathbf{E}$ is symmetric, we bound its spectral norm by the maximal absolute row sum (Gershgorin):

$$\|\mathbf{E}(\gamma)\| \leq \max\{E_{11} + |E_{12}|,\, E_{22} + |E_{12}|\}. \tag{202}$$

Using $b_{2,\text{eff}} \geq b_2 + \frac{a_2}{2}$ and $k_P(\varpi_*) \leq \tilde{k}_P(\varpi_*)$, both row sums admit the same uniform upper bound:

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\left[\tfrac{3}{2}\,\tilde{k}_P(\varpi_*) + \tfrac{1}{2}\,\tilde{k}_P(\varpi_*)\right] = b_{2,\text{eff}} + 2\alpha_*(\gamma)\,\tilde{k}_P(\varpi_*). \tag{203}$$

Recall from Lemma 4.6 that the bounds k_P and $\tilde{k}_P$ depend on $\varpi_*^{-1/2}$. We define the *strictly constant* structural coefficients (independent of ϖ and γ):

$$C_{\text{sing}} := 2(b_1 + b_2)k_{\mathcal{A}} + (a_1 + a_2). \tag{204}$$

Indeed,

$$2\tilde{k}_P(\varpi_*) = 2\Big((b_1 + b_2)k_{\mathcal{A}} + \frac{a_1 + a_2}{2}\Big)\varpi_*^{-1/2} = C_{\text{sing}}\,\varpi_*^{-1/2},$$

so the bound (203) becomes

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\,C_{\text{sing}}\,\varpi_*^{-1/2}. \tag{205}$$

Substituting $\alpha_*(\gamma)$ from (44), the bound becomes

$$\|\mathbf{E}(\gamma)\| \leq b_{2,\text{eff}} + \alpha_*(\gamma)\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2} = B_{\text{pert}}(\gamma), \tag{206}$$

consistent with definition (46).

Friction-dependent threshold. The stability condition $\lambda_{\min}(\mathbf{M}) > 0$ is ensured by $\eta\|\mathbf{E}(\gamma)\| \leq \frac{1}{2}\Lambda_{\text{hypo}}(\gamma)$, which yields the exact matrix threshold

$$\eta^\sharp(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2\|\mathbf{E}(\gamma)\|}. \tag{207}$$

Since $\|\mathbf{E}(\gamma)\| \leq B_{\text{pert}}(\gamma)$, the explicit certified threshold used in Theorem 5.1 is

$$\eta_{\text{mat}}(\gamma) := \frac{\Lambda_{\text{hypo}}(\gamma)}{2B_{\text{pert}}(\gamma)} \leq \eta^\sharp(\gamma). \tag{208}$$

Together with the energy-admissibility condition $\eta \leq \eta_{\text{sym}}(\gamma)$ coming from (149), this gives the theorem threshold

$$\eta^*(\gamma) := \min\{\eta_{\text{sym}}(\gamma), \eta_{\text{mat}}(\gamma)\}. \tag{209}$$

Substituting the explicit forms of Λ_{hypo} (45) and B_{pert} (46) yields

$$\eta_{\text{mat}}(\gamma) = \frac{\frac{1}{4}\,\gamma\lambda_R}{\big(Z(\gamma)^2 + 2k_{\text{reg}}\big)\big(2b_{2,\text{eff}} + 2\alpha_*(\gamma)\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}\big)}. \tag{210}$$

Asymptotic analysis of the certified threshold. We analyze first the scaling of $\eta_{\text{mat}}(\gamma)$ in the limiting regimes. Note that $Z(\gamma) \asymp \gamma$ as $\gamma \to \infty$.

Overdamped Limit ($\gamma \to \infty$). The numerator scales as $\mathcal{O}(\gamma)$. In the denominator, the term $2b_{2,\text{eff}}Z(\gamma)^2$ scales as $\mathcal{O}(\gamma^2)$, while the skew-term scales as $\mathcal{O}(\gamma)$. If $b_{2,\text{eff}} > 0$ (symmetric perturbation present), the $\mathcal{O}(\gamma^2)$ term dominates and $\eta_{\text{mat}}(\gamma) \asymp \gamma^{-1}$. If $b_{2,\text{eff}} = 0$ (purely skew perturbation, i.e. $b_2 = a_2 = 0$), the denominator is dominated by the skew term $\mathcal{O}(\gamma)$ and $\eta_{\text{mat}}(\gamma) \asymp \mathcal{O}(1)$.

Optimal Lift ($\gamma \sim \sqrt{\lambda(Q_O)}$). At the optimal friction γ_{opt}, determined in the optimization step below, we have $Z(\gamma_{\text{opt}}) = \mathcal{O}(1)$. Consider the singular term in the denominator:

$$\gamma_{\text{opt}} \cdot C_{\text{sing}}\lambda(Q_O)^{-1/2} \asymp \sqrt{\lambda(Q_O)} \cdot \lambda(Q_O)^{-1/2} \asymp \mathcal{O}(1). \tag{211}$$

The singularity thus cancels, and at the optimal friction the matrix-controlled threshold is well-behaved:

$$\eta_{\mathrm{mat}}(\gamma_{\mathrm{opt}}) \asymp \sqrt{\lambda(Q_O)}. \tag{212}$$

Since $\eta_{\mathrm{sym}}(\gamma_{\mathrm{opt}}) = +\infty$ when $a_2 = 0$ and $\eta_{\mathrm{sym}}(\gamma_{\mathrm{opt}}) = 2\gamma_{\mathrm{opt}}/a_2 \asymp \sqrt{\lambda(Q_O)}$ when $a_2 > 0$, the overall threshold $\eta^*(\gamma_{\mathrm{opt}}) = \min\{\eta_{\mathrm{sym}}(\gamma_{\mathrm{opt}}), \eta_{\mathrm{mat}}(\gamma_{\mathrm{opt}})\}$ has the same $\sqrt{\lambda(Q_O)}$ scaling.

From coercivity to global decay. Having established the positivity of the dissipation matrix, we convert the algebraic estimate into a global semigroup decay bound. This step involves explicit algebraic substitution to derive the final decay rate, ensuring that the singular scaling of the perturbation is correctly propagated.

Uniform norm equivalence. Recall the distortion parameter

$$\delta_*(\gamma) := 2\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\|. \tag{213}$$

Using $\|\mathcal{A}_{\varpi_*}\| \le k_{\mathcal{A}}\, \varpi_*^{-1/2}$ with $k_{\mathcal{A}} = (2\sqrt{\lambda_R})^{-1}$ and the explicit weight $\alpha_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2+2k_{\mathrm{reg}}}$, we obtain

$$\delta_*(\gamma) = \frac{\gamma\lambda_R}{Z(\gamma)^2 + 2k_{\mathrm{reg}}}\,\frac{1}{\sqrt{\lambda_R\,\varpi_*}} = \frac{\gamma\sqrt{\lambda_R}\,\varpi_*^{-1/2}}{\big(\gamma\Theta\varpi_*^{-1/2} + K_{\mathrm{geom}}\big)^2 + 2k_{\mathrm{reg}}}. \tag{214}$$

Setting $x := \gamma\Theta\varpi_*^{-1/2}$ gives

$$\delta_*(\gamma) = \frac{\sqrt{\lambda_R}}{\Theta}\cdot\frac{x}{(x+K_{\mathrm{geom}})^2 + 2k_{\mathrm{reg}}}. \tag{215}$$

The elementary optimisation

$$\sup_{x>0}\frac{x}{(x+K_{\mathrm{geom}})^2 + 2k_{\mathrm{reg}}} = \frac{1}{2\big(K_{\mathrm{geom}} + \sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}}\big)} \tag{216}$$

together with $M_R \ge \lambda_R$ and $\sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}} \ge \sqrt{2k_{\mathrm{reg}}} \ge \sqrt{2}$ (the latter since $k_{\mathrm{reg}} = \sqrt{M_R/\lambda_R} \ge 1$) therefore yields

$$\delta_*(\gamma) \le \frac{\sqrt{\lambda_R}}{\sqrt{M_R}\big(K_{\mathrm{geom}} + \sqrt{K_{\mathrm{geom}}^2 + 2k_{\mathrm{reg}}}\big)} \le 2^{-1/2}, \tag{217}$$

and therefore

$$\alpha_*(\gamma)\|\mathcal{A}_{\varpi_*}\| = \frac{\delta_*(\gamma)}{2} \le 2^{-3/2} < \frac{1}{2}. \tag{218}$$

Thus Proposition 4.8 applies with $(\alpha, \varpi) = (\alpha_*, \varpi_*)$. Hence the norm-equivalence prefactor satisfies

$$C_*(\gamma) := \sqrt{\frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}} \le 1 + \sqrt{2}. \tag{219}$$

Thus, for all f,

$$\frac{1}{2}(1-\delta_*(\gamma))\|f\|^2 \le \mathcal{E}_{\alpha_*,\varpi_*}(f) \le \frac{1}{2}(1+\delta_*(\gamma))\|f\|^2. \tag{220}$$

Explicit calculation of the decay rate. Set

$$c(\gamma,\eta) := \Lambda_{\mathrm{hypo}}(\gamma) - \eta\,\|\mathbf{E}(\gamma)\|. \tag{221}$$

The algebraic dissipation estimate reads

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le -c(\gamma,\eta)\,\|f_t\|^2. \tag{222}$$

Next we use the norm equivalence from Proposition 4.8 in the convenient form

$$\frac{1}{2}(1-\delta_*(\gamma))\,\|f\|^2 \le \mathcal{E}_{\alpha_*,\varpi_*}(f) \le \frac{1}{2}(1+\delta_*(\gamma))\,\|f\|^2, \tag{223}$$

where $\delta_*(\gamma) := 2\alpha_*(\gamma)\,\|\mathcal{A}_{\varpi_*}\| \in (0,1)$. The upper bound implies

$$\|f\|^2 \ge \frac{2}{1+\delta_*(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(f). \tag{224}$$

Substituting this into the dissipation inequality gives a closed differential inequality for the Lyapunov functional:

$$\frac{d}{dt}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le -\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t). \tag{225}$$

By Grönwall's inequality,

$$\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le \exp\Big(-\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}\,t\Big)\,\mathcal{E}_{\alpha_*,\varpi_*}(f_0). \tag{226}$$

Finally, converting back to the L^2 norm using the two-sided equivalence yields

$$\|f_t\|^2 \le \frac{2}{1-\delta_*(\gamma)}\,\mathcal{E}_{\alpha_*,\varpi_*}(f_t) \le \frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}\,e^{-\frac{2c(\gamma,\eta)}{1+\delta_*(\gamma)}t}\,\|f_0\|^2. \tag{227}$$

Therefore

$$\|f_t\| \le \sqrt{\frac{1+\delta_*(\gamma)}{1-\delta_*(\gamma)}}\,e^{-\lambda_\eta t}\,\|f_0\|, \qquad \lambda_\eta := \frac{c(\gamma,\eta)}{1+\delta_*(\gamma)}. \tag{228}$$

We now substitute the explicit expressions derived in the previous steps. For the unperturbed rate:

$$\Lambda_{\text{hypo}}(\gamma) = \frac{\frac{1}{4}\lambda_R\,\gamma}{Z(\gamma)^2+2k_{\text{reg}}}, \quad \text{with } Z(\gamma) = \gamma\Theta\lambda(Q_O)^{-1/2} + K_{\text{geom}}. \tag{229}$$

For the perturbation norm, we use the sharp bound (205) at $\varpi_* = \lambda(Q_O)$ to handle the macroscopic singularity:

$$\|\mathbf{E}(\gamma)\| \le b_{2,\text{eff}} + \underbrace{\frac{\gamma\lambda_R}{Z(\gamma)^2+2k_{\text{reg}}}}_{\alpha_*(\gamma)}\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}. \tag{230}$$

Substituting these into the expression for λ_η and using the explicit distortion bound on $\delta_*(\gamma)$:

$$\lambda_\eta \ge \frac{\frac{1}{4}\lambda_R\,\gamma}{(1+\delta_*(\gamma))(Z(\gamma)^2+2k_{\text{reg}})} - \frac{\eta}{1+\delta_*(\gamma)}\left(b_{2,\text{eff}} + \frac{\gamma\lambda_R}{Z(\gamma)^2+2k_{\text{reg}}}\,C_{\text{sing}}\,\lambda(Q_O)^{-1/2}\right). \tag{231}$$

Equivalently,

$$\lambda_\eta \ge \frac{\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)}{1+\delta_*(\gamma)} \ge (2-\sqrt{2})\big(\Lambda_{\text{hypo}}(\gamma) - \eta B_{\text{pert}}(\gamma)\big), \tag{232}$$

using $1+\delta_*(\gamma) \le 1+2^{-1/2}$, i.e. $(1+\delta_*(\gamma))^{-1} \ge (1+2^{-1/2})^{-1} = 2-\sqrt{2}$. This is the rigorous lower bound. The potentially singular factor $C_{\text{sing}}\lambda(Q_O)^{-1/2}$ is multiplied by the mixing weight $\alpha_*(\gamma) = \gamma\lambda_R/(Z(\gamma)^2+2k_{\text{reg}})$, which scales as $\sqrt{\lambda(Q_O)}$ at the optimal lift, ensuring the perturbation contribution remains bounded.

Finally, converting back to the L^2 norm via Grönwall's lemma yields the global estimate:

$$\|e^{t\mathcal{L}}f\|_{\mathcal{H}} \le C_*(\gamma)\, e^{-\lambda_\eta t}\|f\|_{\mathcal{H}} \le (1+\sqrt{2})\, e^{-\lambda_\eta t}\|f\|_{\mathcal{H}}. \tag{233}$$

This completes the proof.

C.5. **Optimization over Friction γ: Kinetic Acceleration.** We determine $\gamma_{\rm opt}$ maximising $\Lambda_{\rm hypo}(\gamma)$. From (197), setting $x := \Theta\lambda(Q_O)^{-1/2}\gamma$, the rate function reads

$$\Lambda_{\rm hypo}(\gamma) = \frac{\frac{1}{4}\lambda_R}{\Theta}\sqrt{\lambda(Q_O)}\cdot\frac{x}{(x+K_{\rm geom})^2+2k_{\rm reg}}. \tag{234}$$

Optimal friction. The function $x \mapsto x/[(x+K_{\rm geom})^2+C_0]$ with $C_0 := 2k_{\rm reg}$ is maximised when $(x+K_{\rm geom})^2+C_0 = 2x(x+K_{\rm geom})$, equivalently $x^2 = K_{\rm geom}^2 + C_0$. Translating back:

$$\gamma_{\rm opt} = \frac{\sqrt{K_{\rm geom}^2+2k_{\rm reg}}}{\Theta}\sqrt{\lambda(Q_O)} =: C_{\rm lift}\sqrt{\lambda(Q_O)}, \tag{235}$$

where $C_{\rm lift} := \Theta^{-1}\sqrt{K_{\rm geom}^2+2k_{\rm reg}}$.

Maximal rate. At $\gamma = \gamma_{\rm opt}$ one has $x_* := \Theta\lambda(Q_O)^{-1/2}\gamma_{\rm opt} = \sqrt{K_{\rm geom}^2+C_0}$. Substituting into the denominator: $(x_*+K_{\rm geom})^2+C_0 = 2(K_{\rm geom}^2+C_0+K_{\rm geom}x_*)$. Hence

$$\Lambda_{\max} := \Lambda_{\rm hypo}(\gamma_{\rm opt}) = \frac{\frac{1}{4}\lambda_R}{2\Theta(x_*+K_{\rm geom})}\sqrt{\lambda(Q_O)} \asymp \sqrt{\lambda(Q_O)}. \tag{236}$$

This confirms the kinetic acceleration: $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$, compared to the overdamped ceiling $\Lambda_{\rm hypo}(\gamma) \asymp \lambda(Q_O)/\gamma$ as $\gamma \to \infty$.

Certified threshold at optimal lift. The admissible perturbation range certified by Theorem 5.1 is determined as follows. From (207), the exact matrix threshold depends on $\|\mathbf{E}(\gamma)\|$, while the theorem uses the explicit upper bound $B_{\rm pert}(\gamma)$. The dominant term involves the mixing weight $\alpha_*(\gamma)$. Evaluating at $\gamma_{\rm opt}$:

$$\alpha_*(\gamma_{\rm opt}) = \frac{\gamma_{\rm opt}\lambda_R}{Z(\gamma_{\rm opt})^2+2k_{\rm reg}}. \tag{237}$$

Using $\gamma_{\rm opt} = C_{\rm lift}\sqrt{\lambda(Q_O)}$ and $Z(\gamma_{\rm opt}) = \sqrt{K_{\rm geom}^2+2k_{\rm reg}} + K_{\rm geom} = \mathcal{O}(1)$, the numerator of $\alpha_*(\gamma_{\rm opt})$ is $\gamma_{\rm opt}\lambda_R \asymp \sqrt{\lambda(Q_O)}$ and the denominator $Z(\gamma_{\rm opt})^2+2k_{\rm reg} = \mathcal{O}(1)$ (independent of $\lambda(Q_O)$). Hence $\alpha_*(\gamma_{\rm opt}) \asymp \sqrt{\lambda(Q_O)}$, so $\alpha_*(\gamma_{\rm opt})\cdot\lambda(Q_O)^{-1/2} = \mathcal{O}(1)$. Applying (205):

$$\|\mathbf{E}(\gamma_{\rm opt})\| \le b_{2,\rm eff} + \alpha_*(\gamma_{\rm opt})\, C_{\rm sing}\, \lambda(Q_O)^{-1/2} = \mathcal{O}(1). \tag{238}$$

Since $\Lambda_{\max} \asymp \sqrt{\lambda(Q_O)}$ and $B_{\rm pert}(\gamma_{\rm opt}) = \mathcal{O}(1)$, the matrix-controlled threshold satisfies

$$\eta_{\rm mat}(\gamma_{\rm opt}) = \frac{\Lambda_{\max}}{2B_{\rm pert}(\gamma_{\rm opt})} \asymp \sqrt{\lambda(Q_O)}. \tag{239}$$

Together with $\eta_{\rm sym}(\gamma_{\rm opt}) = +\infty$ when $a_2 = 0$ and $\eta_{\rm sym}(\gamma_{\rm opt}) = 2\gamma_{\rm opt}/a_2 \asymp \sqrt{\lambda(Q_O)}$ when $a_2 > 0$, this yields

$$\eta^*(\gamma_{\rm opt}) = \min\{\eta_{\rm sym}(\gamma_{\rm opt}), \eta_{\rm mat}(\gamma_{\rm opt})\} \asymp \sqrt{\lambda(Q_O)}. \tag{240}$$

This confirms that the accelerated regime retains an $\mathcal{O}(\sqrt{\lambda(Q_O)})$ admissible perturbation range.

Appendix D. Upper bound via singular value gap

This appendix proves Theorem 5.6.

D.1. **Exponential decay implies a singular value-gap bound.** For a (possibly unbounded) closed operator T on $\mathcal{H}_0$, define its injectivity modulus

$$s(T) := \inf_{f\in \mathrm{dom}(T)\cap\mathcal{H}_0\setminus\{0\}} \frac{\|Tf\|}{\|f\|}. \tag{241}$$

Lemma D.1 (From exponential decay to a singular value-gap bound)**.** *Assume* $\ker(\mathcal{L})\cap\mathcal{H}_0=\{0\}$. *Assume moreover that* $e^{t\mathcal{L}}$ *is a contraction semigroup on* $\mathcal{H}_0$*, i.e.* $\|e^{t\mathcal{L}}f\|\le\|f\|$ *for all* $t\ge 0$ *and* $f\in\mathcal{H}_0$ *(equivalently,* $\mathcal{L}$ *is dissipative on* $\mathcal{H}_0$*:* $\mathrm{Re}\langle\mathcal{L}f,f\rangle\le 0$*). If there exist constants* $C\ge 1$ *and* $\nu>0$ *such that*

$$\|e^{t\mathcal{L}}f\|\le C\,e^{-\nu t}\,\|f\|,\qquad \forall t\ge 0,\ \forall f\in\mathcal{H}_0, \tag{242}$$

then 0 *belongs to the resolvent set of* $\mathcal{L}|_{\mathcal{H}_0}$ *and*

$$\nu\le(1+\log C)\,s(\mathcal{L}). \tag{243}$$

Proof. Set $t_*:=(\log C)/\nu\ge 0$ (the time at which $Ce^{-\nu t}$ crosses 1). For $\lambda>0$, define the resolvent regularization

$$u_\lambda := -\int_0^\infty e^{-\lambda t}e^{t\mathcal{L}}f\,dt. \tag{244}$$

This Bochner integral is absolutely convergent because $e^{-\lambda t}\le 1$ and the semigroup is bounded by the piecewise integrable envelope used below. On $[0,t_*]$ use the contraction bound $\|e^{t\mathcal{L}}f\|\le\|f\|$; on $[t_*,\infty)$ use the exponential decay bound $\|e^{t\mathcal{L}}f\|\le Ce^{-\nu t}\|f\|$. Combining these estimates and the inequality $e^{-\lambda t}\le 1$ gives

$$\|u_\lambda\|\le\int_0^{t_*}\|f\|\,dt+\int_{t_*}^\infty Ce^{-\nu t}\|f\|\,dt = t_*\,\|f\|+\frac{C}{\nu}e^{-\nu t_*}\,\|f\| = \frac{\log C}{\nu}\,\|f\|+\frac{1}{\nu}\,\|f\| = \frac{1+\log C}{\nu}\,\|f\|, \tag{245}$$

where we used $Ce^{-\nu t_*}=Ce^{-\log C}=1$. By the Hille–Yosida resolvent formula, $u_\lambda\in\mathrm{dom}(\mathcal{L})$ and

$$(\lambda-\mathcal{L})(-u_\lambda)=f,\qquad \text{equivalently}\qquad \mathcal{L}u_\lambda=f+\lambda u_\lambda. \tag{246}$$

The family $(u_\lambda)_{\lambda>0}$ is uniformly bounded in $\mathcal{H}_0$ by the estimate above and converges, as $\lambda\downarrow 0$, to

$$u := -\int_0^\infty e^{t\mathcal{L}}f\,dt \tag{247}$$

by dominated convergence. Since $\lambda u_\lambda\to 0$ in $\mathcal{H}_0$ and $\mathcal{L}$ is closed, the relation $\mathcal{L}u_\lambda=f+\lambda u_\lambda$ passes to the limit and gives $u\in\mathrm{dom}(\mathcal{L})$ together with $\mathcal{L}u=f$. Thus $(\mathcal{L}|_{\mathcal{H}_0})^{-1}$ exists and satisfies $\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|\le(1+\log C)/\nu$, hence $s(\mathcal{L})=\|(\mathcal{L}|_{\mathcal{H}_0})^{-1}\|^{-1}\ge\nu/(1+\log C)$, i.e. $\nu\le(1+\log C)\,s(\mathcal{L})$. □

Lemma D.2 (Relaxation time is bounded below by the singular value gap)**.** *Assume* $\ker(\mathcal{L})\cap\mathcal{H}_0=\{0\}$. *Assume moreover that* $e^{t\mathcal{L}}$ *is a contraction semigroup on* $\mathcal{H}_0$*, i.e.* $\|e^{t\mathcal{L}}f\|\le\|f\|$ *for all* $t\ge 0$ *and* $f\in\mathcal{H}_0$*. Assume there exist constants* $C\ge 1$ *and* $\nu>0$ *such that*

$$\|e^{t\mathcal{L}}f\|\le C\,e^{-\nu t}\,\|f\|,\qquad \forall t\ge 0,\ \forall f\in\mathcal{H}_0. \tag{248}$$

Define the L^2 *relaxation time*

$$t_{\mathrm{rel}}(\mathcal{L}) := \inf\Big\{t\ge 0\,;\ \|e^{t\mathcal{L}}f\|\le e^{-1}\,\|f\| \text{ for all } f\in\mathcal{H}_0\Big\}. \tag{249}$$

Then

$$t_{\mathrm{rel}}(\mathcal{L})\ge\frac{1}{2\,s(\mathcal{L})}. \tag{250}$$

This L^2 *bound is a continuous-time analogue of discrete-time relaxation estimates based on singular values for non-reversible Markov chains; compare* [10].

Proof. Let $\tau > t_{\mathrm{rel}}(\mathcal{L})$ be such that

$$\|e^{\tau\mathcal{L}} f\| \le e^{-1}\|f\|, \qquad \forall f \in \mathcal{H}_0. \tag{251}$$

Such a τ exists by the definition of the infimum. For any $t \ge 0$ write $t = n\tau + r$ with $n \in \mathbb{N}_0$ and $r \in [0, \tau)$. The contraction property and the semigroup identity yield

$$\|e^{t\mathcal{L}}\| \le \|e^{r\mathcal{L}}\|\, \|e^{\tau\mathcal{L}}\|^n \le e^{-n}. \tag{252}$$

Since $n \ge t/\tau - 1$, this implies

$$\|e^{t\mathcal{L}}\| \le e\, e^{-t/\tau}. \tag{253}$$

We may therefore apply Lemma D.1 with $C = e$ and $\nu = \tau^{-1}$ to obtain

$$\tau^{-1} \le (1 + \log e)\, s(\mathcal{L}) = 2\, s(\mathcal{L}). \tag{254}$$

Hence $\tau \ge (2s(\mathcal{L}))^{-1}$. Letting $\tau \downarrow t_{\mathrm{rel}}(\mathcal{L})$ gives the stated bound. □

D.2. **Test function and completion of the proof.**

Proof of Theorem 5.6. Let $\phi \in \mathrm{dom}(Q_O) \cap \mathcal{H}_S \cap \mathcal{H}_0$ be a normalized eigenfunction of Q_O with

$$Q_O\phi = -\lambda(Q_O)\, \phi, \qquad \|\phi\| = 1. \tag{255}$$

Define the first-order corrector

$$\psi := \mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi \in \mathcal{H}_F, \tag{256}$$

and the test function $f_\gamma := \phi - \gamma^{-1}\psi$. Since $\phi \in \mathcal{H}_S$ and $\psi \in \mathcal{H}_F$ we have $\langle \phi, \psi\rangle = 0$ and therefore

$$\|f_\gamma\|^2 = \|\phi\|^2 + \gamma^{-2}\|\psi\|^2 \ge 1. \tag{257}$$

Using $\mathcal{R}\phi = 0$ and $\mathcal{R}\mathcal{R}^+ = P_F$ on $\mathcal{H}$, together with centered coupling $\mathcal{V}_{\mathrm{skew}}\phi \in \mathcal{H}_F$, we obtain

$$\mathcal{R}\psi = \mathcal{R}\mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi = P_F\mathcal{V}_{\mathrm{skew}}\phi = \mathcal{V}_{\mathrm{skew}}\phi. \tag{258}$$

Consequently,

$$\mathcal{L}f_\gamma = \big(\gamma\mathcal{R} + \mathcal{V}_{\mathrm{skew}} + \eta\mathcal{V}_{\mathrm{pert}}\big)\big(\phi - \gamma^{-1}\psi\big) \tag{259}$$

$$= \underbrace{\mathcal{V}_{\mathrm{skew}}\phi - \mathcal{R}\psi}_{=0} - \gamma^{-1}\mathcal{V}_{\mathrm{skew}}\psi \;+\; \eta\mathcal{V}_{\mathrm{pert}}f_\gamma. \tag{260}$$

We decompose the remaining transport term into slow/fast components:

$$\mathcal{V}_{\mathrm{skew}}\psi = P_S\mathcal{V}_{\mathrm{skew}}\psi + P_F\mathcal{V}_{\mathrm{skew}}\psi. \tag{261}$$

For the slow part, using $Q_O := -P_S\mathcal{V}_{\mathrm{skew}}\mathcal{R}^+\mathcal{V}_{\mathrm{skew}}P_S$ and $P_S\phi = \phi$, we have

$$P_S\mathcal{V}_{\mathrm{skew}}\psi = P_S\mathcal{V}_{\mathrm{skew}}\mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\phi = -Q_O\phi = \lambda(Q_O)\, \phi. \tag{262}$$

For the fast part, apply the flux regularity estimate (25) with the slow input ϕ. Writing $g_\phi := P_F\mathcal{V}_{\mathrm{skew}}P_S\phi = \mathcal{V}_{\mathrm{skew}}\phi$ and noting that $(-\mathcal{R}^+)g_\phi = -\psi$ on $\mathcal{H}_F$, we obtain

$$\|P_F\mathcal{V}_{\mathrm{skew}}\psi\| = \|P_F\mathcal{V}_{\mathrm{skew}}P_F\psi\| = \|P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)g_\phi\| \le \kappa_{\mathrm{ell}}\, \|(-Q_O)\phi\| = \kappa_{\mathrm{ell}}\, \lambda(Q_O). \tag{263}$$

Combining the slow and fast contributions yields

$$\|\mathcal{V}_{\mathrm{skew}}\psi\| \le (1 + \kappa_{\mathrm{ell}})\, \lambda(Q_O). \tag{264}$$

Hence

$$\|\mathcal{L}f_\gamma\| \le \frac{(1 + \kappa_{\mathrm{ell}})\, \lambda(Q_O)}{\gamma} + \eta\, \|\mathcal{V}_{\mathrm{pert}}f_\gamma\|. \tag{265}$$

To bound the perturbation, use Assumption 4.1.4 to estimate

$$\|\mathcal{V}_{\rm pert} f\| \le \|\mathcal{K}_{\rm pert} f\| + \|\mathcal{S}_{\rm pert} f\| \tag{266}$$

$$\le (a_1 + a_2)\, \|(-\mathcal{R})^{1/2} f\| + (b_1 + b_2)\, \|f\|, \qquad f \in \mathrm{dom}((-\mathcal{R})^{1/2}). \tag{267}$$

Since $\phi \in \ker(\mathcal{R})$ and $f_\gamma = \phi - \gamma^{-1}\psi$, we have

$$\|(-\mathcal{R})^{1/2} f_\gamma\| = \gamma^{-1}\, \|(-\mathcal{R})^{1/2}\psi\|. \tag{268}$$

Finally, using $\psi = \mathcal{R}^+ \mathcal{V}_{\rm skew}\phi$ and the definition of Q_O,

$$\|(-\mathcal{R})^{1/2}\psi\|^2 = \langle (-\mathcal{R})\psi, \psi\rangle = \langle \mathcal{V}_{\rm skew}\phi, (-\mathcal{R}^+)\mathcal{V}_{\rm skew}\phi\rangle = -\langle Q_O\phi, \phi\rangle = \lambda(Q_O). \tag{269}$$

Therefore $\|(-\mathcal{R})^{1/2} f_\gamma\| = \sqrt{\lambda(Q_O)}/\gamma$, and

$$\|\mathcal{V}_{\rm pert} f_\gamma\| \le (b_1 + b_2)\, \|f_\gamma\| + (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{270}$$

Dividing by $\|f_\gamma\| \ge 1$ gives

$$\frac{\|\mathcal{L} f_\gamma\|}{\|f_\gamma\|} \le \frac{(1 + \kappa_{\rm ell})\, \lambda(Q_O)}{\gamma} + \eta\, (b_1 + b_2) + \eta\, (a_1 + a_2)\, \frac{\sqrt{\lambda(Q_O)}}{\gamma}. \tag{271}$$

By definition of $s(\mathcal{L})$, this implies $s(\mathcal{L}) \le \|\mathcal{L} f_\gamma\|/\|f_\gamma\|$. Combining with Lemma D.1 yields the claimed bound on ν. □

Appendix E. Proof of Proposition 6.2

We verify Assumptions 4.1.1–4 and 4.3 for a uniformly convex potential $U \in C^\infty(\mathbb{R}^d)$ with $\nabla^2 U \ge m > 0$ and with all derivatives of positive order growing at most polynomially,

$$\mathcal{V}_{\rm pert} f = F(q) \cdot \nabla_p f, \tag{272}$$

with $d \ge 2$, $m \in (0, 1]$, and $F \in C_b^\infty(\mathbb{R}^d; \mathbb{R}^d)$ bounded. We work on the Schwartz core

$$\mathcal{C}_{\rm Lan} := \mathcal{S}(\mathbb{R}^d_q \times \mathbb{R}^d_p) \subset L^2(\mu), \tag{273}$$

where $\mu(dq, dp) = Z^{-1} e^{-\beta(\frac{1}{2}|p|^2 + U(q))}\, dq\, dp$. As in Example 6.2, we only verify below the model-specific quantitative constants. The unbounded common-core and semigroup facts are standard for confining Kramers–Fokker–Planck operators and are quoted later in Proposition E.1. The invariance properties required for Assumption 4.3 are justified at the end of the appendix, in Proposition E.1; for the present structural computations we only use that for every slow test function $\psi = \psi(q)$ one has

$$\mathcal{R}^+ \mathcal{V}_{\rm skew}\psi = -p \cdot \nabla_q \psi \in \mathcal{C}_{\rm Lan}, \tag{274}$$

so all compositions in the definition of Q_O and in the corrector stay on the core. Since

$$Q_O = \beta^{-1}\Delta_q - \nabla_q U \cdot \nabla_q \tag{275}$$

is the weighted Laplacian on $L^2(\mu_q)$ with $\mu_q(dq) \propto e^{-\beta U(q)}\, dq$. We take its non-positive self-adjoint Friedrichs realization, whose form domain is $H^1(\mu_q)$ and whose operator domain is denoted $\mathrm{dom}(Q_O)$. Because U is uniformly convex with $\nabla^2 U \ge m > 0$, the Lax–Milgram construction of the resolvent together with the Friedrichs realization of the weighted Dirichlet form yields that $(\varpi - Q_O)^{-1}$ maps $L^2(\mu_q)$ into $\mathrm{dom}(Q_O)$ for every $\varpi > 0$; moreover $C_c^\infty(\mathbb{R}^d)$ is a core for this realization, hence so is the larger class $\mathcal{S}(\mathbb{R}^d) \subset \mathcal{C}_{\rm Lan} \cap \mathcal{H}_S$. Therefore the slow operator has the exact mapping and core properties required in Assumption 4.3. The explicit differential identities below are first checked on $\mathcal{C}_{\rm Lan}$ and then extended to the relevant weighted Sobolev spaces by closure of the corresponding bilinear forms.

E.1. **Microscopic spectral gap, centering, and slow gap.** The fast dissipator

$$\mathcal{R} = -p \cdot \nabla_p + \beta^{-1}\Delta_p \tag{276}$$

acts only on the Gaussian momentum variable and has spectral gap

$$\lambda_R = 1 \tag{277}$$

on $\mathcal{H}_F = \ker(\Pi_p)$. The Hamiltonian part $\mathcal{V}_{\text{skew}} = p \cdot \nabla_q - \nabla_q U \cdot \nabla_p$ is skew-adjoint in $L^2(\mu)$ because the Hamiltonian flow preserves μ. Since $\int p \, d\mu_p = 0$, one has

$$P_S \mathcal{V}_{\text{skew}} P_S = 0. \tag{278}$$

Also $P_S \mathcal{V}_{\text{pert}} P_S = 0$ because $\nabla_p \phi(q) = 0$ for slow functions.

For $\psi = \psi(q)$,

$$g_\psi := P_F \mathcal{V}_{\text{skew}} P_S \psi = p \cdot \nabla_q \psi. \tag{279}$$

Since $\mathcal{R}(p_i) = -p_i$, one has $(-\mathcal{R}^+) g_\psi = g_\psi$, so the conditioning constant is

$$M_R = 1. \tag{280}$$

The effective slow operator is

$$Q_O \psi = \beta^{-1}\Delta_q \psi - \nabla_q U \cdot \nabla_q \psi. \tag{281}$$

By the Poincaré inequality for the uniformly log-concave measure $\mu_q(dq) \propto e^{-\beta U(q)}\, dq$ with constant m—equivalently, the Brascamp–Lieb/Bakry–Émery spectral-gap estimate associated with $\nabla^2 U \geq m > 0$ [5]—one has

$$-\langle \psi, Q_O \psi \rangle_{L^2(\mu_q)} = \beta^{-1} \|\nabla_q \psi\|^2_{L^2(\mu_q)} \geq m \, \|\psi - \textstyle\int \psi \, d\mu_q\|^2_{L^2(\mu_q)}, \tag{282}$$

so Assumption 4.2 holds with

$$\lambda(Q_O) = m. \tag{283}$$

E.2. **Flux regularity (Assumption 4.1.3).** Define

$$Z_{ij}(p) := p_i p_j - \beta^{-1}\delta_{ij}. \tag{284}$$

Using $(-\mathcal{R}^+) g_\psi = g_\psi$, we compute

$$\begin{aligned} \mathcal{V}_{\text{skew}} g_\psi &= p \cdot \nabla_q (p \cdot \nabla_q \psi) - \nabla_q U \cdot \nabla_p (p \cdot \nabla_q \psi) \\ &= \sum_{i,j=1}^{d} p_i p_j \, \partial_{q_i q_j} \psi - \nabla_q U \cdot \nabla_q \psi. \end{aligned} \tag{285}$$

The p-average of this expression is

$$P_S \mathcal{V}_{\text{skew}} g_\psi = \beta^{-1}\Delta_q \psi - \nabla_q U \cdot \nabla_q \psi = Q_O \psi, \tag{286}$$

so subtracting the slow part gives

$$P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) g_\psi = \sum_{i,j=1}^{d} (p_i p_j - \beta^{-1}\delta_{ij}) \, \partial_{q_i q_j} \psi = \sum_{i,j=1}^{d} Z_{ij}(p) \, \partial_{q_i q_j} \psi. \tag{287}$$

The Gaussian fourth-moment identity

$$\mathbb{E}_{\mu_p}[Z_{ij} Z_{kl}] = \beta^{-2}(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}) \tag{288}$$

therefore gives

$$\|P_F \mathcal{V}_{\text{skew}} P_F (-\mathcal{R}^+) g_\psi\|^2_{L^2(\mu)} = 2\beta^{-2} \, \|\nabla_q^2 \psi\|^2_{L^2(\mu_q)}. \tag{289}$$

To control the Hessian, we use the Bochner identity. Set $L := -\beta Q_O = -\Delta_q + \beta \nabla_q U \cdot \nabla_q$, the weighted Laplacian on $L^2(\mu_q)$. For every $\psi \in \mathcal{S}(\mathbb{R}^d)$, one has the commutator

$$[\partial_{q_i}, L] = \beta \left(\partial^2_{q_i q_k} U\right) \partial_{q_k}, \tag{290}$$

obtained from $[\partial_{q_i}, \nabla_q U \cdot \nabla_q] = (\partial^2_{q_i q_k} U)\partial_{q_k}$ and $[\partial_{q_i}, -\Delta_q] = 0$. Since L is self-adjoint in $L^2(\mu_q)$ and $\langle g, Lg\rangle_{L^2(\mu_q)} = \|\nabla_q g\|^2_{L^2(\mu_q)}$,

$$\begin{aligned}
\|L\psi\|^2_{L^2(\mu_q)} = \langle \psi, L^2\psi\rangle_{L^2(\mu_q)} &= \sum_{i=1}^d \langle \partial_{q_i}\psi, \partial_{q_i}(L\psi)\rangle_{L^2(\mu_q)} \\
&= \sum_{i=1}^d \langle \partial_{q_i}\psi, L(\partial_{q_i}\psi)\rangle_{L^2(\mu_q)} + \beta \left\langle \nabla_q^2 U \, \nabla_q\psi, \nabla_q\psi\right\rangle_{L^2(\mu_q)} \\
&= \|\nabla_q^2\psi\|^2_{L^2(\mu_q)} + \beta \left\langle \nabla_q^2 U \, \nabla_q\psi, \nabla_q\psi\right\rangle_{L^2(\mu_q)}.
\end{aligned} \tag{291}$$

Since U is convex ($\nabla_q^2 U \geq 0$), the second term is non-negative, giving the *Bochner inequality*

$$\|\nabla_q^2\psi\|_{L^2(\mu_q)} \leq \|L\psi\|_{L^2(\mu_q)} = \beta \, \|(-Q_O)\psi\|_{L^2(\mu_q)}. \tag{292}$$

Combining with (289) yields

$$\|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) g_\psi\|_{L^2(\mu)} \leq \sqrt{2} \, \|(-Q_O)\psi\|_{L^2(\mu_q)}. \tag{293}$$

Thus Assumption 4.1.3 holds with

$$\kappa_{\text{ell}} = \sqrt{2}. \tag{294}$$

E.3. **Perturbation bounds (Assumption 4.1.4).** The $L^2(\mu)$-adjoint of $\mathcal{V}_{\text{pert}} = F(q) \cdot \nabla_p$ is

$$\mathcal{V}^\dagger_{\text{pert}} = -F(q) \cdot \nabla_p + \beta F(q) \cdot p, \tag{295}$$

so the skew/symmetric decomposition is

$$\mathcal{K}_{\text{pert}} = F(q) \cdot \nabla_p - \tfrac{\beta}{2} F(q) \cdot p, \qquad \mathcal{S}_{\text{pert}} = \tfrac{\beta}{2} F(q) \cdot p. \tag{296}$$

For $f \in \mathcal{C}_{\text{Lan}}$, the fast Dirichlet form of $\mathcal{R}$ gives

$$\|(-\mathcal{R})^{1/2} f\|^2_{L^2(\mu)} = \langle (-\mathcal{R}) f, f\rangle_{L^2(\mu)} = \beta^{-1} \, \|\nabla_p f\|^2_{L^2(\mu)}, \tag{297}$$

and by density the same identity extends to $f \in \operatorname{dom}((-\mathcal{R})^{1/2})$. Since F is bounded,

$$\|F \cdot \nabla_p f\|_{L^2(\mu)} \leq \|F\|_{L^\infty} \, \|\nabla_p f\|_{L^2(\mu)} \leq \sqrt{\beta} \, \|F\|_{L^\infty} \, \|(-\mathcal{R})^{1/2} f\|_{L^2(\mu)}. \tag{298}$$

For the multiplier part, the Gaussian integration-by-parts estimate in momentum gives

$$\||p| f\|_{L^2(\mu)} \leq \sqrt{2d/\beta} \, \|f\|_{L^2(\mu)} + (2/\beta) \, \|\nabla_p f\|_{L^2(\mu)}. \tag{299}$$

Therefore

$$\|F \cdot p \, f\|_{L^2(\mu)} \leq \|F\|_{L^\infty} \Big(\sqrt{2d/\beta} \, \|f\|_{L^2(\mu)} + \tfrac{2}{\beta} \, \|\nabla_p f\|_{L^2(\mu)} \Big). \tag{300}$$

Combining the last two bounds shows that Assumption 4.1.4 holds with constants (a_1, b_1, a_2, b_2) depending only on $(\beta, d, \|F\|_{L^\infty})$ and independent of m.

E.4. **Semigroup generation and core invariance.** The following proposition, which completes the verification of Assumption 4.3, addresses semigroup generation and justifies the Schwartz-class core.

Proposition E.1 (Semigroup generation and Schwartz core)**.** *Under the hypotheses of Proposition 6.2—in particular, the smooth polynomial-growth assumptions on U from Example 6.2—the closure of $\mathcal{L}|_{\mathcal{C}_{\mathrm{Lan}}}$ generates the standard nonequilibrium Langevin semigroup on $L^2(\mu)$. The abstract passage from core estimates to semigroup decay is the standard DMS closure step* [14, §2]*; the required confining Kramers–Fokker–Planck regularization and common-core facts are classical in* [22, Introduction and §§1–2]*; see also* [21]*. Moreover, $\mathcal{C}_{\mathrm{Lan}} = \mathcal{S}(\mathbb{R}^{2d})$ satisfies all five items of Assumption 4.3.*

Proof. The semigroup-generation statement is standard for confining Kramers–Fokker–Planck operators with smooth polynomial coefficients and bounded first-order perturbations in the momentum variable; the precise references are the ones quoted in the proposition. We therefore record only the concrete core facts used in this paper.

Because the coefficients of $\mathcal{R}$, $\mathcal{V}_{\mathrm{skew}}$, and $\mathcal{V}_{\mathrm{pert}}$ are smooth and have at most polynomial growth, each of these differential operators maps $\mathcal{C}_{\mathrm{Lan}}$ into itself, and the Gaussian conditional expectation operators P_S and P_F preserve $\mathcal{C}_{\mathrm{Lan}}$ as well. The Ornstein–Uhlenbeck operator $\mathcal{R}$ diagonalizes in the Hermite basis in the p-variable, so $\mathcal{R}^+$ preserves the Schwartz class. In particular, for every slow test function $\psi \in \mathcal{S}(\mathbb{R}^d_q)$,

$$\mathcal{R}^+\mathcal{V}_{\mathrm{skew}}\psi = -p \cdot \nabla_q \psi \in \mathcal{C}_{\mathrm{Lan}}. \tag{301}$$

These are exactly the mapping properties needed for item (iii) of Assumption 4.3. The remaining items follow from the cited Kramers–Fokker–Planck common-core results together with the density and continuity arguments already established in the preceding subsections. □

□

Appendix F. Proof of Proposition 6.4

All operators are bounded on the finite-dimensional space $\mathcal{H}$.

(1) Assumption 4.1.1 holds for $\mathcal{R}$ by construction, with $\lambda_R = 4\Gamma$.
(2) Assumption 4.1.2 holds since both $\mathcal{V}_{\mathrm{skew}}$ and $\mathcal{V}_{\mathrm{pert}}$ map $\mathcal{H}_S$ into $\mathcal{H}_F$.
(3) Assumption 4.1.3 and Assumption 4.2 are both verified explicitly. The key improvement over a single-channel coupling (e.g., $H_{\mathrm{int}} = g\sigma_z \otimes \sigma_z$) is that the double commutator $\mathrm{ad}^2_{\sigma_a}$ has kernel $\mathrm{span}\{I, \sigma_a\}$; with two channels,

$$\ker(Q_O) \cap \mathcal{H}_0 = \ker(\mathrm{ad}^2_{\sigma_x}) \cap \ker(\mathrm{ad}^2_{\sigma_z}) \cap \{\mathrm{tr} = 0\} = \{0\}, \tag{302}$$

so Assumption 4.2 holds with $\ker(Q_O) = \mathbb{C}\mathbf{1}$ and $\lambda(Q_O) = 4g^2/\kappa = g^2/\Gamma$; the numerical values in Table 2 simply reproduce this exact formula.

For the flux regularity bound (25), define the bounded operator

$$\mathcal{K} := P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)P_F\mathcal{V}_{\mathrm{skew}}P_S : \mathcal{H}_S \to \mathcal{H}_F. \tag{303}$$

Because $P_S\mathcal{V}_{\mathrm{skew}}P_S = 0$, both $\mathcal{K}$ and $-Q_O$ annihilate the stationary direction $\mathbb{C}\mathbf{1}$. On the finite-dimensional mean-zero slow space $\mathcal{H}_{S,0}$, the kernel computation above shows that $-Q_O|_{\mathcal{H}_{S,0}}$ is positive definite and therefore invertible. We may thus set

$$\mathcal{T} := \mathcal{K}\,(-Q_O|_{\mathcal{H}_{S,0}})^{-1} : \mathcal{H}_{S,0} \to \mathcal{H}_F, \qquad \kappa_{\mathrm{ell}} := \|\mathcal{T}\| < \infty. \tag{304}$$

Writing any slow vector as $\psi = c\mathbf{1} + \psi_0$ with $\psi_0 \in \mathcal{H}_{S,0}$ and using $g_{\mathbf{1}} = 0$, we obtain

$$P_F\mathcal{V}_{\mathrm{skew}}P_F(-\mathcal{R}^+)g_\psi = \mathcal{K}\psi_0 = \mathcal{T}(-Q_O)\psi_0, \tag{305}$$

and hence

$$\|P_F \mathcal{V}_{\text{skew}} P_F(-\mathcal{R}^+) g_\psi\| \le \kappa_{\text{ell}} \|(-Q_O)\psi_0\| = \kappa_{\text{ell}} \|(-Q_O)\psi\|. \tag{306}$$

Thus (25) holds analytically in this finite-dimensional model, and one may take $M_R := \|\mathcal{R}\|$.

(4) For Assumption 4.1.4, decompose $\mathcal{V}_{\text{pert}} = \mathcal{K}_{\text{pert}} + \mathcal{S}_{\text{pert}}$ by $\mathcal{K}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} - \mathcal{V}_{\text{pert}}^\dagger)$ and $\mathcal{S}_{\text{pert}} := \frac{1}{2}(\mathcal{V}_{\text{pert}} + \mathcal{V}_{\text{pert}}^\dagger)$. Since $\mathcal{V}_{\text{pert}}$ is bounded, the energy bounds (28)–(29) hold with $a_1 = a_2 = 0$ and $b_1 = \|\mathcal{K}_{\text{pert}}\|$, $b_2 = \|\mathcal{S}_{\text{pert}}\|$.

□

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